BALIKBAYANI PORTAL: A FILING SYSTEM FOR MIGRANT RECORDS IN THE PROCESSING DIVISION OF THE DEPARTMENT OF MIGRANT WORKERS

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In Partial Fulfillment of the Requirements for the Degree
Bachelor of Science in Information Technology
Specialized in Web and Mobile Application Development

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DEDICATION

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A.N.L

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ABSTRACT

The Processing Division of the Department of Migrant Workers (DMW) Region IV-A manages thousands of records for Overseas Filipino Workers (OFWs), with document handling that often involves slow and time-consuming procedures. To address this challenge, the study titled “BalikBayani Portal: A Filing System for Migrant Records in the Processing Division of the Department of Migrant Workers” aimed to digitize the filing process by integrating document generation and data visualization. The project was developed using the Agile Scrum methodology. System evaluation was conducted using the Technology Acceptance Model (TAM) to assess user perception and ISO 25010 to evaluate technical quality. Results from 104 TAM respondents yielded an overall mean score of 3.78, which was interpreted as high, while evaluations from 3 IT professionals using ISO 25010 received an overall mean score of 4.67, interpreted as excellent, indicating that the system met essential requirements for performance, usability, reliability, and security. Positive user feedback confirms the system’s alignment with user needs, demonstrating its potential as an effective platform for record management. In conclusion, the BalikBayani Portal meets its technical and functional goals and aligns with the Department of Migrant Workers’ mission to promote digital transformation and improve service delivery. With continued improvement and institutional support, the BalikBayani Portal can serve as a model for digital innovation in migrant worker governance.

Keywords: *Agile Scrum, BalikBayani Portal, Data Visualization, Department of Migrant Workers, Document Management, Filing System, ISO 25010, Technology Acceptance Model (TAM)*

TABLE OF CONTENTS

Preliminaries	***Page***
Title Page	i
Approval Sheet	ii
Acknowledgement	iii
Dedication	v
Abstract	vi
Table of Contents	vii
List of Figures	ix
List of Tables	x
Definition of Terms	xii
CHAPTER	
I INTRODUCTION	
Project Context	1
Objectives	4
Project Purpose	5
Scope and Limitations	6
II THEORETICAL FRAMEWORK OF THE PROJECT	
Review of Related Literature, Studies, and Systems	12
Synthesis and Relevance of the Study	24
Literature Matrix	26
Conceptual Framework	28
III METHODOLOGY	
Project Design	30
Software Development Model	33
Product Backlog	35
Sprint Planning	38
Sprint Execution	42
Flowchart	43
Use Case Diagram	46
Materials	47
Software	47
Hardware	48
Project Testing and Evaluation	49
Unit Testing	50
Integration Testing	51
System Testing	52
Compatibility Testing	52
Experimental Setup	53
Evaluation Procedure	54
Population of the Study	56
Sampling Design	57
Data Collection Instrument	57
Statistical Treatment	58

IV RESULTS AND DISCUSSIONS

Results by Objectives 1 61
Results by Objectives 2 68
Results of Unit Testing 71
Results of Integration Testing 78
Results of Compatibility Testing 80
Results of Experimental Setup 83
Results of ISO/IEC 25010 Evaluation 84
Results of TAM Evaluation 86

V SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

Summary 92
Conclusions 94
Recommendations 96

REFERENCES
APPENDICES

Appendix A. Letter of Approval
Appendix B. Figures
Appendix C. Compatibility Result
Appendix D. Survey Questionnaires
Appendix E. TAM and ISO 25010 Survey Result
Appendix F. Sample of Output
Appendix G. Source Code
Appendix H. Formulas
Appendix I. Grammarly Writing Assistant Report
Appendix J. Turnitin Similarity Report
Appendix K Curriculum Vitae
Appendix L. Publishable Paper (IEEE format)

LIST OF FIGURES

No.	Title	Page
1	Conceptual Framework	29
2	System Architecture	32
3	Agile Scrum	34
4	Flow Chart	45
5	Use Case Diagram	46
6	Project Testing and Evaluation	50
7	Unit Testing	51
8	Integration Testing	52
9	Compatibility Testing	53
10	ISO 25010 Software Product Quality Characteristics	55
11	Technology Acceptance Model	56
12	Direct Hire - View Modal	62
13	Balik Manggagawa - View Modal	63
14	Gov-to-Gov - View Modal	64
15	Information Sheet - View Modal	65
16	Monitoring List - Job Fairs View Modal	66
17	Monitoring List - Job Fair Monitoring Summary View Modal	67
18	Super Admin Panel - System Reports	68
19	Dashboard	69
20	Direct Hire Overview	70
21	Balik Manggagawa Overview	70
22	Information Sheet Overview	71

LIST OF TABLES

No.	Title	Page
1	Literature Matrix	26
2	Product Backlog (Admin)	36
3	Product Backlog (Staff)	37
4	Product Backlog (Migrant Worker)	38
5	Sprint Backlog (Admin)	40
6	Sprint Backlog (Staff)	41
7	Sprint Backlog (Migrant Worker)	42
8	Hardware Requirements	48
9	Hardware Requirements for PC/Laptop	49
10	Experimental Setup for Process Cycle Time Simulation	54
11	5-point Likert Scale Method	60
12	4-point Likert Scale Method	60
13	Unit Testing for Direct Hire Module	72
14	Unit Testing for Balik Manggagawa Module	73
15	Unit Testing for Gov-to-Gov Module	74
16	Unit Testing for Information Sheet Module	75
17	Unit Testing for Monitoring List Module	76
18	Unit Testing for User Profile Module	76
19	Unit Testing for Admin Panel Module	77
20	Integration Testing of the System	79
21	Compatibility Testing of the System	80
22	Process Time per Client: Manual vs. System	84
23	Post Evaluation Result	85
24	Perceived Usefulness	87
25	Perceived Ease of Use	88

26	Attitude Towards Using	89
27	Behavioral Intention to Use	90
28	Evaluation of Survey Results	91

DEFINITION OF TERMS

Terms	Definition
Agile Scrum Methodology	A project management framework emphasizing iterative development, collaboration, and continuous feedback for system development.
Attitude Towards Use	A person's overall positive or negative feelings towards using a system or technology.
Balik Manggagawa	OFWs who are returning to continue or extend their employment contract with the same or a different employer or job site.
Behavioral Intention to Use	A person's desire or intent to use a system or technology in the future.
Department of Migrant Worker	A Philippine government agency created to safeguard and advocate for the rights of OFWs and their families.
Direct Hire	The process by which a foreign employer recruits a Filipino worker overseas without the use of PRA.
Gov-to-Gov	Interaction and collaboration between the Philippine government and the government of a partner country, with PRA.
Information Sheet	Centralized repository for managing all records and transactions associated with OFWs.
Interaction Capability	The ability of a system to support effective user-system interaction through user-friendly interfaces.

ISO 25010	An international standard that establishes a thorough quality model for the evaluation of software products.
Overseas Filipino Worker	Referring to Filipino nationals working abroad to improve the living standards of their family in the Philippines.
Perceived Ease of Use	The degree to which a person believes that using a system or technology will take no effort.
Perceived Usefulness	The degree to which a person believes that using a specific system or technology will improve their performance or allow them to complete tasks more effectively.
Performance	The degree to which a system runs efficiently, including its responsiveness and task-handling ability.
Philippine Recruitment Agency (PRA)	Refers to agencies that assist foreign companies in recruiting Filipino workers for overseas employment.
Process Cycle Time	Refers to the total time it takes to complete a process from start to finish.
Reliability	The ability of a system to function consistently and properly throughout time, without failure or error.
Security	The ability of a system to protect data and resources against unauthorized access.
Technology Acceptance Model (TAM)	A framework used to evaluate user acceptance and usability of the system, focusing on factors such as perceived ease of use and perceived usefulness.

CHAPTER I

INTRODUCTION

Project Context

In today's fast-paced and constantly evolving technological landscape, the digital world has become essential to our daily lives (GB Times, 2024). Digital technology has interconnected almost every aspect of our lives, from communication and interaction to business operations. As a result, information became one of the most valuable assets, driving innovations and competence in various fields. EncompaaS (2024) stated that, with the growing reliance on data-driven systems, the proper management of records and documents has become important for any organization, contributing to increased productivity, legal compliance, and informed/evidence-based decision-making. Effective records management also supported accountability and continuity of government operations, especially in agencies that handled high volumes of sensitive information.

One sector that handled a particularly large volume of information was the government, which was responsible for processing important and sensitive data, such as legal documents and personal records. These records were essential for services including identity verification, social welfare distribution, and labor deployment. Inefficient handling of such records often resulted in processing delays, which could negatively affect both service providers and beneficiaries. In response, many government institutions globally had begun shifting towards digital governance systems to improve service delivery.

Within this sector, the processing division of the Department of Migrant Workers (DMW) played a vital role in managing migrant workers' records and ensuring compliance with both local and international regulations. The DMW also managed worker protection and welfare programs and coordinated with foreign labor offices to protect the rights and interests of migrant workers.

Based on 2024 data, the CALABARZON region recorded 229,884 land-based and 106,876 sea-based migrant workers, resulting in an estimated 336,760 migrant worker records processed annually. Notably, CALABARZON has the highest number of deployed migrant workers among all regions in the Philippines, making it a critical area for efficient management of migrant worker records. The growing number of applications further increased the demand for faster processing and accurate documentation.

Despite the scale and importance of its operations, the DMW continued to rely heavily on traditional and manual document processing methods. Records were primarily managed using basic applications such as Microsoft Excel and Word. While these tools were widely accessible, they were not designed for large-scale document management and have limitations in scalability, data security, error prevention, and processing efficiency. Bowers (2022) further emphasized that manual filing became cumbersome because it took up a lot of space, was prone to damage and misplacement, was difficult to modify, was time-consuming, and lacked security measures.

The inefficiency of the current process was evident in the time required to complete transactions. For instance, processing under the Direct Hire category took an estimated 30 minutes per client, while applications under the

Balik-Manggagawa (BM) category took more than one week to complete. These variations in processing time were primarily due to the status and completeness of the worker's application, including the need for additional document verification, compliance checks, and case-specific assessments. Such delays affected not only administrative efficiency but also migrant workers' employment opportunities and welfare.

Given the increasing volume of records and the critical role of timely document processing, there was a clear need to modernize the DMW's existing system. The continued reliance on manual processes highlighted structural inefficiencies that hindered service delivery and limited the agency's capacity to respond effectively to growing demands. These challenges underscored the importance of adopting a centralized, web-based records management system that could standardize procedures and support faster decision-making.

Thus, it was important to focus on the inconsistency in document processing within the Department of Migrant Workers (DMW), as the results determined employment opportunities and migrant workers' well-being. Consequently, this motivated the researchers to conduct a study on the BalikBayani Portal, a filing system for migrant worker records. The study aims to determine how the BalikBayani Portal could improve document processing, minimize human error, and reduce bureaucratic procedures by collecting quantitative data from system testing and user feedback. Moreover, it sought to identify the underlying challenges of manual processing and the experiences of staff handling these documents. This research helps improve

accreditation processes and modernize migrant worker governance, creating a faster, more reliable system. As such, the results were beneficial to the Department of Migrant Workers, migrant workers, employees, researchers, and future researchers.

Project Objectives

The main objective of this capstone project is to develop the BalikBayani Portal, a system designed to digitize the filing process of migrant worker records in the DMW processing division, thereby improving workflow efficiency by reducing manual document processing.

System Development Objectives

Specifically, the study aims to:

1. Develop a filing system that integrates document generation into a printable format and supports quick, accurate retrieval of information.
2. Integrate a data visualization feature that provides representation across the main categories.

Evaluation Objectives

Additionally, the study aims to:

3. Test and evaluate user acceptance of the system using the Technology Acceptance Model (TAM) and the system's effectiveness based on the ISO 25010 software quality model, focusing on performance, interaction capability, reliability, and security.

Project Purpose

The purpose of this study is to improve the filing process of records by developing a digitized system for the Processing Division of the Department of Migrant Workers (DMW). This system intends to help achieve various UN Sustainable Development Goals. It aligns with SDG 8 (Decent Work and Economic Growth) by creating a more organized and secure work environment for staff through modernized processes. It also contributes to SDG 10 (Reduced Inequality) and SDG 16 (Peace, Justice, and Strong Institutions) by ensuring fair and ethical processing and promoting equitable access to services through an accountable, transparent digital system. Additionally, by improving workflows and protecting confidential information, the system supports SDG 9 (Industry, Innovation, and Infrastructure) by developing a reliable, sustainable digital infrastructure to improve public services. This study is highly valuable and beneficial to the following groups:

Migrant Workers. The system increased the speed and accuracy of processing migrant workers' records, leading to better service delivery. This means quicker access to services and a more transparent process for managing and monitoring their documents. Through real-time updates and online submission features, the system allows migrant workers to stay informed about the status of their records without the necessity of frequent in-person visits.

Employees. The new system simplified record-keeping by reducing manual steps and reducing the risk of mistakes. This would enable staff in the Processing Division to be more productive and concentrate on other tasks.

Department of Migrant Workers (DMW). This study has the potential to increase operational efficiency significantly. The system reduced the time and resources required for manual processing and minimized errors. In return, it improves the department's capacity to handle a higher number of migrant workers. Also, the system can be made to grow with future developments and increasing technological needs.

Researchers. As part of the community, the researchers are also among the beneficiaries of the system. The researcher can use their system to test, evaluate, and observe, and to upgrade it to meet user demand. It also benefits the researchers, as it is one of their academic requirements for completing the bachelor's degree.

Future Researchers. This study serves as a starting point for future research or as a basis for evaluating the validity of similar findings. It would be valuable to them when undertaking studies related to this topic, and can also serve as a reference when reviewing data and improving the system's functionality.

Scope and Limitations

This study covers the complete development of the system's core features, which were primarily used by the Processing Division of the Department of Migrant Workers (DMW) Region IV-A, located in Brgy. Parian, Calamba City, Laguna. The Portal digitizes the filing and storage of migrant workers' records, replacing manual processes with a more organized, accessible platform. It was accessible through both desktop and mobile

browsers, providing DMW staff and migrant workers with a user-friendly platform for managing and submitting required documents.

The project consists of the following modules/pages:

Login/Authentication Module. The system implements a login/authentication module that manages both login and access control. It supports role-based authentication, ensuring that only authorized users can access specific features. Migrant workers register by filling out a sign-up form that collects their personal information, and once registered, they can access their account using their email and password. On the other hand, the administrator creates an account for DMW staff, assigning them a temporary username and password. When the system detects a first login, a form is displayed prompting the staff to provide their basic information and update the temporary password with a more secure one.

Dashboard Module. This module serves as the landing page for admin and staff after creating or logging in to their account. It generates statistical and analytical reports based on stored data. It provides insights into the number of clients served, sex distribution, and processing volume across the different categories.

Direct Hire Module. This handles the applications of workers who are either going abroad or joining a company for the first time. It allows applicants to enter the required information and upload the necessary documents via a file attachment feature. DMW staff then verify and review the submitted records. Once the applicant’s information and documents have been checked and approved, the system issues an official Direct Hire clearance. If any

information is found to be incorrect or if any uploaded document has issues, the staff would notify the applicant through their account and email to correct the specific issue, allowing them to update the necessary information or documents without resubmitting the entire application.

Balik Manggagawa Module. This module manages applications from overseas workers who are temporarily on vacation and are set to resume employment. It allows returning migrant workers to enter the required information and upload the necessary documents via a file attachment feature. DMW staff then verify and review the records until the application is approved. If an issue arises with the worker's status, a clearance form is required. Clearances apply to cases such as Critical Skill, For Assessment Country, No Verified Contract Clearance, Non-Compliant Country Clearance, Seafarer's Position, Watchlisted Employer Clearance, and Watchlisted OFW. These must be processed and approved before the worker can proceed.

Gov-to-Gov Module. The Gov to Gov Module manages records of migrant workers processed directly between the Philippine government and the government of a partner country through PRA. This module allows migrant workers to enter the required information and upload the necessary documents via a file attachment feature. DMW staff then verify and review the submitted records. Once the applications are validated, the staff can endorse the migrant workers for deployment under gov-to-gov programs.

Information Sheet Module. This module serves as a centralized repository for managing all records and transactions related to OFWs. The system features summary statistics and serves as a repository for all

issuances of previously released documents or records about OFWs. Every transaction is carefully documented, containing the particular documents obtained and their specified purpose.

Notification. The system automatically notified migrant workers via email and the portal about their application updates. Staff received notifications in the portal whenever workers complied or when action was needed, ensuring timely processing and a smooth workflow.

Monitoring List (List of Job Fairs) Module. This module provides updated records of scheduled job fairs to serve as a reference for upcoming employment opportunities. The creation includes information such as the date, venue, office head, email address, and contact numbers to ensure organized scheduling and proper dissemination of information.

Monitoring List (PESO Contact Info) Module. This module displays the list of Public Employment Service Office (PESO) contact information of Region IV-A (CALABARZON). The directory includes the details such as the province, PESO office, office head, contact number/s, and email address.

Monitoring List (PRAs Contact Info) Module. This module stores the contact information of verified Private Recruitment Agencies (PRA) for recruitment-related communication. The monitoring table shows the PRA name, contact person, office head, email address, and contact number (s).

Monitoring List (Job Fair Monitoring Summary) Module. This module generates summary reports and statistics of completed job fairs. Each record includes the job fair date, venue, number of invited agencies, number

of invited agencies with Job Fair Authority (JFA), total number of applicants by gender, total number of applicants, and the number of DMW staff assigned to the job fair.

Super Admin Panel (User Management) Module. This module allows the Super Admin to oversee all registered users in the system. It also allows the Super Admin to view user activity status (active or inactive), manage roles, and, if necessary, set access permissions. In this module, access can be restricted to specific categories, such as Direct Hire, Balik Manggagawa, Gov-to-Gov, Information Sheet, and Monitoring List.

Super Admin Panel (Data Backups) Module. This module allows the administrator to manually back up, import, export, restore, or delete system files. It ensures that records are securely stored and can be recovered in the event of system errors. Regular use of this module helps maintain system continuity and protect migrant worker data.

Super Admin Panel (System Reports). This page allows the administrator to generate a Certificate of Utilization. The generated report includes monthly, quarterly, and annual summaries of the total number of clients served by the DMW.

User Profile Module. The User Profile Module allows system users to manage their personal account information and security. It allows them to view and update their personal information and account details, perform security verification, and change their passwords. This module ensures that users' data remain accurate and current, supporting smooth processing of applications and records.

Limitations

While the system provides an organized and accessible platform for managing migrant worker records, certain limitations must be considered. The system is designed strictly for division-level operations within the Processing Division of DMW Region IV-A. It does not replace or directly integrate with national DMW databases or centralized government systems. It functions solely as an operational support system to assist in local record processing and monitoring.

In terms of data privacy, the system handles only records necessary for processing division transactions and operates under controlled access via role-based authentication. The study does not cover system interoperability with external agencies or departments outside the DMW. For administrators and staff, the system is intended to operate within an intranet environment, allowing secure access through authorized computers connected to the internal network. This setup helps maintain data confidentiality and prevents unauthorized external access.

On the other hand, Overseas Filipino Workers (OFWs) require a stable internet connection to submit applications and upload documents online. All data transactions and updates are stored and synchronized within the local server to preserve system integrity. Despite the availability of online submissions, migrant workers may still be required to appear in person for identity verifications or other procedural requirements mandated by DMW policies.

CHAPTER II

THEORETICAL FRAMEWORK OF THE PROJECT

Review of Related Literature, Studies, and Systems

This chapter presents related literature, studies, and existing systems that are useful in comprehending the key concepts and issues relevant to the development and implementation of the BalikBayani Portal.

E-Governance

The use of information and communication technologies (ICT) has been a means of developing a country's democratic system and raising public engagement in various decision-making processes (Sharma et al., 2021). One such ICT-enabled effort that governments around the world have embraced is e-government. It involves providing or permitting citizens to access a wide range of internet services and government decision-making procedures (Fan & Yang, 2015). By digitizing government services, e-governance reduces the reliance on physical offices and face-to-face transactions, which can improve efficiency and accessibility, particularly in geographically dispersed regions.

Recently, e-governance has evolved into more interactive forms, such as online platforms and social media channels, enabling individuals to participate in governance processes actively (Manoharan et al., 2022). These interactive platforms allow citizens not only access services but also to provide feedback, track service requests, and report issues, thereby strengthening public accountability and responsiveness. A research study conducted in India shows that citizens’ adoption of e-government portals

depends on the clarity of information, relevance of content to their needs, and reliability (Singh et al., 2022). These findings emphasize that intuitive navigation and timely updates are critical to the success of digital government initiatives. Additionally, real-time notifications and automated updates can further improve user engagement and trust.

Carter and Bélanger (2005) contended that perceived usability and trust significantly influenced people's desire to use online government services. This implied that both public trust and technological excellence influenced the success of e-governance. Effective online systems can reduce administrative delays and improve service delivery for citizens. Ensuring reliability and security in digital platforms encourages sustained engagement and fosters positive perceptions of government efficiency.

E-governance is associated with accountability, transparency, and accessibility of services. As stated by Almarabeh & AbuAli (2010), e-government programs offered a way to reduce administrative delays and the potential for corruption by automating transactions and ensuring that documents are digitally traceable. The use of ICT in public service promoted good governance, particularly in poorer nations where institutional incompetence was evident. Several studies also found that e-governance not only made services more accessible but also improved efficiency, security, transparency, and overall service delivery (Branofino, 2024; Sharmin & Chowdhury, 2025; Kader et al., 2025).

Globally, the United Nations E-Government Survey (UN DESA, 2020) highlighted that digital governance helped achieve the Sustainable

Development Goals (SDGs) by encouraging innovations in service delivery and increasing citizen involvement. Several studies stated that e-governance should have prioritized public empowerment through participatory platforms that promoted communication and collaborative decision-making, in addition to digitizing administrative services. Programs and initiatives such as the Philippine e-Government Master Plan (EGMP), the DFA Passport Appointment System, SSS Online Registration, and BIR eFPS helped modernize government processes and promote inter-agency collaboration. Nonetheless, despite these advancements, global and local studies showed issues such as digital literacy gaps, unequal access to ICT between urban and rural areas, cybersecurity threats, and resistance to technological change (Zou et al., 2023; Kader et al., 2025).

Department of Migrant Workers

Republic Act No. 11641, also known as “An Act Creating the Department of Migrant Workers,” was signed into law by former President Rodrigo Duterte on December 30, 2021. This act established the DMW, defined its powers and functions, and rationalized the organization and roles of government agencies involved in overseas employment and labor migration. The DMW was responsible for safeguarding the rights and welfare of Overseas Filipino Workers (OFWs), providing legal support, and carrying out a full-cycle national reintegration program. The implementing rules and regulations of RA 11641 were promulgated on April 4, 2022, to establish the department's structure and operating guidelines. The DMW was also responsible for processing legal documents for OFWs, including recruitment

requirements, overseas employment contracts, and resolving disputes that arose during the immigration process (Supreme Court E-Library, n.d.; DMW, n.d.).

Given the crucial role of the DMW in processing the legal documents of migrant workers, it was essential to understand what constituted a document. The Oxford Dictionary (n.d.) defined a document as an official paper, electronic file, or record that provided information or served as proof of something. For migrant workers, such documents are necessary because they validate their identity, qualifications, and employment terms. These included essential requirements such as employment contracts, passports, work visas or permits, medical clearances, and certification documents (Behnke, 2025). Without these documents, migrant workers became vulnerable to human trafficking, forced labor, and violations of labor laws (Fabbri et al., 2023). For this reason, the role of the DMW was particularly evident, as it ensured that all required documents were properly processed before migrant workers left the country. By verifying and organizing these documents, the DMW played a key role in safeguarding migrant workers and ensuring compliance with legal and employment regulations.

Migrant Worker

Migration has been an ongoing global phenomenon driven by better economic opportunities and improved living conditions. Many individuals traveled outside their home countries in search of jobs, higher wages, and a better standard of living. IOM UN Migration (2024) reported that the number of international migrants increased from 128 million in 2000 to 281 million in

2020, representing 3.6 percent of the global population. As migration continues to rise, governments are challenged to strengthen policies and systems that ensure the protection proper management of migrant workers

The Philippines had become one of the world's largest labor-exporting countries. According to a Social Weather Stations (SWS) survey, 7% of adult Filipinos were actively seeking employment abroad. The Philippine Statistics Authority reported that by 2021, there were 1.83 million Overseas Filipino Workers (OFWs), and about 800,000 additional workers left for work in 2022. Among the 17 regions in the country, CALABARZON consistently served the largest number of OFWs (PSA, 2024).

OFWs were a vital component of the national economy. Approximately 75% of households received remittances, totaling $32.14 billion in 2022 and contributing 8.9% to the country’s GDP. These remittances supported families, financed education, and stimulated local businesses, leading to the recognition of OFWs as modern-day heroes and contributors to national economic stability. The survey further reveals that the top destination countries for Filipino workers include Canada, Saudi Arabia, Kuwait, and the United States (Agcaoili, 2023; Filipino Journal, 2023).

The COVID-19 pandemic disrupted worldwide migrant labor markets and exposed vulnerabilities and insufficient worker protections. During this period, the Philippines experienced a 75 percent decrease in overseas deployment. They faced an unprecedented number of repatriations, as many Overseas Filipino Workers (OFWs) lost their jobs, suffered income reduction, or returned home due to sudden contract terminations (IOM, 2020). Despite

these challenges, Filipino workers remained in high demand across sectors such as healthcare, domestic work, and construction (Eugenio, 2023). These events highlighted the need for digital system that enable real-time monitoring and secure management of migrant worker records to reduce delays and improve protection during crises.

Migrant Worker Records

Digital platforms have become increasingly important in managing migrant worker records to improve service accessibility and transparency. E-governance initiatives in record management simplified the processes of collecting, storing, and retrieving data, allowing governments to deliver services promptly and reduce administrative constraints (Bhatnagar, 2014). Additionally, using ICT tools in the context of labor migration enabled agencies to track deployments, maintain up-to-date worker profiles, and ensure access to welfare benefits and services.

Several studies also highlighted the need for secure, reliable record-keeping methods to protect the information of foreign workers. Vassilakopoulou and Hustad (2021) pointed out that data integrity and privacy protections built user trust and encouraged the adoption of e-government systems. Secure digital solutions reduced the risks of data loss and fraud for migrant workers, whose employment and personal information frequently cross international borders.

Several government agencies in Asia have implemented digital platforms designed specifically for migrant workers. In particular, India's e-Migrate system connected government officials, employers, and recruitment

agencies on a single platform, simplifying the registration and supervision of foreign workers (Ministry of External Affairs, 2015). This system was shown to reduce abuse by unregulated intermediaries and to increase transparency in the recruitment process (Khadria, 2017). Similarly, Singapore's Work Permit Online (WPOL) system reduced processing time and paperwork by enabling organizations to manage foreign worker records and complete applications digitally (Ng & Tan, 2018). Meanwhile, in the Philippines, the Overseas Workers Welfare Administration (OWWA) and the Department of Migrant Workers (DMW) have implemented systems to oversee the welfare and deployment services provided to overseas Filipino workers (OFW).

Records Management

A record's "lifecycle" referred to the processes a record underwent from creation to disposal (Tab, 2019). Shepherd and Yeo (2004) emphasized the significance of record management in the context of rapidly evolving information and communication technologies. They argue that as digital records became more prevalent, organizations needed to implement robust records management strategies to ensure the trustworthiness, reliability, and usefulness of records going forward. The shift from traditional paper-based systems to electronic records presents both opportunities and challenges, requiring a strategic approach to managing records in line with legal requirements and industry guidelines.

Moreover, effective records management serves as a bridge between technological advancement and organizational governance by ensuring that digital records remain authentic, accessible, and secure throughout their

lifecycle. It supports informed decision-making by providing accurate and timely information, while also promoting accountability among personnel through clearly defined record-keeping responsibilities. In the public sector, integrating records management practices into daily operations helps institutions respond more efficiently to stakeholder needs, reduce administrative delays, and reinforce transparency.

In the public sector, Tajudin and Shukor (2024) stated that record management was an essential tool for enhancing good governance, especially in addressing prevalent problems such as corruption, fraud, and record alteration. Furthermore, records management was viewed as an essential function within the organization. Based on their study, a conceptual framework that embraces professionalism, accountability, transparency, and effective service delivery was proposed to strengthen the governance practices. Moreover, as noted by Demirtel and Bayram (2014), proper record management is crucial for institutions to achieve their administrative and operational goals. Public institutions use ERM systems to increase the efficiency of these processes and to enhance the management of these records. The results indicated that the use of ERM applications enhanced organizational efficiency, improved information flow, and increased compliance with legal and regulatory requirements.

Information Management System

The introduction of computers into information technology significantly increased organizations' information demands; however, system performance largely depended on the underlying knowledge base. A database was defined

as a structured collection of data, typically organized to model relevant aspects of reality and support systems that required such information (Kathuria, 2020). Modern database management systems now leverage advanced algorithms and cloud architecture to ensure that this vast amount of structured data remains both accessible and scalable for real-time decision-making.

Database Management Systems (DBMS) were applications designed to interact with databases, other programs, and users to collect and manage data. A general-purpose database management system (DBMS) was described as a software program that facilitated the definition, creation, querying, updating, and administration of databases (Ahishakiye et al., 2020).

Database Management Systems (DBMSs) could communicate with each other through standards such as SQL, JDBC, or ODBC, though databases were generally not directly transferable between systems. Wahome (2018) described a DBMS as a common information processing or organizational information system that manages people's data. In addition, World Net defined an information system (IS) as a system that gathers and organizes data and delivers it to managers at all levels for planning, decision-making, program implementation, and control. An information system was also described as a network of communication channels within an organization, comprising both hardware and software components.

In the context of immigration, an information system is intended to support management in collecting and reporting data while improving the accuracy and quality of information provided to all stakeholders.

Computerization was described as a societal process that involved providing individuals with access to computers and making decisions about investment levels, knowledge controls, and equipment selection. It also referred to the use of computer-based tools for tasks such as file processing, education bookkeeping, writing, designing, and technical operations.

Filing System

Fenwick and Brownstone (2003) defined electronic filing as the submission of records in electronic rather than paper form. A filing system was described as a systematic approach to organizing and managing information. Effective filing systems helped organizations retrieve and maintain records effortlessly (Harbor, 2022). By digitizing these processes, organizations can implement automated indexing and search functionalities that drastically reduce the time spent on manual record-keeping.

A filing system could be either traditional or electronic. Traditional filing provided familiarity, personal engagement, and direct communication, making it suitable for organizations that require face-to-face clarification. However, its primary disadvantage was that accessing records was time-consuming, and files could be misplaced and/or disorganized, leading to operational downtime. Electronic filing, on the other hand, offered convenience, faster processing, reduced error rates, and cost savings. It converted traditional files to digital formats, allowing records to be sorted, retrieved, and shared remotely. Proper file tagging and naming were considered essential, as neglecting these practices reduces retrieval efficiency (Ocampo, 2023; Perez, 2024). Furthermore, Tang (2025) stated that e-filing simplified the submission and

maintenance of legal documents and enabled future technological developments, such as virtual courtrooms and AI-assisted legal research.

Web-Based

The state of computer-based information technology in education had advanced significantly and had begun to influence information delivery systems. However, Herawati et al. (2021) stated that, in practice, many businesses had not fully utilized technology for its intended purposes. Pasaribu (2021) noted that technological advancements brought by globalization created multiple advantages that affected various aspects of life. Individuals used information technology to assist in accomplishing work tasks through system implementation. By automating routine tasks, these systems allowed employees to focus on more strategic and decision-making activities.

Organizations improved performance by leveraging rapidly evolving information technology systems. The processing of inventory data served as an example of an information technology application. Sales-oriented organizations require accurate inventory management to predict demand and prepare sufficient product quantities. Advances in information technology systems enabled the distribution of product information and coordination data to managers. The implementation of computerized systems reduced operational time while using less energy to produce accurate data outputs.

Sufaidah and Putri (2021) established the need for a program to support the proper execution of campus activities and proposed a web application as a solution. The term “web” referred to the World Wide Web, a system of interconnected Internet servers that stored documents written in

Hypertext Markup Language. Businesses also use web-based platforms for marketing purposes.

Zhao and Watterston (2021) noted the growing importance of web-based systems in education, particularly during the COVID-19 pandemic, when online learning became the norm. They also noted the declining relevance of attendance as a measure of student engagement. Similarly, Saracostti et al. (2021) recommended the use of integrated systems to assess, monitor, and enhance school engagement to prevent dropout and promote positive learning pathways. They emphasized the use of strong technological tools to reduce teaching burdens. The Web-Based System for the Measurement, Monitoring, and Promotion of School Engagement Strategies (SIESE), developed by Saracostti et al., served as an example, providing automated reporting and intervention tools for student engagement and contextual factors across educational transitions.

Management Information System

A Management Information System (MIS) consisted of interconnected components that worked together to collect, store, and distribute data for managerial planning, control, coordination, analysis, and decision-making (Mugan, 2019). MIS served as a valuable tool for managers, providing a competitive advantage (Stair, 2017). It had been recognized as an essential component of effective business decision-making. Kenneth et al. (2000) described MIS as a managerial-level information system that generated routine summary and exception reports to support planning and control functions.

According to Garrick (n.d.), MIS was a general term used to describe computer systems organizations that provided information about their operations. The term also referred to the individuals responsible for managing these systems. In larger organizations, MIS, or the MIS department, was a centralized unit that managed computer expertise and resources, including mainframe systems and extensive computer networks (Garrick, n.d.).

Synthesis and Relevance of the Study

Based on data from related literature and studies, information systems played an important role in improving productivity and modernizing record-keeping, particularly in government institutions. Existing systems such as Web-based Records Management Systems, eMigrate, and similar platforms were developed to address the inefficiencies of manual record-keeping by enhancing data accessibility and enabling electronic processing of records.

The implementation of these information systems transformed traditional record management practices by improving operational efficiency, data accuracy, and security. The transition from paper-based processes to digital platforms also contributed to increased transparency and accountability in managing individual records and departmental transactions. These developments demonstrated the value of digital solutions in supporting public service operations.

However, despite these advancements, a notable research gap remained. While existing studies primarily focused on general document management systems or large-scale, national-level migrant information platforms, limited research addressed division-specific filing systems that

integrated document generation, process monitoring, and performance visualization within the Philippine Department of Migrant Workers (DMW) context. Most existing systems were either too broad in scope or not tailored to the operational needs of local processing divisions, leading to continued reliance on manual procedures at the regional level. This gap highlighted the need for a localized, role-based system such as the BalikBayani Portal.

The relevance of the present study lies in its direct response to this gap by proposing a system specifically designed for the Processing Division of DMW Region IV-A. Unlike generalized systems, the BalikBayani Portal focused on digitizing divisional-level workflows, automating record filing, monitoring application progress, and generating performance-based visual reports to support administrative decision-making.

Furthermore, the study contributed to the body of knowledge on digital systems in public service by demonstrating how localized information systems enhanced data security and transparency in the handling of sensitive records. The findings of this research served as a reference for other government agencies seeking to implement technology-driven solutions to improve internal workflows. Ultimately, the study supported the DMW's ongoing efforts to modernize records management, enhance service delivery, and improve the overall experience for both migrant workers and department staff.

Literature Matrix

Table 1 showed that, by reviewing related studies and literature, the current research gained a broader understanding of the factors influencing technology acceptance and their impact on system adoption, providing a well-

rounded basis for the study's objectives. These insights also guided the identification of relevant variables and informed the design and evaluation of the proposed system.

Table 1. Literature Matrix

Author(s)/Year	Title	Findings
Ahishayike, Taremwa & Omulo, E, D (2020)	Web-based Records Management Systems	A Web-based system improves record management, streamlines decision-making, and maintains data integrity.
Mugan (2019)	Management Information System (MIS)	MIS supports administrative responsibilities and helps in decision-making, making it essential to organizations.
Singh S., Kumar, V., Paliwal, M., Verma, P., & Rajak, B. (2022)	A citizen-centric approach to understand the effectiveness of e-government web portals: Empirical evidence from India	The findings emphasize that the government has designed its e-government portal to improve user goals, citizen satisfaction, and perceptions of e-government quality.
Ministry of External Affairs, Government of India (n.d.)	eMigrate System	The findings show that eMigrate is a successful model of e-governance for labor migration, enhancing accountability, reliability, and transparency while protecting migrant workers through inclusive, secure record management.
International Labour Office (ILO) (2018)	Digitalization to Promote Decent Work for Migrant Workers in ASEAN	Digitalization improves labor migration governance by providing migrant workers with better access to services while reducing the risks associated with unsafe migration. Public and private organizations use digital platforms to improve

		workflow and safety. The study shows that digital tools enhance employment outcomes and migrant worker protection services throughout ASEAN by suggesting more integrated systems and coordination. However, the analysis identifies two main risks: unequal access to technology and the need for proper implementation to achieve beneficial results.
Manilyn Joy P. Velita, Aristeo C. Salapa (2023)	Benefits and Effectiveness of Digitalization of Government Services to OFWs	The study found that both employees and OFWs benefited from quicker, cheaper, and improved services resulting from the DMW's digitization of the OEC process. Although respondents expressed great satisfaction with the system, problems such as the digital divide persist, underscoring the need for improved access to ICT services.
L P Hasugian & T M Rahayu (2018)	Requirement Analysis of Monitoring Information System for Indonesian Migrant Workers Protection	The study underscores the challenges faced by Indonesian migrant workers, highlighting gaps in stakeholder coordination and in government protection initiatives. The current responses are mainly legalistic and lack real-time monitoring or proactive support. The study, through a SWOT analysis, indicated the need for a protection information system to ensure regular communication between IMWs and other relevant stakeholders. The system allows IMWs to submit regular reports, monitors their well-being, and alerts the authorities when necessary. This model is intended to improve migrant workers' safety, ensure the government can respond promptly, and increase trust in protective measures.

Conceptual Framework

The researchers created a graphical representation to depict the system's framework. This diagram illustrates the project's overall structure and highlights the primary components. It outlines the steps and considerations required for successful completion of the research. By following this framework, the researchers can stay on track and ensure that all aspects of the project are thoroughly analyzed.

Figure 1 presents the conceptual framework of the BalikBayani Portal, illustrating the interactions among migrant workers, administrators/staff, and system processes. On the left side, migrant workers can access the BalikBayani Portal using web and mobile browsers. The primary features include registration, which requires creating an account by entering personal information. Upon registration or login, migrant workers can proceed to the application process under three distinct categories, such as direct hire, balik manggagawa, and gov-to-gov, based on their employment type. The application process involves filling out forms and uploading supporting documents. The status-tracking feature enables real-time monitoring of application progress, ensuring transparency and reducing the need for frequent office visits.

On the opposite side are the administrator and staff, who access the system through desktop computers. Their main tasks include verifying personal information and reviewing digital files submitted by migrant workers. Once the information and documents are validated, the system enables them to generate official documents such as clearances. Data visualization tools

provide summary views and statistics that support decision-making for report preparation and application monitoring.

The framework's central portion consists of the front end, back end, and database. The front end deals with the user interface and experience for both migrant workers and staff, ensuring accurate input capture. The back-end processes inputs, enforces rules according to the system modules, and interacts with the database. The database stores the records of migrant workers. This configuration provides synchronized, consistent data handling across all components.

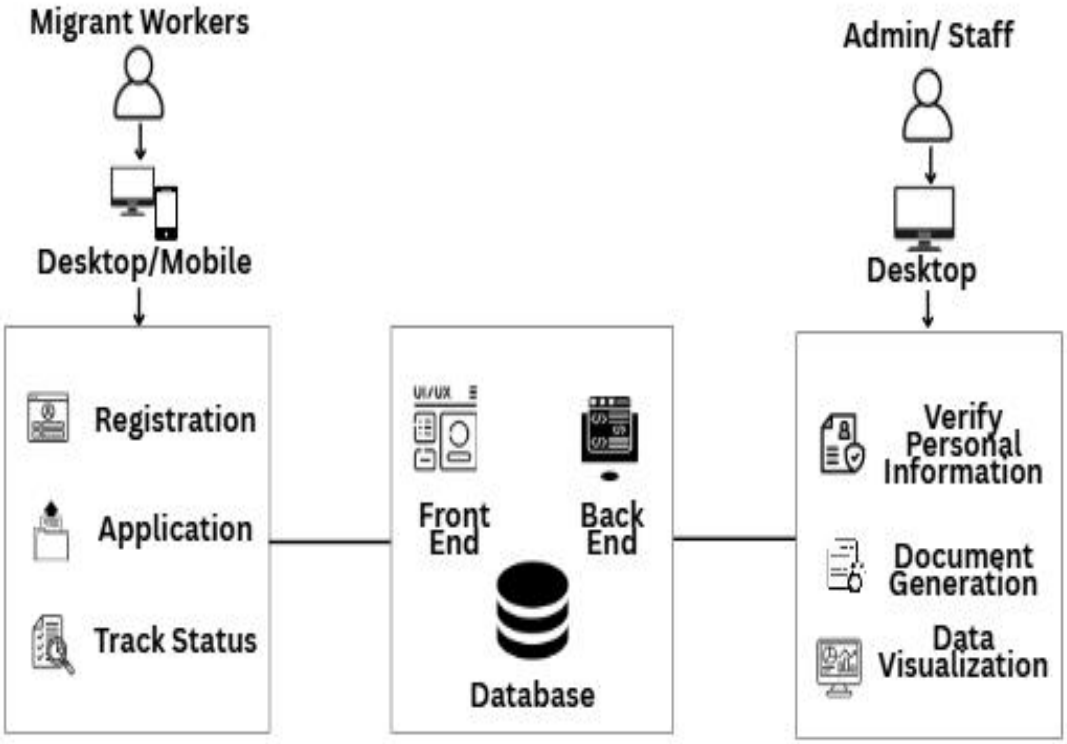


Figure 1. Conceptual Framework

CHAPTER III

METHODOLOGY

This study employed a developmental, quantitative research approach and used the Agile Scrum methodology for system design and development. This chapter also presented the system's software and hardware requirements and discussed the evaluation models, study population, and the instruments and methods used to interpret and analyze the results.

Project Design

Figure 2 shows the system architecture of the BalikBayani Portal. It illustrated the fourteen modules of the system, namely the Login/Authentication Module, Dashboard Module, Direct Hire Module, Balik Manggagawa Module, Gov-to-Gov Module, Information Sheet Module, Monitoring List (List of Job Fairs) Module, Monitoring List (PESO Contact Information) Module, Monitoring List (PRAs Contact Information) Module, Monitoring List (Job Fair Monitoring Summary) Module, Notification Module, Admin Panel with System Reports, User Management Module, Data Backups Module, and User Profile Module. Each of these modules performed a specific function to support the Department of Migrant Workers' (DMW) processing needs.

The architecture highlighted three main user roles: admin, staff, and migrant workers. Each role had a specific set of modules that it was authorized to access. Upon accessing the system, users logged in through their designated interfaces: the admin user interface for administrators and

staff, and the standard user interface for migrant workers. Once authenticated, users were directed only to the modules aligned with their assigned roles.

The admin role had the highest level of access and full control over the system. Administrators were able to manage all modules, including the User Management Module, System Reports, and Data Backups Module. They managed user activity status, administered user roles, established access permissions, generated system reports, and performed data backups to prevent the loss of important information. These permissions enabled administrators to oversee and manage the entire platform and ensure smooth system operation.

The staff role, on the other hand, had more limited privileges. Staff members were allowed to access only the modules directly related to their assigned tasks. They assisted with migrant worker processing and reviewed required information, but lacked the authority to manage high-level administrative functions. This separation of privileges ensured that only authorized personnel accessed sensitive system functions, thereby protecting the system from unauthorized modifications.

The migrant worker role provided users with access to features that supported application processing and status tracking. Migrant workers used the Direct Hire, Balik Manggagawa, and Gov-to-Gov modules depending on their classification. They also accessed the Notification Module to receive updates regarding application progress and the User Profile Module to manage and update personal information. These modules were designed to

provide migrant workers with a convenient and transparent way to monitor their records.

By clearly defining the responsibilities and access levels for each user role, the BalikBayani Portal established a secure, reliable system for managing migrant worker data. Only authorized individuals could interact with specific modules, helping maintain data accuracy, confidentiality, and integrity. This structured approach enabled DMW personnel to process information efficiently while providing migrant workers with an accessible, user-friendly platform to guide them through their applications.

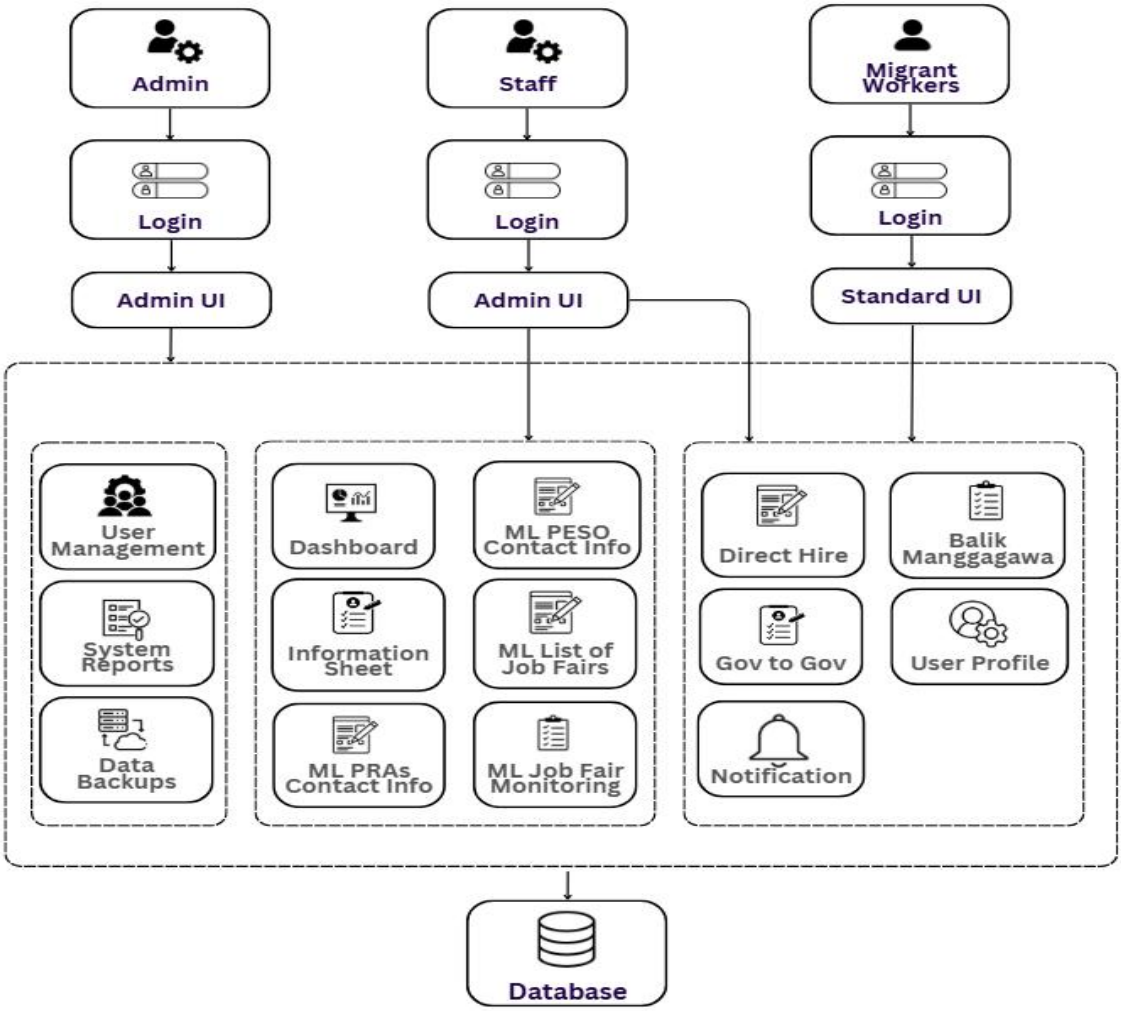


Figure 2. System Architecture

Software Development Model

Figure 3 shows the Agile Scrum methodology. Scrum is a widely used Agile framework for improving, delivering, and maintaining complex products, particularly in software development. It emphasized iterative and incremental development and was valued for its simplicity, flexibility, teamwork orientation, and reliance on empirical process control (Iqbal, 2023; Irsyad, 2022).

In developing the BalikBayani Portal, the Agile Scrum methodology was adopted as the guiding framework. Instead of treating the system as a single, large process, the development work was divided into smaller increments via sprints, enabling continuous refinement and early resolution of challenges.

The process began with the creation of user stories that described the system's functions from the users' perspectives. These user stories were placed in the product backlog, where tasks were organized and prioritized. Each sprint began with a planning session to determine which features would be developed, ensuring that every cycle produced a usable system increment.

The Scrum roles were applied throughout the project. The Regional Director of the Department of Migrant Workers (DMW) communicated the project vision and ensured alignment with departmental needs. The Scrum Master guided project progress, facilitated regular discussions, resolved blockers, and ensured adherence to Scrum principles. The Development Team handled coding, design, and testing activities, working collaboratively to deliver the system incrementally.

Online and face-to-face meetings were conducted twice a week to share accomplishments, identify upcoming tasks, and address emerging issues. These meetings helped the team detect problems early and make timely adjustments. At the end of each sprint, a sprint review was conducted to present progress, followed by a sprint retrospective to identify areas for improvement.

This iterative process ensured flexibility throughout development. When new requirements or issues arose, such as adjustments in data handling or interface design, they were incorporated without disrupting the overall project workflow. By progressing through successive sprints, steady development was maintained, and project goals remained on track.

Through this approach, Scrum served as the framework for the development process, enabling the team to manage system complexity while maintaining collaboration, transparency, and adaptability throughout the project.

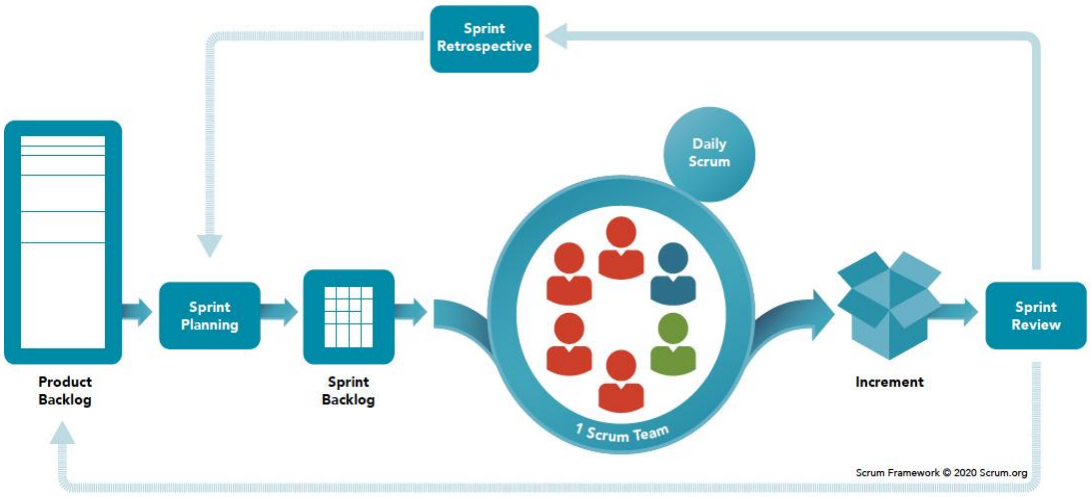


Figure 3. Agile Scrum

Product Backlog

The product backlog was a crucial component in the development process, as it defined the core requirements and features needed for building the BalikBayani Portal. It served as a prioritized list of development tasks that guided the team toward a structured path to achieving the project objectives. Through the product backlog, system goals were broken down into clear and actionable items, making them easier to plan and monitor. Each item, referred to as a user story, specified functionality from the end user's perspective, ensuring that development aligned with actual user needs. Moreover, it enhanced workflow management by allowing the development team to focus on high-priority features that delivered the most value to users. The backlog was continuously reviewed and refined to reflect user feedback and system improvements, ensuring that the development process remained organized and responsive to evolving requirements.

Table 2 shows the product backlog for the BalikBayani Portal admin. The table presented the prioritized user stories, including the functionalities the Admin was expected to perform, along with the estimated time per day required to complete each task. These user stories served as a guide for the system's development, ensuring that the Admin could effectively manage the Portal. The administrator played a major role within the system, as they had full access and control over all functionalities. The administrator was generally responsible for managing user accounts, monitoring records, generating reports, and backing up data. Unlike staff users, admins had higher privileges, which allowed them to oversee and manage the entire platform.

Table 2. Product Backlogs for Administrator

PRIORITY	STORY	Estimated Time/Day
1	As an admin, I want to create an account and log in.	2 hrs
2	As an admin, I want to view the analytics so I can easily visualize trends.	1hr
3	As an admin, I want to monitor and verify direct hire applicants so I can manage them easily.	4hrs
4	As an admin, I want to track Balik Manggagawa records/cases so I can monitor their status.	3hrs
5	As an admin, I want to access and review Government-to-Government deployment records.	3hrs
6	As an admin, I want to view the summaries of all submitted records.	2hrs
7	As an admin, I want to create job fair information including PESO IV-A and PRA contact details, the list of job fairs, and the job fair monitoring summary.	1hr
8	As an admin, I want to manage all user accounts and modify staff access.	2hrs
9	As an admin, I want to perform system back-ups.	1hr
10	As an admin, I want to generate system reports.	1hr

Table 3 shows the product backlog specifically intended for the staff users, who primarily support the processing and monitoring of migrant worker records. It presented the essential tasks and user stories that were developed to support the functional responsibilities of DMW staff members. Each user

story was organized by priority level and included the estimated time required for completion.

Table 3. Product Backlogs for Staffs

PRIORITY	STORY	Estimated Time/Day
1	As a staff, I want to create an account and log in.	2 hrs
2	As a staff, I want to view the analytics so I can easily visualize trends.	1hr
3	As a staff, I want to monitor and verify direct hire applicants so I can manage them easily.	4hrs
4	As a staff, I want to track Balik Manggagawa record/cases so I can monitor their status.	3hrs
5	As a staff, I want to access and review Government-to-Government deployment records.	3hrs
6	As a staff, I want to view the summaries of all submitted records.	2hrs
7	As a staff, I want to create job fair information including PESO IV-A and PRA contact details, the list of job fairs, and the job fair monitoring summary.	1hr

Table 4 displayed the product backlog for the migrant worker or applicant users. It included the prioritized tasks and the expected time for each, focusing on the most important functionalities required for effective system interaction. Primary tasks included creating an account and logging in, filling out forms, uploading documents, tracking application status, and managing personal information. These features enabled applicants to access

the system securely, and by focusing on these tasks, the BalikBayani Portal provided a user-centered experience that supported the efficient and independent management of applications.

Table 4. Product Backlogs for Migrant Worker

PRIORITY	STORY	Estimated Time/Day
1	As an applicant, I want to create an account and log in.	2 hrs
2	As an applicant, I want to submit my application	4hrs
3	As an applicant, I want to track my application status	2hrs
4	As an applicant, I want to update my personal information in my user account.	1hr

Sprint Planning

Sprint planning served as a fundamental development step, helping the team organize priorities, define sprint goals, and identify the collaborative tasks needed to achieve key milestones. Through this process, the development team established clear sprint directions, ensuring that everyone understood their focused objectives for delivering specific features and functionalities. In addition, sprint planning allowed the team to review system requirements and align them with development timelines. It also helped identify task dependencies and potential risks before implementation began. Regular planning discussions encouraged collaboration and shared understanding among team members. This approach ensured that each sprint

contributed directly to the overall project objectives. Sprint planning also supported better decision-making by setting clear expectations for deliverables and timelines.

The project was made more manageable through sprint-based breakdowns, which enabled the team to maintain consistent progress while addressing changing requirements and project challenges. The team developed a sprint backlog containing the tasks to be completed in each sprint. Tasks were assigned to individual members based on their expertise and availability, maximizing resource use and maintaining balanced workloads. This breakdown helped prevent task overload and ensured that responsibilities were evenly distributed. It also allowed the team to track progress more effectively and adjust workloads when necessary. Sprint backlogs provided a clear reference for monitoring task completion during development. Through this structure, the team was able to maintain steady progress while adapting to technical or requirement-related changes.

Table 5 presents the essential features and tasks required for the successful development of the system. It highlighted the highest-priority tasks for the administrator modules through key monitoring and management functions. Each sprint item was broken down into specific development tasks with estimated working hours, showing that the team had outlined the necessary functionalities and workflow for efficient system implementation. The table served as a guide for prioritizing development activities based on system importance and complexity. Overall, the sprint backlog table ensured a structured and organized approach to administrator module development.

Table 5. Sprint Backlogs for Administrator

PRIORITY	ITEMS	TASK	ESTIMATED HOURS/DAY
1	As an admin, I want to monitor and verify direct hire applicants so I can manage them easily.	- Design UI for tracking and monitoring clearance - Develop backend for status updates	4 hrs
2	As an admin, I want to track Balik Manggagawa records/ cases so I can monitor their status.	- Build monitoring interface - Add status filters	3.5 hrs
3	As an admin, I want to access and review Government-to-Government deployment records.	- Gather requirements for the endorsement process - Develop the form submission UI - Implement data validation and backend for storage	3 hrs
4	As an admin, I want to view the summaries of all submitted records.	- Design info sheet summary layout - Fetch and display submitted records - Implement export option	3 hrs
5	As an admin, I want to manage all user accounts and modify staff access.	- Design UI for user account listing and role management - Develop backend for CRUD operations - Implement role-based control	4 hrs

The Sprint Backlog in Table 6 outlined the priority list for the staff user role, listing the functionalities to be developed and executed in the corresponding sprint. It provided a clear development plan by breaking down the staff product backlog items into specific tasks with estimated completion

times. This backlog ensured that staff functionalities, such as monitoring, managing, and processing records, were implemented systematically.

Table 6. Sprint Backlogs (Staff)

PRIORITY	ITEMS	TASK	ESTIMATED HOURS/DAY
1	As a staff, I want to monitor and verify direct hire applicants so I can manage them easily.	- Design UI for tracking and monitoring clearance - Develop backend for status updates	4 hrs
2	As a staff, I want to track Balik Manggagawa record/cases so I can monitor their status.	- Build monitoring interface - Add status filters	3.5 hrs
3	As a staff, I want to access and review Government-to-Government deployment records.	- Gather requirements for the endorsement process - Develop the form submission UI - Implement data validation and backend for storage	3 hrs
4	As a staff, I want to view the summaries of all submitted records.	- Design info sheet summary layout - Fetch and display submitted records - Implement export option	3 hrs
5	As a staff, I want to create job fair information including PESO IV-A and PRA contact details, the list of job fairs, and the job fair monitoring summary.	- Design UI for job fair listings - Develop backend for event storage - Implement UI for listing with filtering and search functionality	3 hrs

Table 7 shows the sprint backlog specifically intended for migrant worker or applicant users, featuring the core tasks required to implement essential system functionalities. Each user story was broken down into actionable tasks with estimated hours per day for completion. The tasks generally focused on designing and developing the user interface, implementing validation of user inputs, submitting applications, and testing system features to ensure they performed as intended.

Table 7. Sprint Backlogs (Migrant Workers)

PRIORITY	ITEMS	TASK	ESTIMATED HOURS/DAY
1	As an applicant, I want to create an account and log in.	- Design UI for registration/login - Test login and registration functionality	2 hrs
2	As an applicant, I want to submit my application	- Design UI for application form - Implement form validation	4 hrs
3	As an applicant, I want to track my application status.	- Design UI for status tracking - Implement notification for status updates	2hrs
3	As an applicant, I want to update my personal information in my user account.	- Design UI for user profile	1 hr

Sprint Execution

Sprint execution began after the development team completed the sprint backlog for the proposed system. At this stage, the team focused on

implementing key features, including Direct Hire, Balik Manggagawa, Government-to-Government, Information Sheet, and Monitoring List. The sprint was carried out with minimal interruptions, allowing the team to focus on completing the planned tasks efficiently. Any urgent modifications or unexpected challenges that arose were noted and scheduled for the next sprint cycle.

After completing each sprint, the team presented the implemented features to stakeholders, including the Processing Division of the Department of Migrant Workers (DMW), to gather feedback. This feedback was analyzed to determine necessary improvements or refinements. The team ensured that critical adjustments were prioritized in the following sprint while maintaining the overall project timeline. This iterative approach guaranteed that the system remained functional, user-friendly, and aligned with the needs of the DMW staff.

Flowchart

Figure 4 illustrates how users initially accessed the BalikBayani Portal. Staff received a temporary username and password from the Admin for their first login. Once authenticated, the system detected that it was an initial login and automatically displayed a personal information form. This form required the staff to provide complete details, including creating new login credentials to replace the temporary ones. Completing this form was mandatory. After submission, the system logged the staff out, and the newly created credentials had to be used for future logins. Migrant workers or applicants, meanwhile, had to first register by providing their basic information. After

submitting their details, the system sent a verification code to their email. Once verified, the migrant worker could log in using their email address and the password they had set up during registration.

Migrant workers submitted application forms and uploaded the required documents, with personal information automatically populated from registration, completing only the necessary fields. They submitted applications under the Direct Hire, Balik Manggagawa, and Gov-to-Gov categories. Staff users then reviewed the submitted records and updated each applicant's status. If any part contained errors, the staff notified the migrant worker to correct only the specific section with the issue. Any updates to a migrant worker's status made by staff are automatically notified to the migrant worker. For direct-hire applications, the system generated a clearance once the application was approved. For Balik Manggagawa applications, a clearance form was generated only if there were issues with the worker’s status. Staff could also soft-delete records, which would archive them rather than permanently remove them. Staff users could not view or restore archived records.

For Admin and staff, the Dashboard served as the main interface after login, displaying current statistics such as sex distribution across different categories and the application trend graph. From the Dashboard, users navigated to different modules to manage migrant worker information. Within the Monitoring List section of the system, there were four modules. The List of Job Fairs module allowed users to record job fair events. The Job Fair Monitoring Summary module helped track the results of those activities,

including the number of applicants served. The PESO IV-A Contact Information module stored important contact details for coordination with regional partners. The PRAs' Contact Information module kept a record of recruitment agencies for easier communication. These modules supported monitoring efforts and helped ensure organized coordination with external offices.

Admin users had more control than staff users. Aside from managing active and archived records, they also managed user accounts, generated system reports, and performed backups. This setup helped protect data and ensured that users accessed only the functions relevant to their responsibilities.



Figure 4. System Flowchart

Use Case Diagram

Figure 5 shows the use case diagram of the BalikBayani Portal, illustrating how the system interacts with its different users. The Admin had the highest level of access and could perform all system functions. However, having access did not necessarily mean that the Admin performed all these tasks routinely, as they primarily oversaw and ensured that the system and users operated smoothly. Staff users had more limited access and primarily assisted in managing applicant records. They could also submit applications on behalf of migrant workers in special cases, such as when a user was unable to do so due to a lack of technical knowledge or the need for additional assistance. Migrant workers, meanwhile, primarily interacted with the Portal to submit applications and track their status.

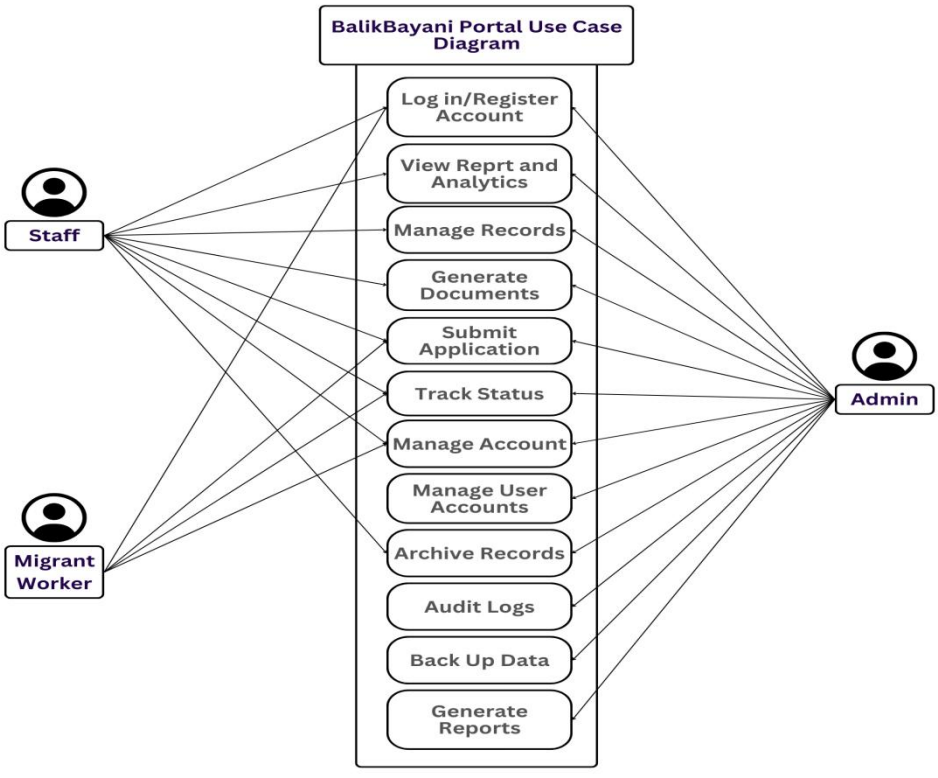


Figure 5. Use Case Diagram

Materials

This section discusses the software and hardware resources used in developing the BalikBayani Portal. These resources were essential for the system's design, development, implementation, and testing, ensuring that it operated properly. The software tools supported programming, database management, UI design, and overall system integration, while the hardware components provided the computer specifications and storage required to run the system.

Software

The developers used the following software tools to develop the BalikBayani Portal:

Visual Studio Code. It served as the integrated development environment (IDE), providing support for building, debugging, and running the web application.

Next.js. Next.js powers both the React UI and the API layer from a single codebase, enabling server components, routing, and server-side rendering for faster page loading.

React. eact rendered the interface and enabled client interactivity. Its hooks and concurrent features helped build complex dashboards, forms, and modals with predictable state updates.

TypeScript. ypeScript was used alongside React to provide static typing, improve code reliability, and enhance development tooling. Its type-

checking capabilities reduced runtime errors and improved maintainability of the front-end codebase.

PostgreSQL. PostgreSQL stores structured program data, including applications, users, and reports. The pg driver ensured reliable transactional access and maintained integrity for government-grade records.

Tailwind CSS. Tailwind CSS was used as the styling framework for the user interface, enabling consistent, responsive, and utility-first design without writing custom CSS from scratch.

Git and GitHub. Git provided version control for the source code, while GitHub served as the remote repository for collaboration, code backups, and change tracking.

Hardware

Table 8 shows the specifications of the computer used to develop the BalikBayani Portal. The development machine ran on Windows 11 Pro with 32 GB RAM and an AMD Ryzen 5 processor. This hardware provided the necessary performance to support coding, testing, and debugging activities.

Table 8. Hardware Requirements

Component	Specification
Random Access Memory (RAM)	32GB
CPU	AMD Ryzen 5 5600
Storage	256 GB SSD
Operating System (OS)	Windows 11 Pro

Table 9 shows the minimum system requirements for desktop or laptop computers to run the BalikBayani Portal smoothly. These specifications ensured stable usability and compatibility, particularly when users submitted migrant records.

Table 9. Minimum Requirements for PC/Laptop

Component	Specification
Random Access Memory (RAM)	4GB
CPU	At least Intel I3 / Ryzen 3
Storage	At least 8GB storage
Operating System (OS)	Windows 10 or later (32-bit/64-bit)

Project Testing and Evaluation

Multiple tests were conducted to ensure the quality and reliability of the BalikBayani Portal. As shown in Figure 6, evaluation and testing were performed at different levels, including Unit Testing, Integration Testing, System Testing, and the Evaluation Procedure.

Unit Testing focused on individual modules, verifying that each function produced the expected results. Integration Testing examined module interactions to confirm seamless communication and correct data flow throughout the system. System Testing evaluated the complete system to ensure it met the intended requirements and performed reliably under realistic conditions.

The Evaluation Procedure included a simulation comparing the new system to the previous manual process. A staff member used the BalikBayani

Portal at a single counter, and process times were recorded for each transaction. These times were compared to the historical process times to measure improvements in efficiency, accuracy, and overall workflow.

This comprehensive testing and evaluation helped identify areas for refinement, ensured that the system met the needs of the Department of Migrant Workers staff, and confirmed that the BalikBayani Portal provided a functional, user-friendly, and reliable solution for managing migrant records.

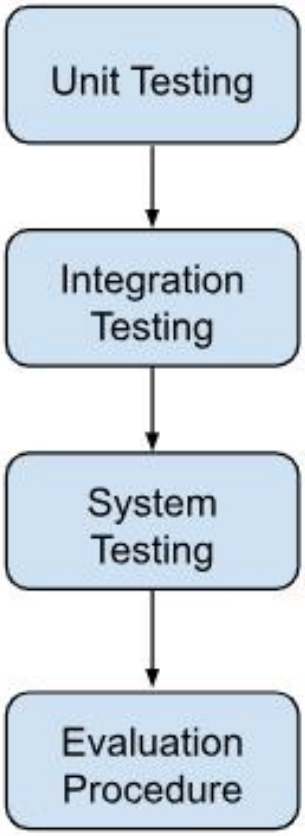


Figure 6. Project Testing and Evaluation

Unit Testing

Figure 7 shows the unit testing conducted on the system. Unit testing involves testing individual software modules or components, focusing on

specific units of code, such as functions, methods, and classes, to ensure that each performs correctly and produces the expected results. The process began with checking out the code from the repository, followed by making any necessary changes or updates. Unit tests were then carried out to confirm that each component performed as expected. If any errors were found, they were fixed immediately, and the unit tests were rerun to verify that the problems had been resolved. After passing all tests, the code underwent a code review, where team members checked its quality and logic. Lastly, the verified code was checked back into the repository to complete the cycle.

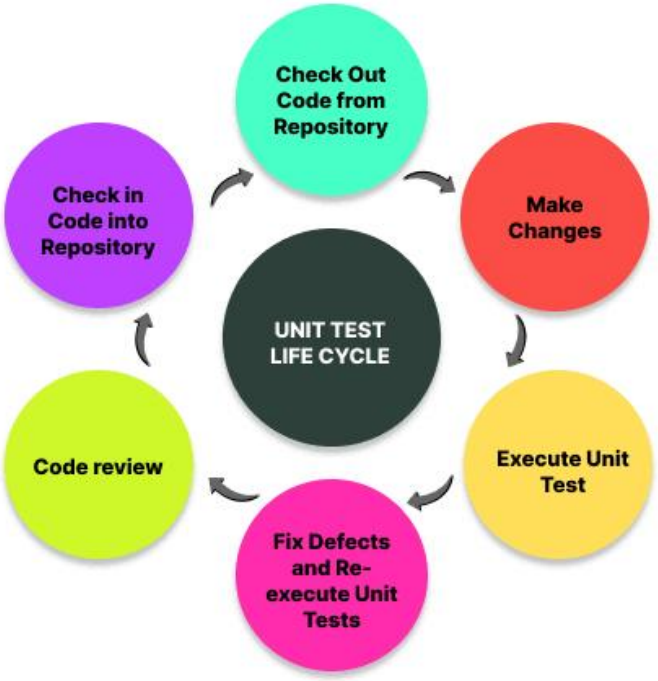


Figure 7. Unit Testing

Integration Testing

Figure 8 shows the integration testing conducted on the system. Integration testing involved combining software modules or units and verifying

that they worked together as expected. Interactions between modules were checked to ensure they functioned correctly, and data flowed properly across the integrated system.

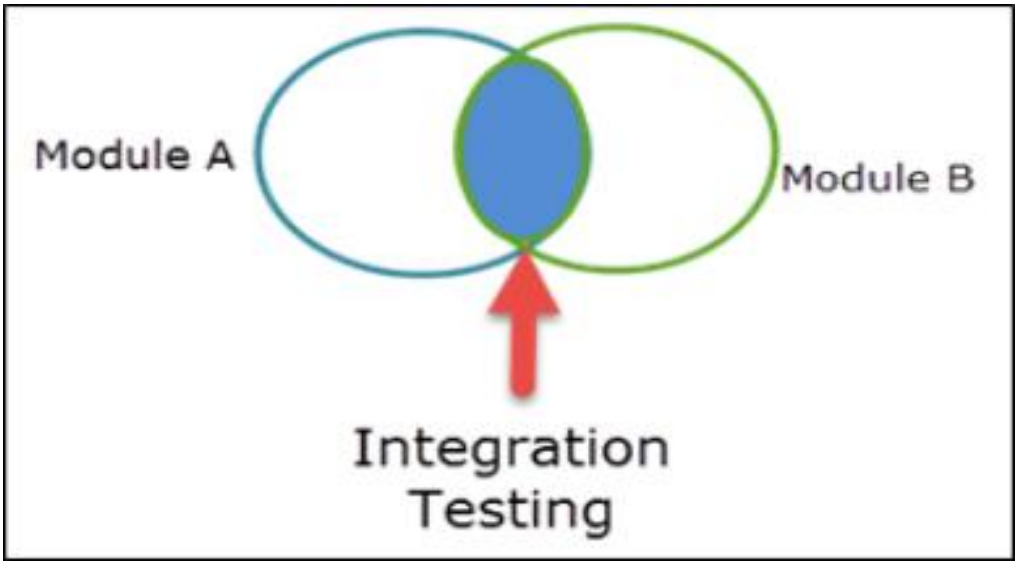


Figure 8. Integration Testing

System Testing

The BalikBayani Portal was tested as a complete system to ensure that all features functioned in accordance with the project objectives. This testing involved verifying that each module and component worked seamlessly together, that the system met functional requirements, and that it provided a reliable, user-friendly experience for the DMW Processing Division staff. Any issues or unexpected behavior were documented and addressed to maintain overall system quality.

Compatibility Testing

Figure 9 shows the compatibility testing conducted on the BalikBayani Portal. The system's compatibility was tested with different browsers,

including Chrome, Microsoft Edge, and Opera GX, to ensure consistent performance and usability across platforms.

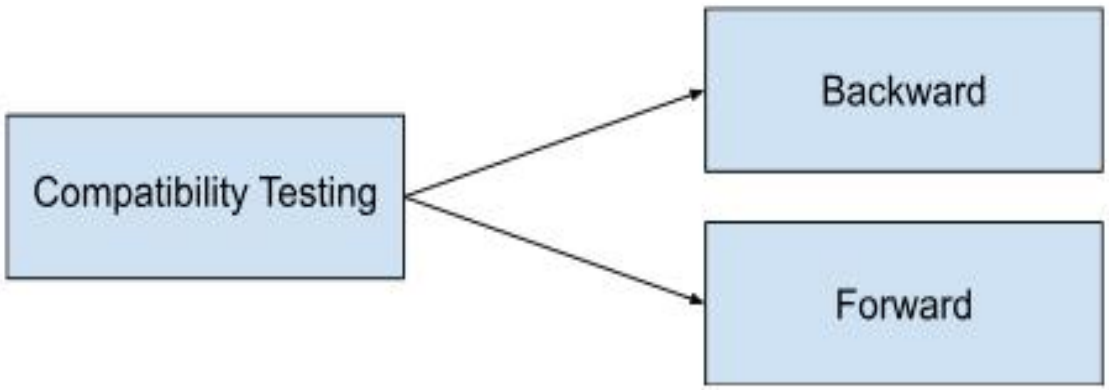


Figure 9. Compatibility Testing

Experimental Setup

To evaluate the efficiency of the BalikBayani Portal compared to the previous manual process, a simulation was conducted at the Balik Manggagawa counter of the Department of Migrant Workers (DMW) Processing Division. During the simulation, a staff member performed standard tasks using the system, and the Portal automatically recorded the time required for each transaction. This setup allowed observation of the system under real working conditions and captured precise process times.

For comparison, manual process times were obtained from historical records collected by the DMW the previous week. These records documented the start and end times for each transaction, including creating a BM application, generating clearances, and completing approvals or advising resubmission. This comparison allowed assessment of the system’s efficiency relative to the previous workflow.

It was noted that differences in workload, applicant complexity, or staff performance between the historical data and the simulation could have affected the observed process times.

Table 10. Experimental Setup for Process Time Simulation

TASK/ TRANSACTION	STEPS INVOLVED	MEASUREMENT METHOD
Create BM application	Input worker's name, sex, destination, position, job type, employer, monthly salary, and currency	Manual (historical data) / System (automatic log)
Generate clearance for compliance	Produce clearance if applicable	Manual / System
Approve or advise resubmission	Review documents and instruct applicant if follow-up is needed	Manual / System
Full transaction cycle	Complete all above steps	Manual / System

Evaluation Procedure

The evaluation of the BalikBayani Portal was conducted to measure both its technical quality and user acceptance. This ensured that the system was not only functional but also user-friendly and practical for the Department of Migrant Workers (DMW) staff. Two evaluation models were applied: the Technology Acceptance Model (TAM) to assess user perceptions and acceptance, and the ISO/IEC 25010 Software Quality Model to evaluate the system's technical quality across various characteristics, including functionality, reliability, usability, and maintainability.

ISO/IEC 25010 Software Product Quality

The ISO/IEC 25010 standard framework was used to assess and measure the software quality of the BalikBayani Portal through defined characteristics and sub-characteristics. This framework was applied to evaluate the system's performance, usability, reliability, and security, following the guidelines of the ISO/IEC 25010 Software Product Quality Model shown in Figure 10. This evaluation ensured that the system met the operational needs of the DMW Processing Division and provided a reliable and user-friendly solution for managing migrant records.

SOFTWARE PRODUCT QUALITY								
FUNCTIONAL SUITABILITY	PERFORMANCE EFFICIENCY	COMPATIBILITY	INTERACTION CAPABILITY	RELIABILITY	SECURITY	MAINTAINABILITY	FLEXIBILITY	SAFETY
FUNCTIONAL COMPLETENESS FUNCTIONAL CORRECTNESS FUNCTIONAL APPROPRIATENESS	TIME BEHAVIOUR RESOURCE UTILIZATION CAPACITY	CO-EXISTENCE INTEROPERABILITY	APPROPRIATENESS RECOGNIZABILITY LEARNABILITY OPERABILITY USER ERROR PROTECTION USER ENGAGEMENT INCLUSIVITY USER ASSISTANCE SELF-DESCRIPTIVENESS	FAULTLESSNESS AVAILABILITY FAULT TOLERANCE RECOVERABILITY	CONFIDENTIALITY INTEGRITY NON-REPUDIATION ACCOUNTABILITY AUTHENTICITY RESISTANCE	MODULARITY REUSABILITY ANALYSABILITY MODIFIABILITY TESTABILITY	ADAPTABILITY SCALABILITY INSTALLABILITY REPLACEABILITY	OPERATIONAL CONSTRAINT RISK IDENTIFICATION FAIL SAFE HAZARD WARNING SAFE INTEGRATION
iso25000.com								

Figure 10. ISO/IEC 25010 Software Product Quality Characteristics

Technology Acceptance Model

The Technology Acceptance Model (TAM), shown in Figure 11, was applied to understand user adoption and acceptance of the BalikBayani Portal. This framework was used to evaluate the system’s functionality, Usefulness, reliability, and ease of use. Data were collected from selected respondents,

primarily system users in the DMW Processing Division, using a questionnaire survey. Responses were scored on a four-point Likert scale, and the gathered data were analyzed to assess how well the system met user needs and expectations.

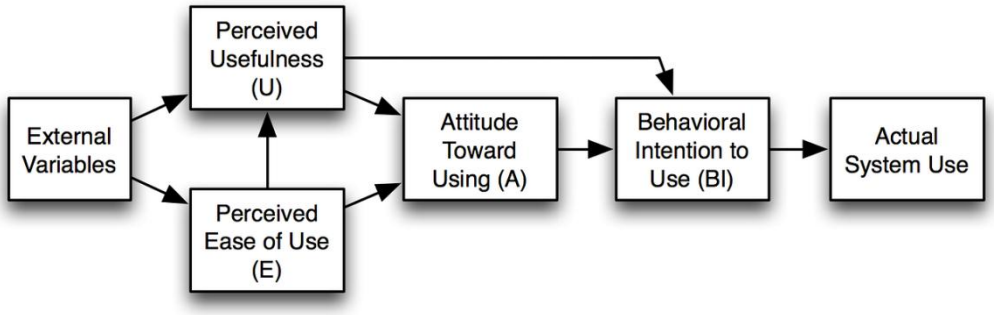


Figure 11. Technology Acceptance Model

Population of the Study

The population of the study consisted of one hundred four (104) system users who participated in the alpha testing of the BalikBayani Portal and three (3) IT professionals who conducted the beta testing. The system users served as respondents for the Technology Acceptance Model (TAM) evaluation and were guided in testing the system to ensure it functioned correctly and operated as intended.

The inclusion of 104 TAM respondents was considered adequate, as TAM-based studies focus on user perceptions and acceptance from actual system users rather than on a fixed minimum sample size. These respondents represented the intended users of the BalikBayani Portal, allowing the study to obtain reliable measures of TAM criteria.

Meanwhile, the three (3) IT professionals evaluated the system using the ISO 25010 software quality model. Three evaluators were deemed sufficient in accordance with ISO-based software quality assessment practices, which commonly use expert judgment from a small number of qualified professionals to assess software quality attributes. The combined feedback from system users and IT experts was essential for identifying areas for improvement and ensuring that the system met both user acceptance and technical quality requirements prior to full deployment.

Sampling Design

Purposive sampling was employed in this study, selecting respondents based on their relevance to the system's intended use and their ability to provide accurate, experience-based feedback. A total of 107 respondents were involved, comprising 104 individuals directly engaged in customer service, filing, processing, and oversight of migrant worker records within the Department of Migrant Workers (DMW), and three (3) IT professionals qualified to assess the system’s technical functionality.

Data Collection Instrument

Data collection for the BalikBayani Portal was conducted using several instruments to gain a comprehensive understanding of the system requirements and user needs. Data for the system was gathered through interviews with the processing division staff to gain insights into their filing procedures, existing problems or challenges, and their user expectations. These interviews were conducted both face-to-face and online. In addition, the current tools and materials, such as Excel sheets and clearance

documents, were collected and analyzed to determine which data needed to be digitized and the need for effective record management. Daily routines were also observed to identify redundant procedures and potential areas for improvement.

For the ISO evaluation, IT professionals were provided with printed survey questionnaires to assess the system's effectiveness against the nine software quality characteristics defined in ISO 25010. It includes Functional Suitability, Performance Efficiency, Compatibility, Interaction Capability, Reliability, Security, Maintainability, Flexibility, and Safety. Responses were recorded on a five-point Likert scale with the following values: 1 - Poor, 2 - Needs Improvement, 3 - Fair, 4 - Good, and 5 - Excellent.

The Technology Acceptance Model (TAM) was used to measure user acceptance of the system. Online survey questionnaires were distributed via Google Forms, focusing on Perceived Usefulness, Perceived Ease of Use, Attitude Towards Using, and Behavioral Intention to Use. Responses were recorded using a 4-point Likert scale, with the following values: 1 – Very Low, 2 – Low, 3 – Moderate, and 4 – High.

Statistical Treatment of Data

The researcher employed a purposive sampling technique to select respondents who were directly involved and knowledgeable about the system being evaluated. Data gathered from the target respondents were organized and processed in Microsoft Excel, where the mean and standard deviation were computed for further interpretation of the results. Descriptive statistical tools, specifically the mean and standard deviation, along with their

corresponding Likert scale verbal interpretations, were used to analyze the collected data. The mean was used as the primary statistical measure to determine the average level of agreement among respondents for each questionnaire item. It presented a numerical summary of respondents' overall perceptions. A higher mean value indicated a more positive opinion of the system, while a lower mean suggested aspects that may require improvements or further attention.

The standard deviation, on the other hand, measures the degree of variability or dispersion in respondents' answers. A low standard deviation indicated that the respondents shared similar perceptions, reflecting a strong level of agreement. In contrast, a high standard deviation suggested varied viewpoints among respondents, indicating differences in experiences or perspectives.

For the interpretation of responses, the study used two different Likert-scale formats: the Technology Acceptance Model (TAM) and ISO 25010, as each framework served a distinct evaluation purpose and required a scale that best fit its measurement objectives. The system's effectiveness and quality were evaluated using the ISO 25010 software quality model, which utilized a 5-point Likert scale to assess technical quality and performance characteristics. The inclusion of a neutral midpoint allows evaluators, particularly IT professionals, to provide a more balanced, nuanced assessment when certain quality attributes are neither fully positive nor negative. This scale supports a more detailed and objective evaluation of software quality criteria. The numerical ratings, corresponding verbal

interpretations, and computed ranges used in analyzing the mean scores are presented in Table 11.

Table 11. 5-point Likert Scale Method

RATING	INTERPRETATION	RANGE
5	Excellent	4.21-5.00
4	Good	3.41-4.20
3	Fair	2.61-3.40
2	Needs Improvement	1.81-2.60
1	Poor	1.00-1.80

Meanwhile, the Technology Acceptance Model (TAM) variables were measured using a 4-point Likert scale, as presented in Table 12. This scale intentionally eliminates a neutral option to encourage respondents to make a clear judgment regarding their acceptance of the system. Since TAM focuses on subjective user perceptions, the forced-choice format helps reduce indecisive responses and provides a more accurate indication of system acceptance. Table 12 presents the numerical ratings, corresponding verbal interpretations, and the computed ranges used to interpret respondents' levels of acceptance.

Table 12. 4-point Likert Scale Method

RATING	INTERPRETATION	RANGE
4	High	3.25-4.00
3	Moderate	2.50-3.24
2	Low	1.75-2.49
1	Very Low	1.00-1.74

CHAPTER IV

RESULTS AND DISCUSSION

This chapter presents the results based on the project objectives, followed by the evaluation outcomes and system testing results.

Results by Objectives

Develop a filing system that integrates document generation into printable formats and supports quick, accurate retrieval of information.

Figures 12 to 15 illustrate the Record Details View feature, designed to improve the BalikBayani Portal's ability to manage and generate documents effectively. It served as an essential interface for displaying all encoded information, uploaded requirements, and system-generated documents for each migrant worker. Instead of manually searching across multiple spreadsheets, users could retrieve a complete record with a single query. The document generation capability automatically produced official forms in printable formats directly from validated data. This reduced redundant encoding, minimized the chance of human error, and accelerated document preparation. These improvements directly addressed limitations in the previous process, which required multiple manual steps for record lookup and form preparation.

Figure 12 showed the view modal for the Direct Hire Module, which consolidated essential data such as the applicant's basic information, employment details, application status, uploaded documents, and system-generated files into a single, organized view. By centralizing all relevant data,

the feature enabled the staff to review and verify information without navigating multiple files or external spreadsheets.

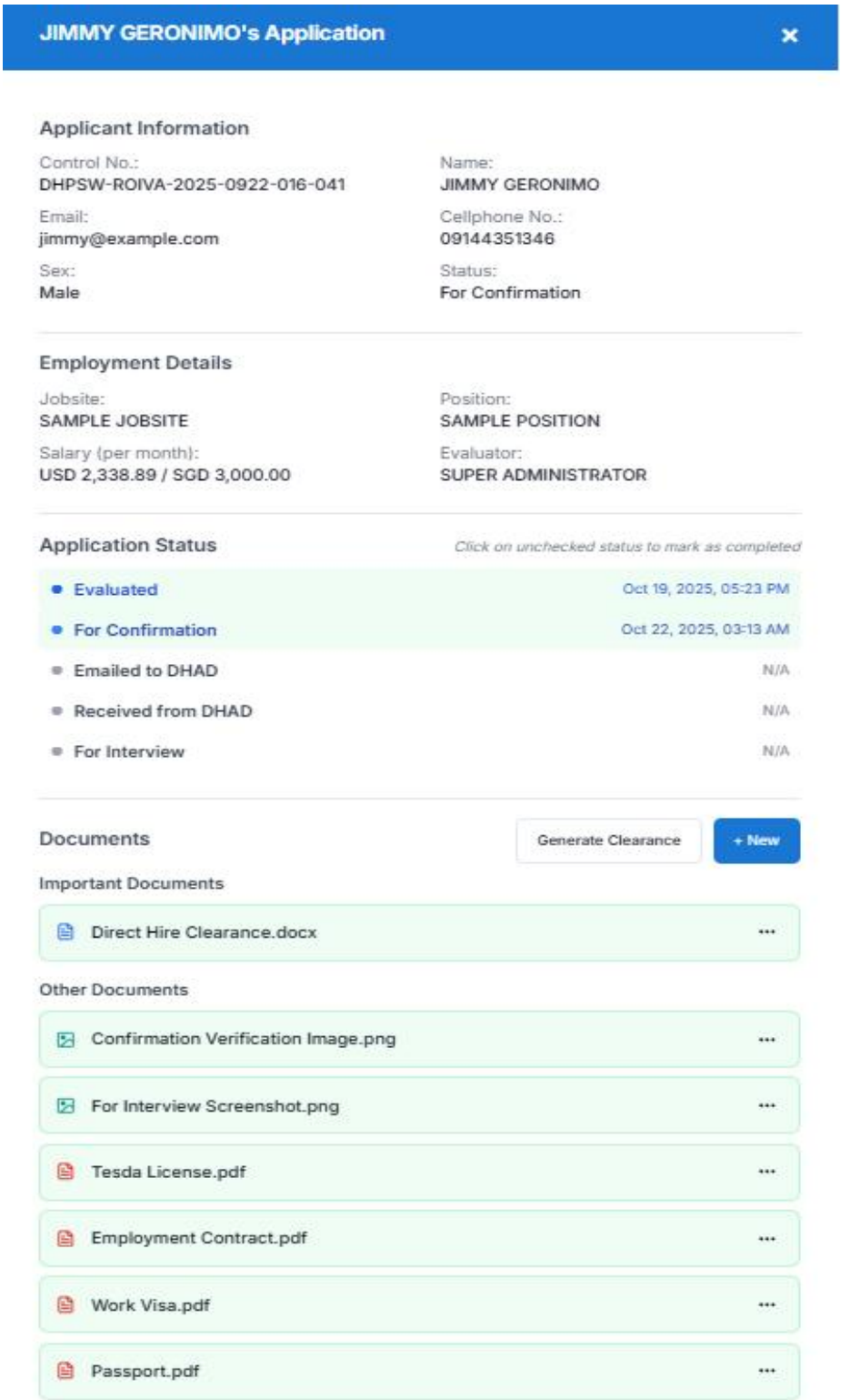


Figure 12. Direct Hire - View Modal

Figure 13 presents the View Modal for the Balik Manggagawa module. The interface consolidated the worker's profile and all system-generated documents into a single modal, eliminating the need for staff to cross-check multiple spreadsheets or physical records manually. Clearances were automatically generated based on the OFW's classification and application requirements, removing the previous dependency on manual identification and encoding.

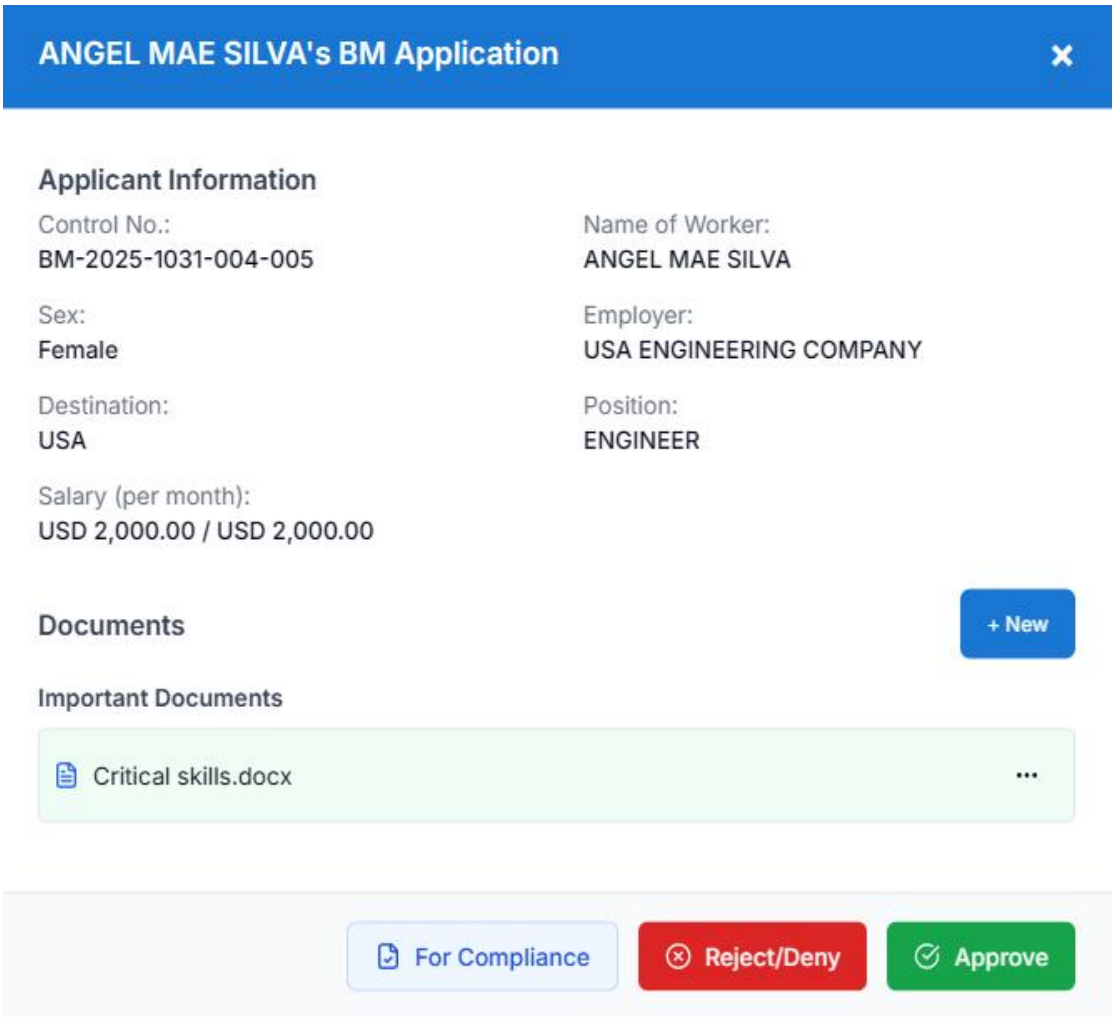


Figure 13. Balik Manggagawa - View Modal

Figure 14 presents the View Model for the Government-to-Government (Gov-to-Gov) module. The interface consolidated the applicant's personal

data, passport information, work experience, and uploaded verification documents into a single view. This eliminated the need for staff to cross-check multiple sources manually, reducing processing time and the risk of errors. By providing an organized, centralized record, the system ensured accurate data retrieval and enabled faster, more reliable decision-making compared to the prior manual process.

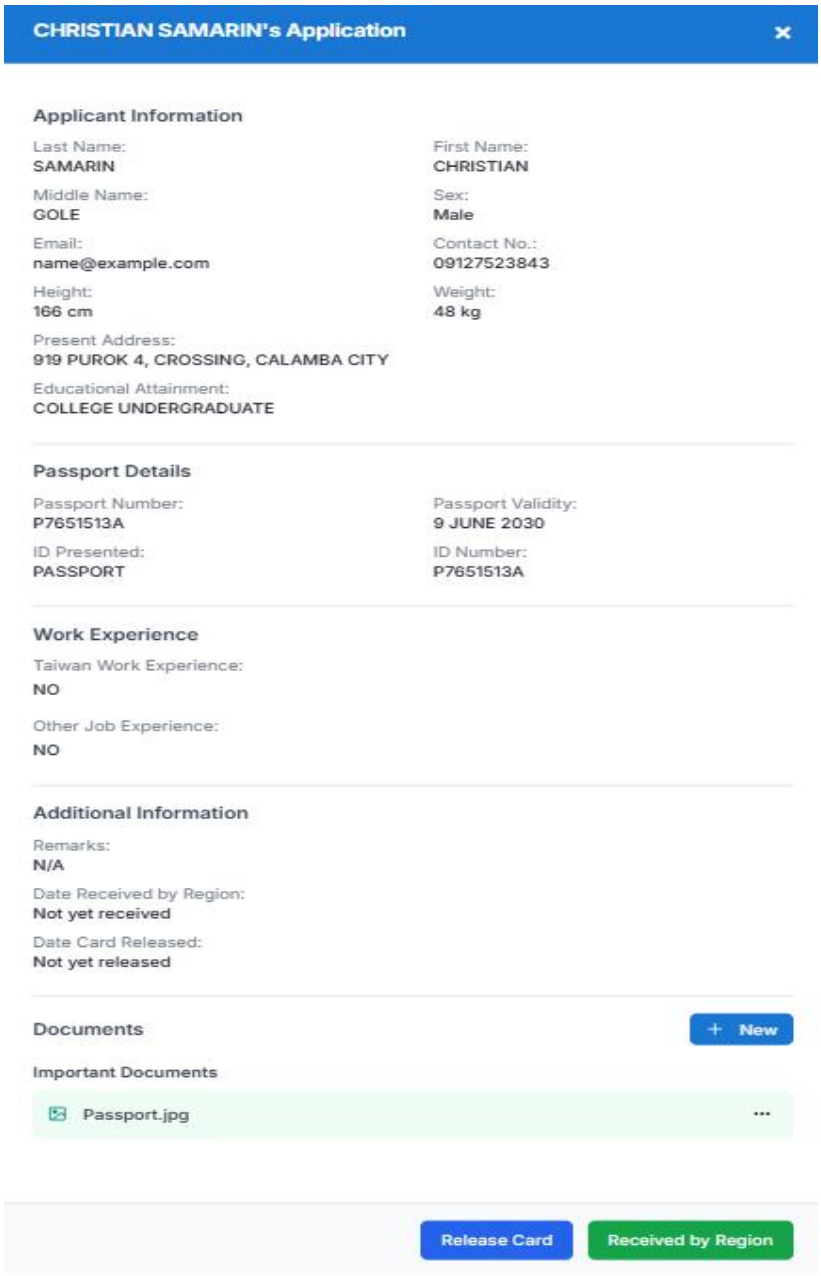


Figure 14. Gov to Gov - View Modal

Figure 15 presents the View Modal for the Information Sheet module. This interface consolidated the applicant's personal information, request details, and uploaded verification documents into a single view. By centralizing all relevant data, the system minimized the need for manual cross-checking, reduced errors, and enabled faster information retrieval, improving the overall efficiency and reliability of the DMW Processing Division's operations compared to the previous manual workflow.

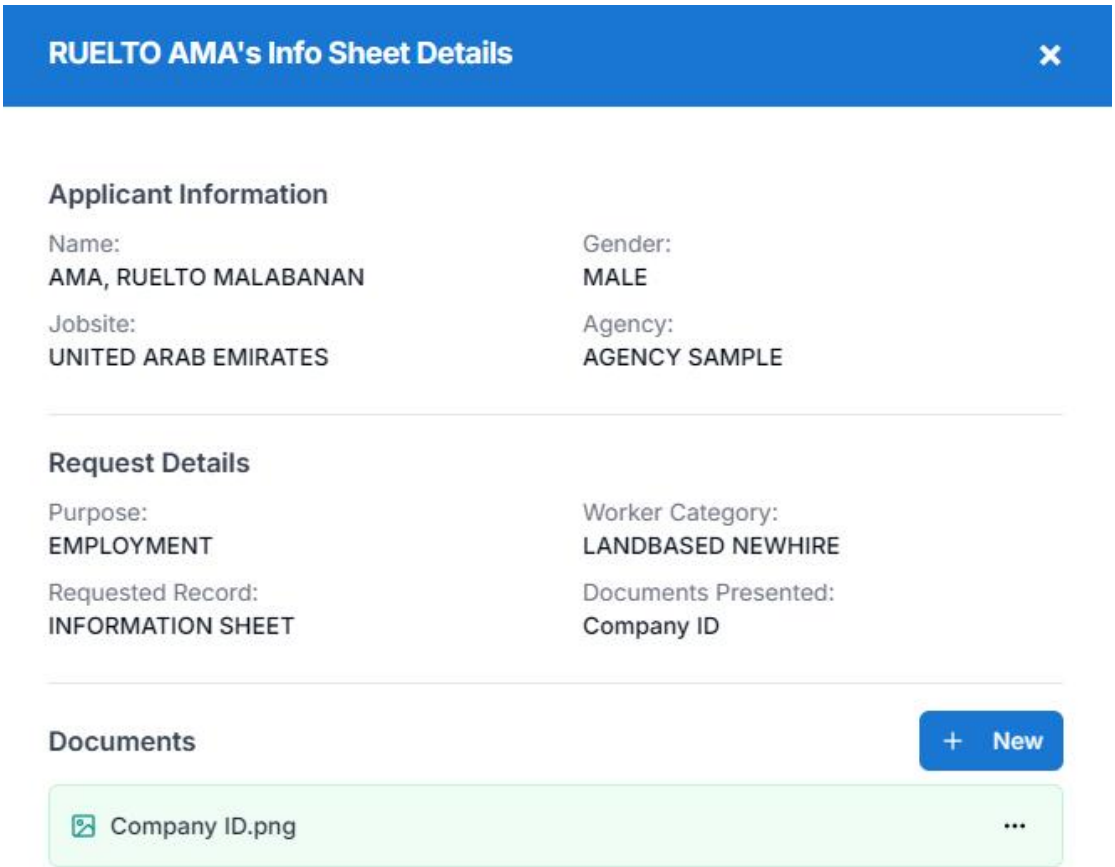


Figure 15. Information Sheet - View Modal

Figure 16 presents the Job Fair View Modal under the Monitoring List module. This interface consolidated information for each job fair, including the date, venue, office head, email address, and contact numbers. By centralizing

these details, the system allowed users to quickly access and verify event information, supporting coordination and collaboration among departments and agencies. Additionally, the view facilitated monitoring and updating schedules, such as rescheduling or venue changes, improving accuracy and reducing administrative delays compared to the previous manual process.

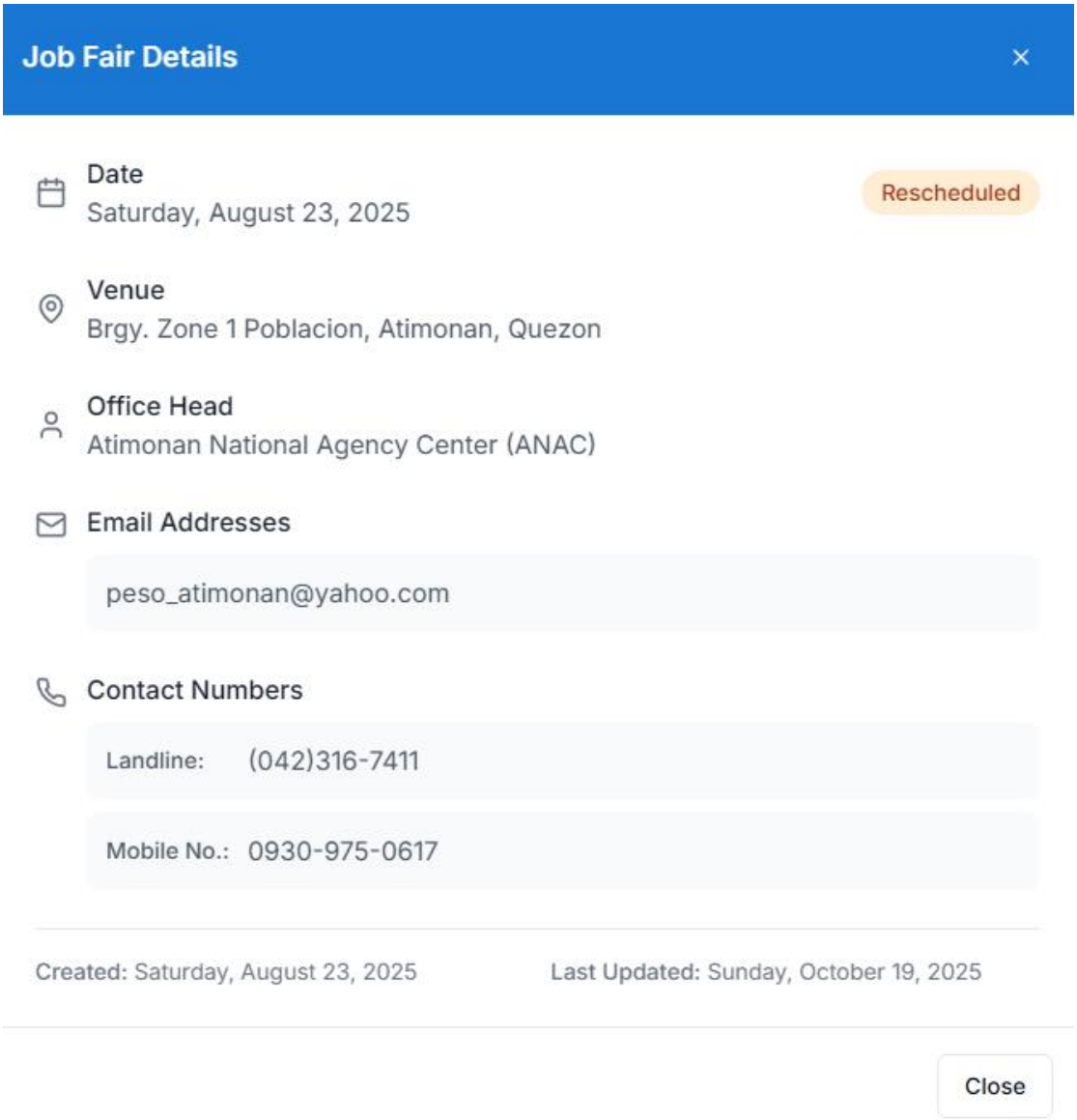


Figure 16. Monitoring List - Job Fairs View Modal

Figure 17 presents the Job Fair Monitoring Summary View under the Monitoring List module. This interface provided a consolidated overview of job

fair details, including the date, venue, participating agencies, applicant statistics, and assigned DMW staff for oversight. The system enabled accurate monitoring, accountability, and evaluation of each event's outcomes. Users were able to efficiently assess the success of the job fair and maintain reliable records, reducing the need for manual processes and minimizing errors compared to the previous workflow.

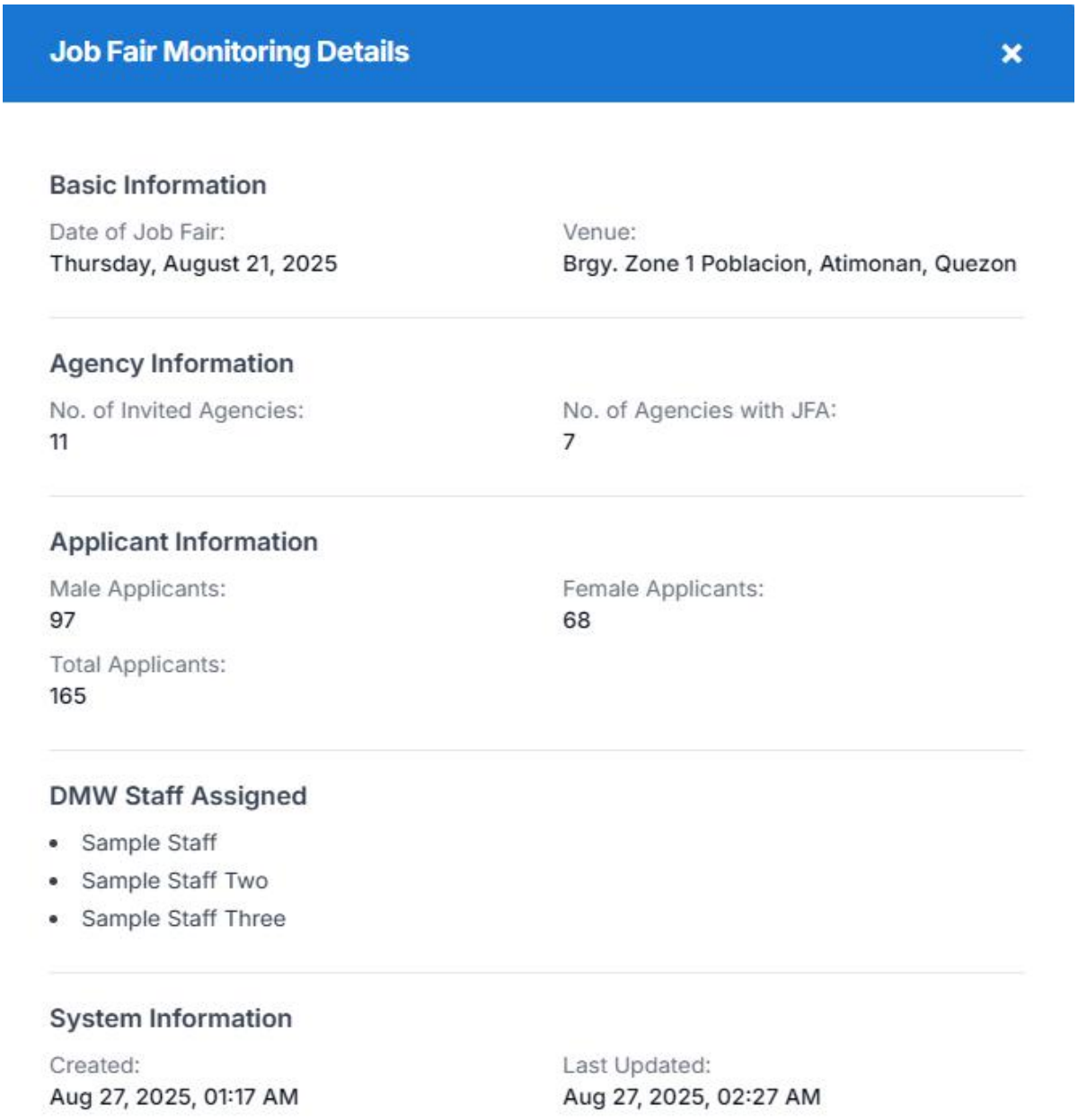


Figure 17. Monitoring List - Job Fair Monitoring Summary View Modal

Figure 18 shows the system reports page under the Super Admin Panel module. This interface displayed the system's generated reports, containing monthly, quarterly, and yearly summaries of processed documents and served clients. This page primarily served for monitoring and reporting, helping the Admin quickly track activities and ensure that all records were properly documented.

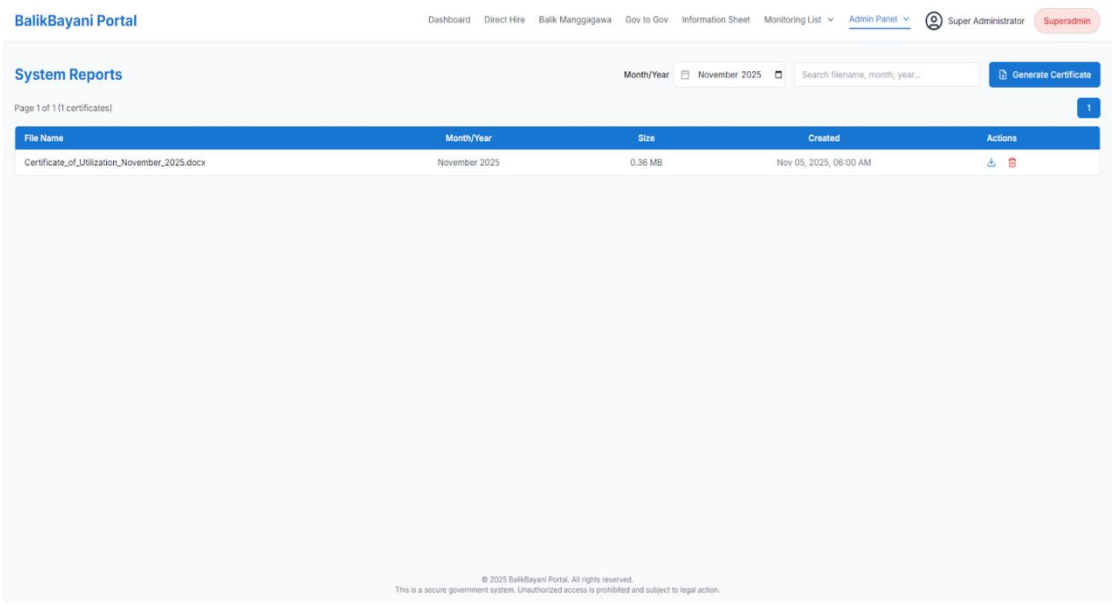


Figure 18. Super Admin Panel - System Reports

Integrate a data visualization feature that provides representation across the main categories.

The BalikBayani Portal achieved the second objective by integrating data visualization across the main categories. Figures 19 to 22 demonstrated the system's visual analytics in the form of charts and graphs that summarized records and activities stored in the database. These visual representations made it easier for admins and staff to understand and interpret data without manually reviewing large volumes of raw records,

enabling users to accurately monitor activities and identify patterns for performance monitoring and annual reports.

Figure 19 presented the main Dashboard, displaying the total number of applications in the Direct Hire, Balik Manggagawa, Gov-to-Gov, and Information Sheet categories. The Dashboard included a sex distribution filter (All, Male, Female) and a gender-ratio chart showing the proportions of male and female applicants across all categories. Additionally, it featured an application trend graph showing the number of processed applications over the past six months for monitoring and analysis.

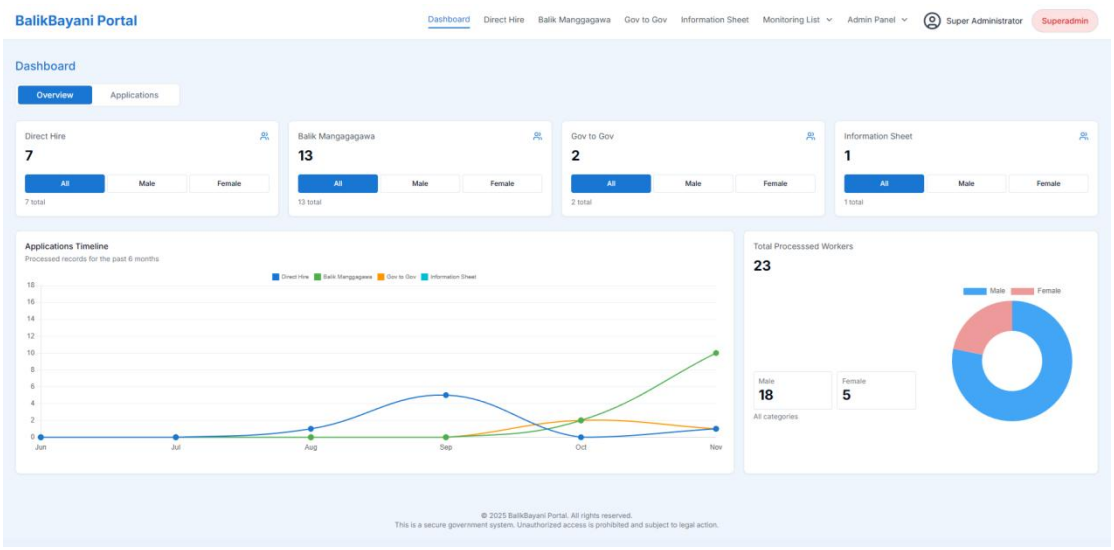


Figure 19. Dashboard Module

Figure 20 presents the Direct Hire module's data visualization panel, summarizing application statuses including For Evaluation, Evaluated, For Confirmation, Emailed to DHAD, Received from DHAD, For Interview, and Finished. This visualization allowed users to efficiently monitor application progress and manage tasks accurately, eliminating the need for manual record checks and reducing the risk of oversight.

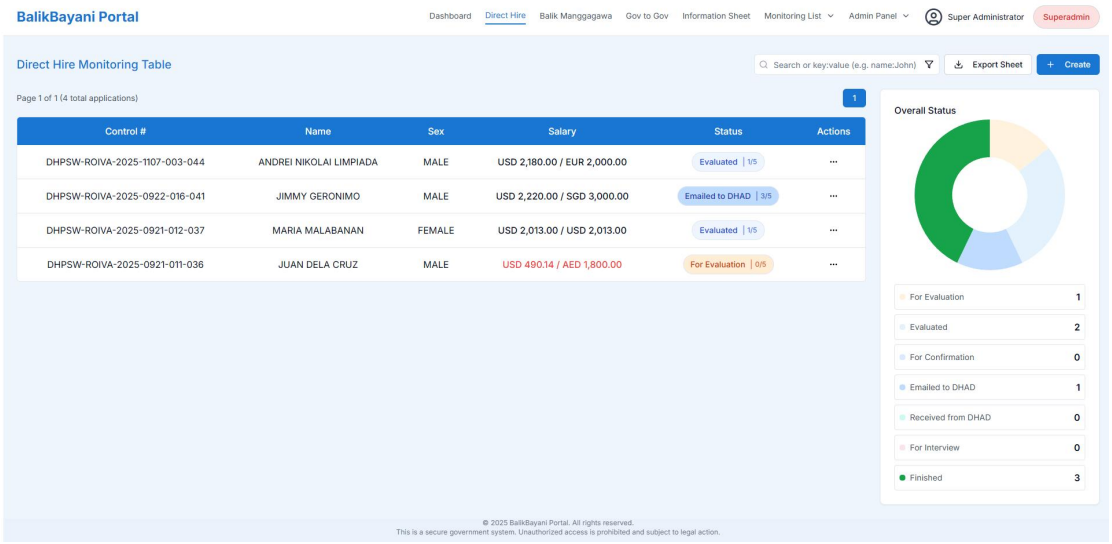


Figure 20. Direct Hire Module

Figure 21 showed the Balik Manggagawa module, featuring a monitoring table and an analytical panel that summarized application statuses, including For Compliance, For Approval, Approved, and Rejected. This visualization enabled users to track application progress efficiently and quickly identify which applications required action or verification, streamlining workflow and reducing manual oversight.

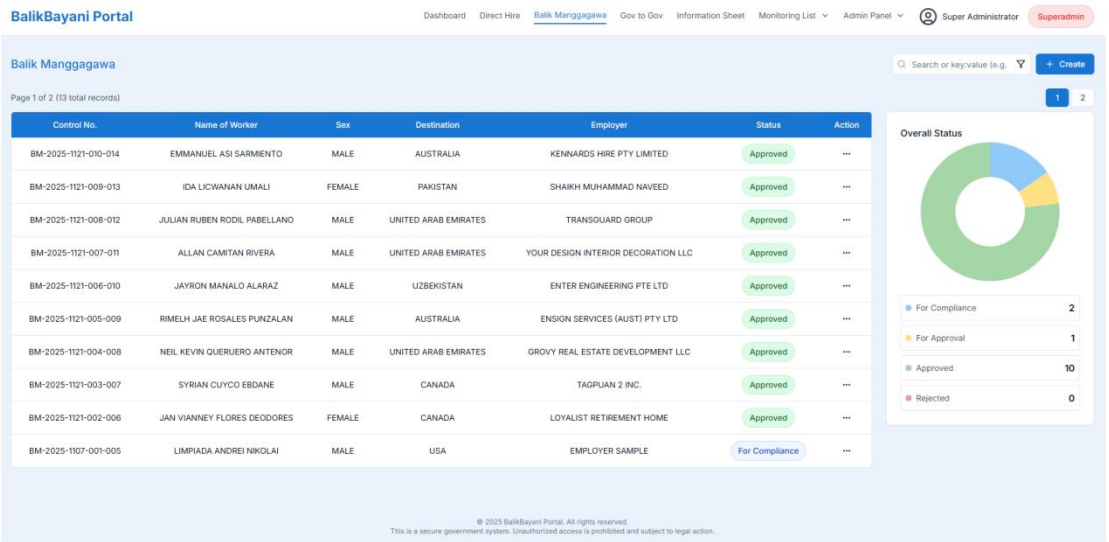


Figure 21. Balik Manggagawa Module

Figure 22 presents the Information Sheet module, featuring a monitoring table and an analytical panel. The panel summarized the overall purposes of requests and highlighted the most frequently submitted records, providing users with a clear overview of trends and supporting efficient management of incoming documents.

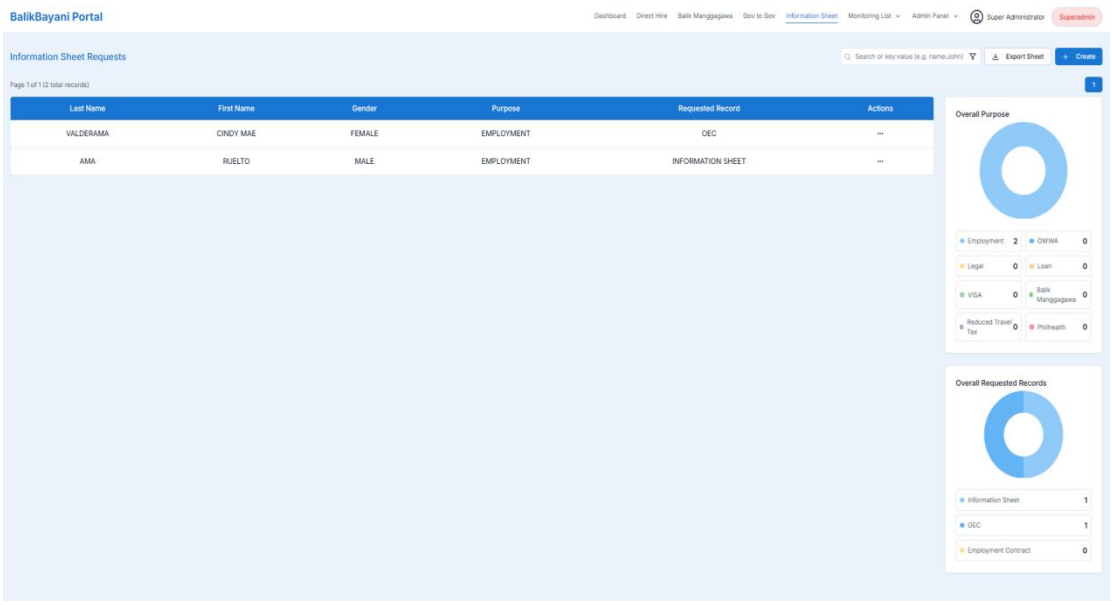


Figure 22. Information Sheet Module

Unit Testing Result

Table 13 displayed the results of unit testing for the Direct Hire Module. The results indicated that the module worked as intended and met the system's functional requirements, as all tested cases produced the expected outcomes. However, during initial testing, minor errors were identified, particularly in form behavior: some input fields did not reset properly after submission. These issues were reviewed and debugged, and the system was retested. After the necessary fixes, the Direct Hire Module successfully passed unit testing.

Table 13. Unit Testing for Direct Hire Module

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT/RESPONSE TIME	REMARKS
Add new DH applicant	System users should be able to create and submit a new DH application for an applicant and attach supporting documents, with the system accepting multiple file formats.	The system successfully allowed the users to create and submit a new DH application. File uploads in different formats were accepted. (1-3 sec)	Passed
Edit DH application	System users can edit/update the applicant's information.	Users successfully updated the applicant's record, and the modal saved the updated information correctly. (1-3 sec)	Passed
Archive application	Administrators can archive applicants' information without deleting them from the database.	The admin successfully archived the record, which was removed from the active monitoring list and moved to the archived list. (1-3 sec)	Passed
Update DH application status	System users can update the application status of an applicant.	The users successfully updated the application status, and the change was reflected in the monitoring table and the overall status chart. (1-3 sec)	Passed
Generate clearance and manage documents	The system should generate a clearance based on input data and allow system users to upload additional documents and download files as needed.	The system generated a compiled file named "Direct Hire Clearance.docx" using the applicant's input data. (1-3 sec)	Passed

Table 14 presents the unit testing results for the Balik Manggagawa Module. During initial testing of this module, the same minor issue previously identified in the Direct Hire Module was observed: the input fields were not cleared after form submission. Additionally, the visual analytics feature did not update as expected. The issue was reviewed and resolved through debugging, and subsequent testing confirmed that all assessed functions worked as intended. Each tested function performed accurately throughout the testing.

Table 14. Unit Testing for Balik Manggagawa Module

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT/RESPONSE TIME	REMARKS
Add new BM applicant	Users should be able to create and submit a new BM application.	The system successfully added the new applicant. The record appeared in the monitoring table. (1-5 sec)	Passed
Edit BM Applicant	System users should be able to edit/update the applicant's information.	The user successfully updated the applicant's details. The modal saved the new information correctly. (1-3 sec)	Passed
Update BM application status	Users should be able to update the application status ("For Compliance", "Approval", "Rejected").	The system successfully updated the application status. When "For Compliance" was selected, the compliance categories appeared, required details were prompted, and the correct compliance form was generated. (1-5 sec)	Passed

Table 15 displayed the results of unit testing for the Gov-to-Gov Module. The results indicated that the module fulfilled all specified functional requirements. All tested cases worked as expected, with the system consistently saving and reflecting data updates without encountering errors.

Table 15. Unit Testing for Gov-to-Gov Module

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT/RESPONSE TIME	REMARKS
Add G2G applicant	Users should be able to add new G2G applicants.	The applicant was successfully added. All details from the view modal were saved correctly. (1-3 sec)	Passed
Edit G2G applicant	System users should be able to edit/update the applicant's information.	The users successfully updated the applicant's details. The modal saves the modified info correctly. (1-3 sec)	Passed
Update G2G application Status	The system should allow the users to update the "Date Card Released" and "Received by Region" status.	The users successfully updated the status. The change was saved. (1-5 sec)	Passed

Table 16 displays the results of the unit testing for the Information Sheet Module. The results indicate that the module operated seamlessly across all essential functions. Each test case provided feedback within an acceptable response time and produced expected and predictable results.

However, during initial testing, minor issues were observed. The input fields were not cleared after form submission, and the visual analytics feature

did not update as expected. These issues were reviewed and resolved through debugging, and subsequent testing confirmed that all assessed functions worked as intended. Each tested function performed accurately throughout the testing process

Table 16. Unit Testing for Information Sheet Module

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT/RESPONSE TIME	REMARKS
Add new info sheet request	The system should allow users to create a new info sheet request.	The new info sheet request was successfully created and appeared in the monitoring table. (1-3 sec)	Passed
Upload supporting documents	Users should be able to upload supporting documents for the request.	Documents were uploaded successfully. (1-3 sec)	Passed
Edit applicant's info sheet request	System users should be able to edit/update the applicant's information.	The users successfully updated the applicant's information. The modal saved the new details correctly. (1-3 sec)	Passed

Table 17 shows the unit testing results for the Monitoring List Module, including the tested functions, expected outputs, actual system responses, and response times. The results indicate that the module performed as intended, with all tested functions passing the unit testing phase. Each operation produced accurate outputs within an acceptable response time, demonstrating system good performance and reliability.

Table 17. Unit Testing for Monitoring List Module

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT/RESPONSE TIME	REMARKS
Add new job fair event	Users should be able to add new job fair events.	The new job fair was added successfully and appeared on the monitoring list. (1-3 sec)	Passed
View job fair details	Clicking “view” on a specific job fair event should open a modal displaying all event details correctly.	The job fair details modal opened and displayed all information correctly, including contact numbers and the “rescheduled” tag. (1-3 sec)	Passed

Table 18 presents the User Profile Module testing, indicating that users can successfully manage their account information without errors. The process of editing personal details and passwords was simple, with changes instantly and precisely reflected in the system. This demonstrated that the module was functional and user-friendly, enabling end users to manage their profiles while maintaining system integrity.

Table 18. Unit Testing for User Profile Module

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT/RESPONSE TIME	REMARKS
Edit account information	System users should be able to update their personal information and account details.	The user successfully updated their personal information and password. All changes were saved. (1-3 sec)	Passed

Table 19 displayed the unit testing for the Super Admin Panel Module, covering User Management, Data Backup, and System Reports. It included

expected, actual, and response times. The results indicated that the module worked properly, with all features successfully passing unit testing. Response time varied across functions due to factors such as internet speed and file size. Initial testing found multiple errors in the User Management feature. These issues were reviewed and debugged, resulting in a successful retest and passing unit tests.

Table 19. Unit Testing for Admin Panel

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT/RESPONSE TIME	REMARKS
Add new user	The admin should be able to create a new user account with a specific role.	A new user account was created successfully. The new account appeared in the user list. (1-3 sec)	Passed
Edit user settings	The admin should be able to update a user role and page permission.	The super admin and admin successfully updated the user role and page permission. (1-3 sec)	Passed
Deactivate user	The admin should be able to deactivate a user account, preventing login without deleting the user's record.	The user's status changed to "inactive." A login attempt with the account failed as expected. (1-3 sec)	Passed
Generate certificate/ report	The system should generate a certificate report for the selected month/year.	Clicking "generate report" for a specific month and year successfully created and listed Certificate_of_Utilization_Month_Year.docx. (1-10 sec)	Passed
Download report	Admin should be able to download the generated report file.	The document was downloaded successfully. (1- 5 sec)	Passed
Delete report	Admin should be able to delete the generated report from the list.	When the admin clicked the delete icon and confirmed the action, the report was successfully removed from the list. (1-5 sec)	Passed

Manual Backup	The system admin should be able to create a manual backup of the database.	The system processed the request and created a new backup file. A success message was displayed. (30-120 sec)	Passed
Import Backup	The system admin should be able to import a previously created backup file.	the system processed the request, successfully imported the backup file, and displayed a confirmation message. (15-30 sec)	Passed
Download backup	The admin should be able to download a previously created backup file from the list.	The file downloaded successfully to the local machine. (1-10 sec)	Passed
Restore backup file	The admin should be able to select a backup file and restore the database to that state.	The system successfully restored the database, and all data matched the backup point. (15-60 sec)	Passed

Integration Testing

Table 20 presents the integration testing results for the different system modules, including their expected and actual outputs. The remarks indicated that the modules worked as intended, with all features successfully passing integration testing. However, during the initial testing phase, various issues were identified in the analytical panels. In the Direct Hire Module, the status counts were inconsistent; specifically, when a status was updated, the previous count remained unchanged, and subsequent updates resulted in duplicate counts. The pie graphs in the Balik Manggagawa and Information Sheet modules experienced glitches, as they visibly refreshed or reloaded while information was being entered. These issues were reviewed and debugged, resulting in a successful retest and the passing of integration testing.

Table 20. Integration Testing of the System

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT	REMARKS
Dashboard	The system should properly integrate and display all summary data from different modules in real time.	The dashboard successfully integrated data from all modules and displayed accurate, real time summaries and details.	Passed
Direct Hire	The system should display added applications in the monitoring table. Upon selecting the view, the complete applicant information should appear accurately, and the record should be reflected in the dashboard and analytical panel.	The system accurately displays applications in the monitoring table. The view function showed the complete and correct applicant details, and data were properly reflected in the dashboard and analytical panel.	Passed
Balik Manggagawa	The created record must reflect in the dashboard and the analytical panel must also be updated.	The dashboard showed the correct sex and record count, and the analytical panel displayed the accurate applicant status.	Passed
Gov-to-Gov	The system should automatically add newly created G2G records to the correct sec category in the dashboard and update the overall count of records.	The dashboard displayed the correct sex and count for all new entries.	Passed
Information Sheet	The system should automatically update the dashboard with the correct sex category and record count, and the analytical panel must also be updated.	The dashboard displayed the correct sex and record count, and the analytical panel was updated accordingly.	Passed
System Reports	The system should gather data from all processing modules (DH, BM, G2G, IS) to generate accurate monthly, quarterly, and yearly utilization reports.	The system successfully pulled data from all processing modules. The generated report accurately reflected the total processed documents and clients served for the selected period.	Passed

Compatibility Testing

The compatibility testing of the BalikBayani Portal was conducted using Google Lighthouse, a computerized tool for evaluating website quality. In terms of performance, as shown in Table 21, Google Chrome showed the most consistent and stable results, followed by Microsoft Edge and lastly Opera GX. In terms of accessibility, both Chrome and Edge obtained excellent scores, while Opera GX produced acceptable but slightly lower results for certain modules. For best practices, all three browsers received ratings of 96%-100% across almost all modules. Meanwhile, in SEO, each browser earned an exceptional 100% score.

Among the tested browsers, Google Chrome demonstrated excellent performance, showing consistent and high scores across all categories, particularly in Performance and Accessibility. Microsoft Edge also demonstrated high compatibility and consistent results, whereas Opera GX excelled in Best Practices and SEO but showed minor performance declines in certain modules. These results indicated that the BalikBayani Portal was fully optimized for Chrome and Edge, with only minor browser-specific adjustments needed to improve performance consistency in Opera GX further.

Table 21. Compatibility Testing of the System

Browser	Web Page / Module	Performance	Accessibility	Best Practice	SEO
Chrome	Login	100	90	100	100
Chrome	Dashboard	98	84	96	100

Chrome	Direct Hire	88	84	100	100
Chrome	Balik Manggagawa	100	90	100	100
Chrome	Gov to Gov	84	90	96	100
Chrome	Information Sheet	96	90	96	100
Chrome	ML - Lists of Job Fairs	87	83	100	100
Chrome	ML - PESO Contact Info	100	80	100	100
Chrome	ML - PRAs Contact Info	100	90	100	100
Chrome	ML - Job Fair Monitoring Summary	97	84	100	100
Chrome	AP- User Management	90	90	96	100
Chrome	AP - Data Backups	89	86	100	100
Chrome	AP - System Reports	89	89	96	100
Chrome	User Profile	97	87	100	100
Microsoft Edge	Login	100	90	100	100
Microsoft Edge	Dashboard	99	84	96	100
Microsoft Edge	Direct Hire	98	84	100	100
Microsoft Edge	Balik Manggagawa	100	90	100	100
Microsoft Edge	Gov to Gov	99	90	96	100
Microsoft Edge	Information Sheet Sheet	87	90	96	100

Microsoft Edge	ML - Lists of Job Fairs	100	83	100	100
Microsoft Edge	ML - PESO Contact Info	100	80	100	100
Microsoft Edge	ML - PRAs Contact Info	100	90	100	100
Microsoft Edge	ML - Job Fair Monitoring Summary	100	84	100	100
Microsoft Edge	AP- User Management	100	76	96	100
Microsoft Edge	AP - Data Backups	100	86	100	100
Microsoft Edge	AP - System Reports	98	75	96	100
Microsoft Edge	User Profile	97	87	100	100
Opera Gx	Login	99	90	100	100
Opera Gx	Dashboard	97	84	96	100
Opera Gx	Direct Hire	98	84	100	100
Opera Gx	Balik Manggagawa	82	80	100	100
Opera Gx	Gov to Gov	70	90	96	100
Opera Gx	Information Sheet	57	90	96	100
Opera Gx	ML - Lists of Job Fairs	97	83	100	100
Opera Gx	ML - PESO Contact Info	96	80	100	100
Opera Gx	ML - PRAs Contact Info	99	90	100	100
Opera Gx	ML - Job Fair Monitoring	86	84	100	100

Summary

Opera Gx	AP- User Management	90	90	96	100
Opera Gx	AP - Data Backups	99	86	100	100
Opera Gx	AP - System Reports	98	75	96	100
Opera Gx	User Profile	99	87	100	100

Results of Experimental Setup

Table 22 showed the process durations of the manual and system-based processes recorded during the simulation. The manual processing times were obtained from official records collected during the previous week, while the BalikBayani Portal automatically logged the system processing times during the experimental run.

The results demonstrated a clear improvement in processing efficiency when using the BalikBayani Portal. The manual process recorded an average transaction time of 1.83 minutes per client, whereas the system-based process reduced the average time to 1.00 minutes per transaction. This reduction indicated that the system streamlined key processing steps, minimized delays associated with manual handling, and optimized transaction flow.

The decrease in processing time suggested a reduced administrative workload for staff and faster service delivery for migrant workers. By shortening the process cycle time, the system enabled personnel to accommodate more clients within the same operational period while

maintaining accurate and organized records. These findings confirmed that the BalikBayani Portal enhanced operational efficiency compared to the previous manual procedure, supporting its effectiveness as a digital solution for record processing and transaction management.

Table 22. Process Time per Client: Manual vs. System

CLIENT	MANUAL PROCESS (IN–OUT)	MANUAL DURATION (MINUTES)	SYSTEM PROCESS (IN–OUT)	SYSTEM DURATION (MINUTES)
1	12:36–12:39	3 mins	12:41–12:42	1 min
2	1:03–1:04	1 min	1:03–1:04	1 min
3	1:37–1:39	2 mins	1:37–1:39	2 mins
4	1:46–1:48	2 mins	1:47–1:47	< 1 min
5	1:40–1:42	2 mins	1:54–1:55	1 min
6	2:05–2:06	1 min	2:28–2:29	1 min
AVERAGE		**1.83 mins**		**1.00 min**

Results of ISO/IEC 25010 Evaluation

Table 23 showed the post-evaluation results of the BalikBayani Portal based on the ISO 25010 quality standard, as assessed by three (3) IT professionals using a 5-point Likert scale. It obtained an outstanding overall weighted mean of 4.67 and was interpreted as “Excellent.” Among the criteria, Safety obtained the highest composite mean of 4.93, indicating the system’s ability to protect sensitive migrant worker data from risks and harm. Functional Suitability and Security both received 4.78, indicating that the system successfully met user requirements while providing secure and reliable data

handling, which was important in a record management platform. Maintainability received a rating of 4.73, indicating that updates and corrections could be implemented properly. Flexibility (4.67) and Interaction Capability (4.63) showed that the system could adapt to different user needs and had a user-friendly interface for easy navigation. Performance Efficiency scored 4.56, showing that the system responded quickly under typical workloads. While Compatibility and Reliability both received a 4.50, confirming that the system worked well in a variety of environments and consistently delivered stable performance. Overall, the high scores confirmed that the system achieved one of its intended objectives, providing its readiness for real-world implementation and potential for long-term use.

Table 23. Post Evaluation Result

Criteria	Composite Mean	SD	Interpretation
Functional Suitability	4.78	±0.33	Excellent
Performance Efficiency	4.56	0	Excellent
Compatibility	4.50	0	Excellent
Interaction Capability	4.63	±0.27	Excellent
Reliability	4.50	0	Excellent
Security	4.78	±0.30	Excellent
Maintainability	4.73	±0.26	Excellent
Flexibility	4.67	±0.29	Excellent
Safety	4.93	±0.26	Excellent
OVERALL WEIGHTED MEAN	**4.67**	**±0.14**	**Excellent**

N=3 Legend: *4.21 – 5.00 = Excellent, 3.41 – 4.20 = Good, 2.61 – 3.40 = Fair, 1.81 – 2.60 = Needs Improvement, 1.00 – 1.80 = Poor*

Results of Technology Acceptance Model Evaluation (TAM)

This section presents the system's acceptability, as evaluated using the Technology Acceptance Model (TAM). A total of 104 respondents, consisting of staff members from the DMW Processing Division and other individuals working in related customer interaction fields, assessed the system using a 4-point Likert scale across four main criteria: Perceived Usefulness, Perceived Ease of Use, Attitude Towards Using, and Behavioral Intention to Use. The data gathered from these assessments was analyzed to determine the mean scores for each criterion, providing a quantitative measure of the system's potential for successful integration

Table 24 showed consistent results, obtaining a composite mean score of 3.82 with a High interpretation under the Perceived Usefulness criterion. The statement with the highest rating (mean = 3.88) indicated that users trusted the system to produce accurate and reliable outputs. The statement related to task efficiency also received a high mean of 3.84, reflecting that users felt the system helped them perform tasks faster and with less effort. The productivity improvement statement, although slightly lower at 3.74, was still interpreted as high, showing that users recognized the system’s contribution to improving work productivity within the processing division. The standard deviation values, ranging from ±0.32 to ±0.48, indicated that the respondents mostly shared similar perceptions, with only slight differences in their evaluations.

These findings indicate that the BalikBayani Portal improved task efficiency, productivity, and output quality, demonstrating that the system is

perceived as a valuable tool that effectively supports tasks and enhances overall operational performance.

Table 24. Perceived Usefulness

Criteria	Mean	SD	Interpretation
Task Efficiency. The system enables users to complete tasks faster and minimizes the time and effort required for document processing.	3.84	±0.40	High
Productivity Improvement. The system contributes to improved work productivity within the processing division.	3.74	±0.48	High
Quality of Output. The BalikBayani Portal produces accurate and reliable outputs.	3.88	±0.32	High
Composite Mean	**3.82**	**±0.08**	**High**

n=104

Table 25 showed users' perceptions of how easy the system was to use, with all statements receiving a High interpretation. The composite mean of 3.72 indicated that users generally considered the system user-friendly. The sub-criteria of System Navigation and Task Simplicity received the highest mean rating of 3.75, reflecting users' agreement that the Portal's layout was clear and facilitated task completion with minimal steps. Learnability obtained a mean score of 3.67 and had a relatively higher standard deviation, suggesting that while most users found the system easy to learn, some users required additional assistance. These findings indicated that the BalikBayani Portal was both effective and accessible, making it

practical for everyday use and allowing it to be efficiently integrated into daily operations with minimal training requirements for users.

Table 25. Perceived Ease of Use

Criteria	Mean	SD	Interpretation
System Navigation. The layout is clear and easy to navigate.	3.75	±0.46	High
Learnability. Users can easily remember system functions without frequent assistance, making it user-friendly for first-time users.	3.67	±0.57	High
Task Simplicity. The process of completing tasks in the BalikBayani Portal is simple and minimal.	3.75	±0.52	High
Composite Mean	**3.72**	**±0.05**	**High**

n=104

Table 26 presented the respondents' attitudes toward using the BalikBayani Portal. Based on the results, the respondents expressed a high level of positive attitude toward the system, as indicated by the composite mean of 3.79. The statement with the highest rating (mean = 3.88) indicated that users felt confident using the portal due to its fast and stable operation. The statement related to overall satisfaction also received a high mean of 3.77, demonstrating that users felt satisfied with the system as it met their expectations. This reflected that the system effectively aligned with user needs and contributed to a favorable attitude. User experience, on the other hand, obtained a mean of 3.72, which, while slightly lower than the other sub-criteria, still fell within the high range, indicating that users generally found the

system design and interface visually appealing. Overall, these findings showed that the BalikBayani Portal was perceived at a high level across all sub-criteria and demonstrated strong acceptance and a favorable attitude toward system use.

Table 26. Attitude Towards Using

Criteria	Mean	SD	Interpretation
System Performance. Users feel confident using the BalikBayani Portal as it responds quickly and runs smoothly.	3.88	±0.35	High
User Experience. Users find the system design and interface visually appealing.	3.72	±0.45	High
Overall Satisfaction. Users feel satisfied using the system as it meets their expectations.	3.77	±0.47	High
Composite Mean	**3.79**	**±0.06**	**High**

n=104

Table 27 showed respondents' behavioral intention to use the BalikBayani Portal. The system received a high level of behavioral intention, as reflected by the composite mean of 3.79. This indicated that users were likely to continue using the system and recommend it to others, reflecting high acceptance and intention for future use. The technology acceptance indicator received the highest mean score of 3.81, suggesting that users perceived the system as essential for record management. The recommendation indicator, with a mean of 3.80, further showed that users were willing to recommend the system because they believed it provided potential benefits. The statement

related to future use obtained a mean of 3.76, which, while slightly lower than the other sub-criteria, still fell within the high range. These findings indicated that the BalikBayani Portal demonstrated high behavioral potential for continued use, reflecting users' strong willingness to adopt and sustain system usage in their record processing tasks.

Table 27. Behavioral Intention to Use

Criteria	Mean	SD	Interpretation
Future Use. Users find the BalikBayani Portal beneficial for long term use.	3.76	±0.48	High
Technology Acceptance. Users perceived the BalikBayani Portal as essential for record management.	3.81	±0.44	High
Recommendation. Users believe that others would benefit from using the system and are willing to recommend the BalikBayani Portal to others.	3.80	±0.40	High
Composite Mean	**3.79**	**±0.04**	**High**

n=104

Table 28 presents the overall results of the Technology Acceptance Model (TAM) evaluation of the BalikBayani Portal based on responses from 104 participants. The system obtained a composite mean of 3.78, interpreted as High, indicating strong user acceptance across all TAM dimensions. All criteria received high mean scores, showing that users viewed the system as useful, easy to use, and suitable for continued adoption.

Perceived Usefulness recorded the highest mean score of 3.82, indicating that users believed the system improved their work performance. This high rating was primarily attributed to key system features such as the Dashboard, which provided an immediate overview of transaction status; automated record and report generation, which reduced manual effort and errors; and monitoring and tracking functions, which enabled efficient management and retrieval of records. These features streamlined workflows and improved output accuracy.

The consistently high scores in Perceived Ease of Use, Attitude Towards Using, and Behavioral Intention to Use further supported the usefulness findings, as users were more likely to adopt a system that was easy to navigate and met their operational needs. Overall, the results indicated that users found the BalikBayani Portal useful because it simplified record processing, enhanced efficiency, and supported effective monitoring of transactions.

Table 28. Evaluation of Survey Results

Criteria	Mean	SD	Interpretation
Perceived Usefulness	3.82	±0.08	High
Perceived Ease of Use	3.72	±0.05	High
Attitude Towards Using	3.79	±0.06	High
Behavioral Intention to Use	3.79	±0.04	High
Overall Weighted Mean	**3.78**	**±0.02**	**High**

n=104

CHAPTER V

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

This chapter contains a summary of the capstone project, which includes a discussion of the conclusions found during the study and the recommendations that make the project successful.

Summary

This capstone project, titled “BalikBayani Portal: A Filing System for Migrant Records in the Processing Division of the Department of Migrant Workers,” aimed to address the challenges associated with the existing manual workflow in migrant record processing. Traditionally, the encoding and handling of migrant worker documents involved repetitive paperwork and redundant procedures, resulting in slow, time-consuming processes. The division relied heavily on Microsoft Excel to manage migrant records, which compromised data accuracy and documentation organization, particularly as the volume of records increased daily. Recognizing these issues, the researchers sought to design and develop a centralized digital solution to simplify the division's entire record-processing workflow.

The web-based system included a record management platform that supported document generation and data visualization. Authorized users could encode and upload supporting documents while monitoring application progress in real time. By replacing manual spreadsheets, the Portal ensured faster processing and more efficient results. Monitoring of each worker’s documentation status. The intuitive interface and structured workflow reduced

repetitive tasks and minimized issues associated with manual record management.

This project was closely aligned with the Sustainable Development Goals (SDGs), specifically SDG 8: Decent Work and Economic Growth, SDG 9: Industry, Innovation, and Infrastructure, SDG 10: Reduced Inequality, and SDG 16: Peace, Justice, and Strong Institutions. By promoting digital transformation in record management, the system supported more transparent workflows and strengthened data security. Beyond technological innovation, the initiative reflected the DMW's broader mission to empower migrant workers through digital transformation, thereby providing sustained ease of doing business.

The system's scope primarily focused on migrant record management within the Processing Division of the DMW. It featured key modules such as the Login/Authentication Module, Dashboard Module, Direct Hire Module, Balik Manggagawa Module, Gov-to-Gov Module, Information Sheet Module, Monitoring List modules (Job Fairs, PESO Contact Information, PRA Contact Information, and Job Fair Monitoring Summary), Super Admin Panel modules (User Management and Data Backup), and the User Profile Module. Each component played a vital role in creating a user-friendly and efficient processing experience for staff. While the system introduced significant improvements, it had limitations, including its dependency on stable internet connectivity and restricted accessibility within the DMW. Nevertheless, these limitations could be addressed through continuous system enhancement and infrastructure upgrades. Overall, the BalikBayani Portal bridged the gap

between manual and digital record-keeping, fostered accountability among staff, and strengthened the DMW's commitment to delivering transparent and reliable services to Filipino migrant workers.

Conclusion

The development and implementation of the BalikBayani Portal demonstrated the significant impact of digital innovation on record management. The project achieved its primary objective by creating a web-based system that simplified and digitized the filing process for migrant worker records, thereby improving organizational effectiveness.

Develop a filing system that integrates document generation into a printable format and supports quick, accurate retrieval of information.

The BalikBayani Portal functioned as a digital filing system that allowed users to generate printable documents and retrieve information accurately and efficiently. The record details view and document generation features ensured that applicant records were properly managed and presented. The ability to create official documents and clearances reduced manual encoding and made it easier for both administrators and staff to access records, addressing many challenges associated with traditional manual methods.

Integrate a data visualization feature that provides representation across the main categories.

The integration of data visualization proved to be an essential feature of the BalikBayani Portal. Through analytical panels and a visual dashboard,

the system enabled administrators and staff to monitor activities, track application progress, and identify patterns across major categories without manually reviewing individual records. This capability reduced time and effort while improving accuracy in monitoring and reporting. The integration of data visualization confirmed that the system enabled more informed, timely decision-making in the management of migrant worker applications.

Test and evaluate user acceptance of the system using the Technology Acceptance Model (TAM) and the system's effectiveness based on the ISO 25010 software quality model focusing on performance, interaction capability, reliability, and security.

The system was evaluated using the Technology Acceptance Model (TAM) and the ISO 25010 software quality standards. The findings revealed a high level of user acceptance, as reflected in the high ratings for perceived Usefulness, perceived ease of use, attitude toward using, and behavioral intention to use. Positive user feedback confirmed the system's alignment with user needs and expectations. In terms of technical effectiveness, the ISO 25010 evaluation showed excellent performance across all major criteria, particularly in performance efficiency, interaction capability, reliability, and security, meeting the most critical system requirements. Overall, the findings demonstrated that the system was both highly accepted by users and technically effective, confirming its full potential as a reliable platform for record management and application processing.

In conclusion, the BalikBayani Portal met its technical and functional objectives and aligned with the Department of Migrant Workers' mission of

promoting digital transformation and improving service delivery. It also aligned with several United Nations Sustainable Development Goals: SDG 8 by creating a more organized and secure work environment through modernized processes; SDG 9 through the development of reliable and sustainable digital infrastructure for improved public service delivery; and SDGs 10 and 16 by promoting fair, ethical, and transparent processing and equitable access to services through an accountable digital system. With continued improvement and institutional support, the BalikBayani Portal could serve as a model for digital innovation in migrant worker governance.

Recommendation

Based on the findings of this project, the following recommendations are highly suggested:

1. Implement the System in Regular Operations. Given its success, the BalikBayani Portal should continue to be used by the Processing Division in daily transactions to reduce processing time and lessen the manual workload. Other DMW regional offices may also adopt the system to promote a more unified and standardized workflow.

2. Strengthen Technical Infrastructure. Reliable internet connectivity, updated hardware, and knowledgeable IT personnel are essential to ensure smooth system performance and prevent delays during system operations and workflow execution.

3. Provide User Training and Support. Regular workshops and orientations should be conducted for DMW staff to familiarize them with the

system’s features. This will ensure proper transaction administration, maximize utilization, and reduce resistance to digital adoption. Clear operation guidelines and readily available technical support will further minimize errors during implementation.

4. Include a Feedback Mechanism. Built-in feedback form should be integrated into the BalikBayani Portal to allow staff to report issues and suggest improvements. This mechanism can guide future updates to the system’s interface and functionality.

5. Ensure Regular Maintenance and Security Measures. The DMW should conduct routine system maintenance, updates, and performance checks to maintain stability. In addition, strict data protection policies must be enforced to safeguard the confidentiality of migrant workers' information.

6. Establish Clear Implementation Ownership and Sustainability Planning. For long-term sustainability, the Processing Division should designate a system administrator to manage system operations, user access, and updates. Furthermore, annual resources should be allocated for system maintenance, technical support, and compliance with data security and privacy standards.

Through these recommendations, the DMW can continue its digital transformation initiatives and ensure that overseas Filipino workers’ records are processed accurately and efficiently, upholding the agency’s mission to provide timely and accessible services to all Filipinos working abroad.

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APPENDICES

APPENDIX A

Letter of Approval

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

CERTIFICATE OF APPROVAL

This is to certify that after having satisfactorily completed all the requirements of the capstone project entitled:

"BALIKBAYANI PORTAL: A FILING SYSTEM FOR MIGRANT RECORDS IN THE PROCESSING DIVISION OF THE DEPARTMENT OF MIGRANT WORKERS"

ANDREI NIKOLAI LIMPIADA, ANGELO M. MANCERA, and FATIMA L. EVANGELISTA have been granted permission to proceed with their proposal presentation.

As their Research Adviser, I confirm that the proponents have met the academic standards and requirements set forth by the department, and therefore, I approve their request to conduct the proposal presentation.

CRISANTO F. GULAY
Capstone Project Adviser

APPROVAL SHEET

The Capstone Project entitled **"BALIKBAYANI PORTAL: A FILING SYSTEM FOR MIGRANT RECORDS IN THE PROCESSING DIVISION OF THE DEPARTMENT OF MIGRANT WORKERS"** prepared and submitted by **Andrei Nikolai Limpiada, Fatima L. Evangelista, and Angelo M. Mancera** in partial fulfillment for the degree Bachelor of Science in Information Technology specialized in Web and Mobile Application Development, is hereby recommended for approval and acceptance.

CRISANTO F. GULAY
Adviser

Approved and Accepted by the Committee on Oral Examination with a grade of _____ .

GENE MARCK B. CATEDRILLA
Member

CAROLINA R. JOVAL
Member

MYRA V. PAICAGLINO
Member

JEFFERSON L. LERIOS
Member

Accepted in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology, specialized in Web and Mobile Application Development.

LOYD S. ECHALAR
Associate Dean, Colllege of Computer Studies

OFELIA B. MANINGAS, EdD.
Chairperson, Research and Development Office

_____
Date Signed

RESEARCH CONTRIBUTION NO.	

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

BAGONG PILIPINAS

October 25, 2024

ATTY. APRIL C. CASABUENA
Director, Department of Migrant Workers 4-A

Dear Director Casabuena,

Greetings from Laguna State Polytechnic University – Los Baños Campus!

We are pleased to inform you that the College of Computer Studies formally accepts your request for assistance in the development and digitization of the Process Division of DMW Region IV-A. We are committed to supporting your office in enhancing your operational efficiency through this system development initiative.

To ensure the smooth execution of the project, we hereby designate the following faculty members and student developers who will work directly with your team:

Faculty-in-Charge:
- **Crisanto F. Gulay**
- **Merardo Arenas Camba, Jr.**

Student Developers (BSIT Program):
- **Andrei Nikolai Limpiada**
- **Angelo M. Mancera**
- **Fatima L. Evangelista**

Our team will coordinate closely with the Process Division to gather requirements, conduct system analysis and design, and carry out the development and implementation activities necessary for this automation project.

We look forward to a productive collaboration and to contributing to the advancement of DMW Region IV-A's service delivery.

Thank you very much, and we remain committed to supporting your office in this endeavor.

Respectfully,

DR. LOYD S. ECHALAR
Associate Dean, College of Computer Studies
Laguna State Polytechnic University – Los Baños Campus

Republic of the Philippines
Department of Migrant Workers
Ople Building, Ortigas Avenue cor. EDSA, Mandaluyong City 1550
Website: www.dmw.gov.ph | Email: feedback@dmw.gov.ph | Hotlines: 8722-1144, 8722-1155, 8724-1665, 8722-1165
BAGONG PILIPINAS

October 21, 2024

ATTY. RUSHID JAY S. SANCON
Campus Director
Laguna State Polytechnic University

Thru: **Ms. Rachel Gallano**
Chairperson, Extension and Training Services
LSPU-LBC

Dear **Atty. Sancon,**

Warm greetings from the Department of Migrant Workers, Regional Office IV-A (CALABARZON)!

We are writing to express our admiration for the exceptional role that Laguna State Polytechnic University – Los Baños Campus (LSPU-LBC) plays in providing impactful extension services and community engagement programs. Your institution's unwavering commitment to community development is truly commendable and has greatly inspired us.

At DMW Regional Office IV-A, we continually seek ways to enhance the services we offer to migrant workers and their families. In line with this, we are eager to explore potential collaboration with your esteemed institution, particularly in areas such as training and software development. We believe that LSPU-LBC's expertise and resources could significantly strengthen our efforts to serve the community more effectively.

With this in mind, we kindly request your support in facilitating a training program or the development of software solutions tailored to our needs. We are confident that the knowledge and skills shared by your faculty and programs will be instrumental in advancing our services.

Additionally, we would like to extend a formal invitation to **Ms. Rachell Gallano**, Chairperson of Extension and Training Services, and **Mr. Merardo Arenas Camba, Jr.,** IT Extension Coordinator, to participate in a discussion with our team. Their insights and expertise will be invaluable as we explore how we can work together to create meaningful impact.

We look forward to the opportunity to meet with you and your team to discuss these possibilities further. Please let us know your availability for a meeting at your earliest convenience.

Thank you for your consideration. We are excited about the prospect of collaborating with LSPU-LBC to benefit the community.

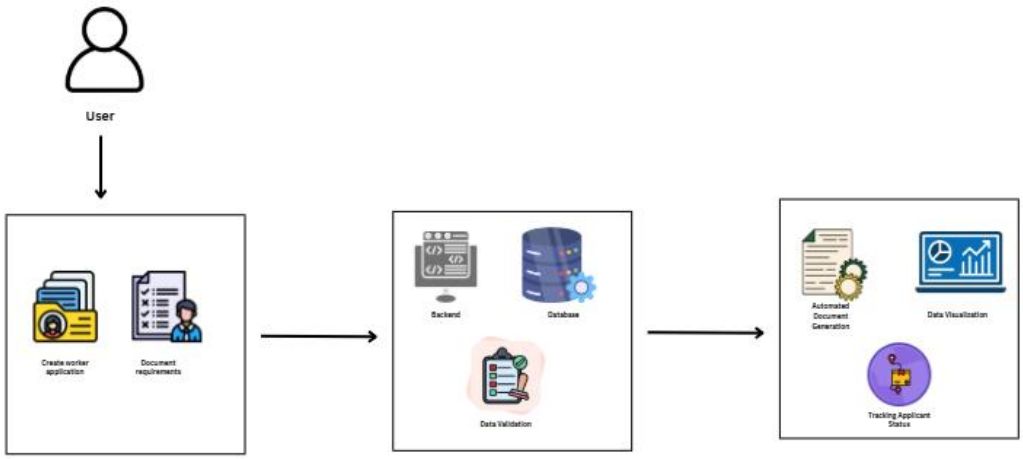
Respectfully,

ATTY. APRIL R. CASABUENA
Regional Director

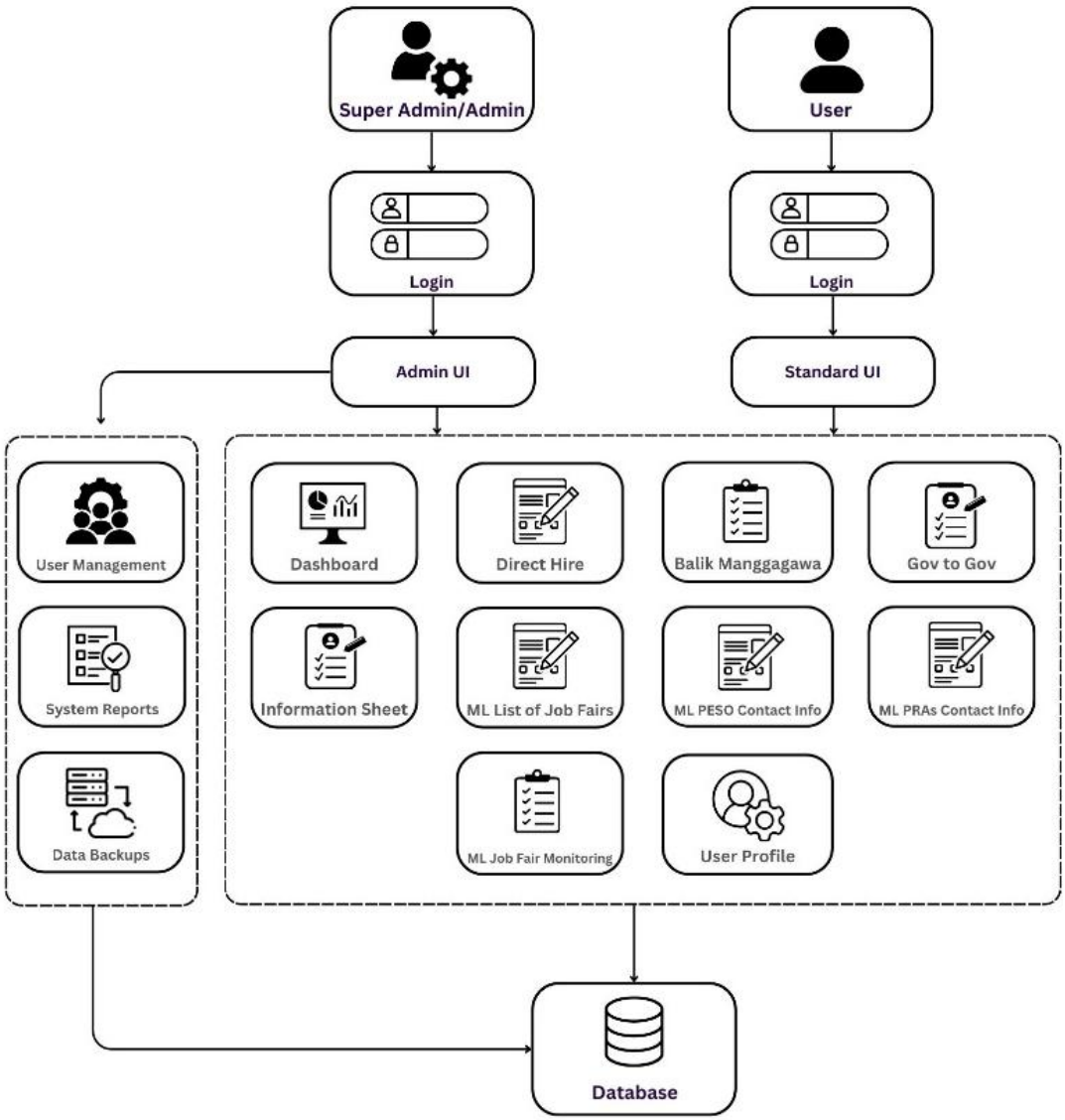
APPENDIX B

Figures

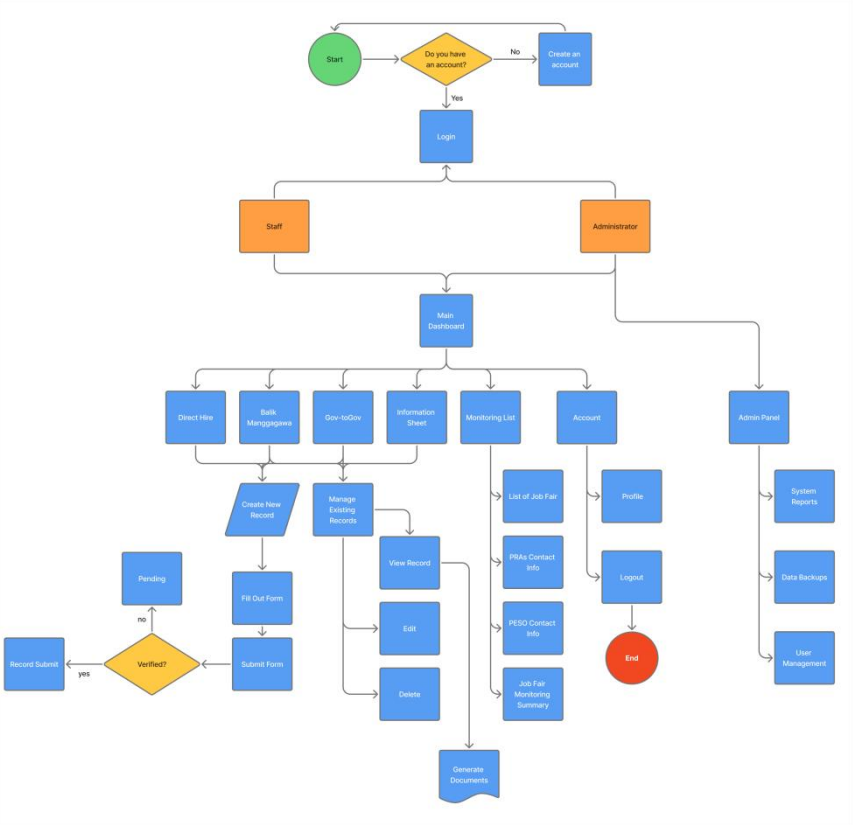
Conceptual Framework



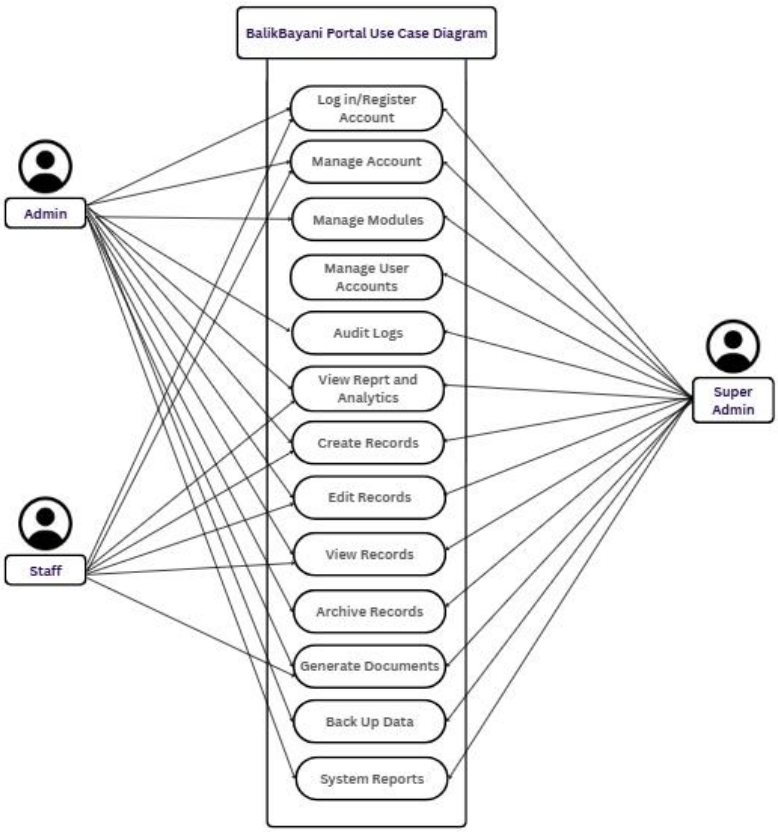
System Architecture



System Flowchart



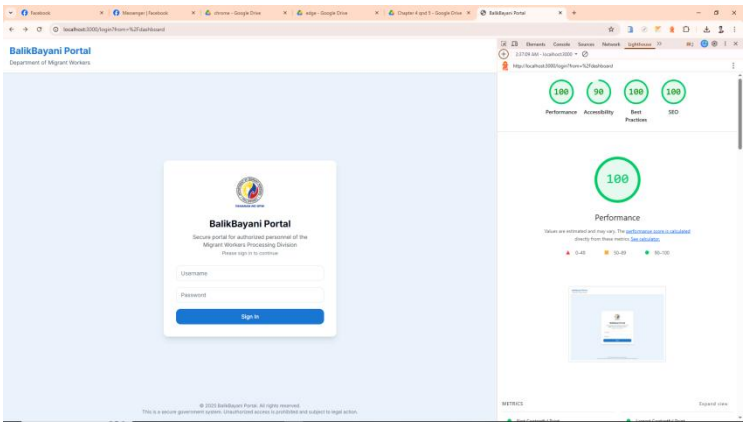
Use Case Diagram

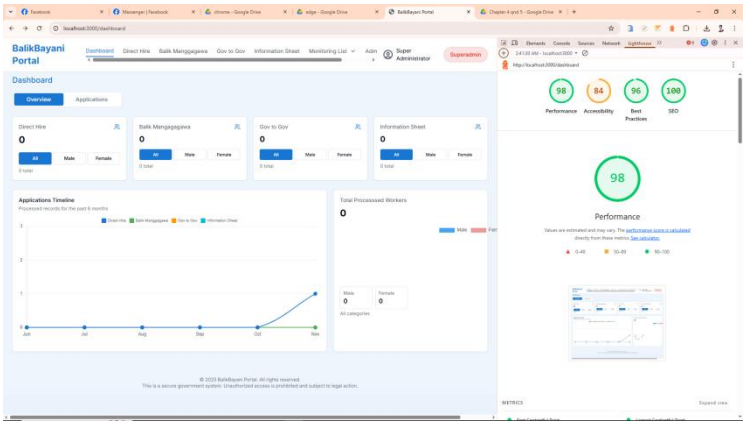


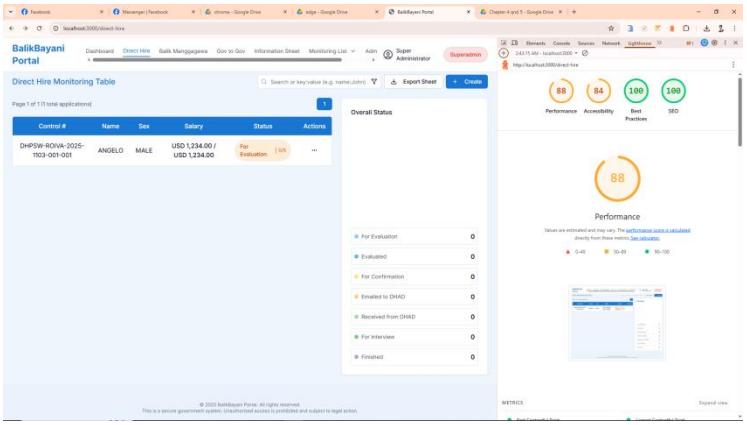
APPENDIX C

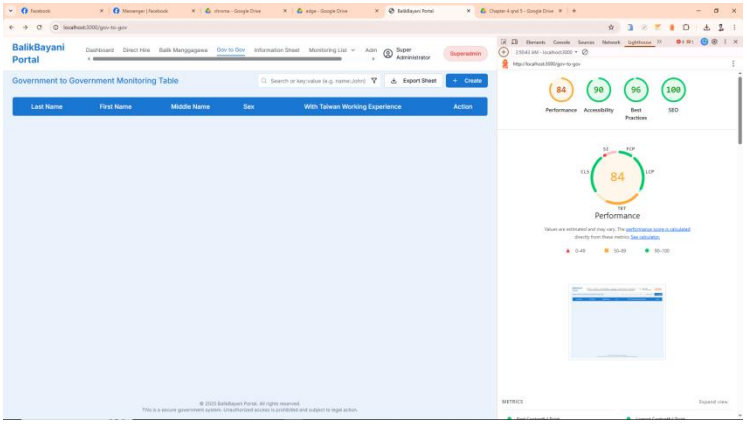
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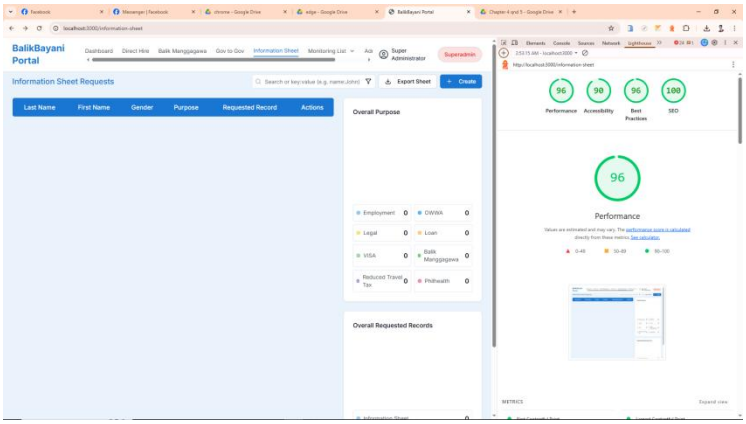
Google Chrome browser

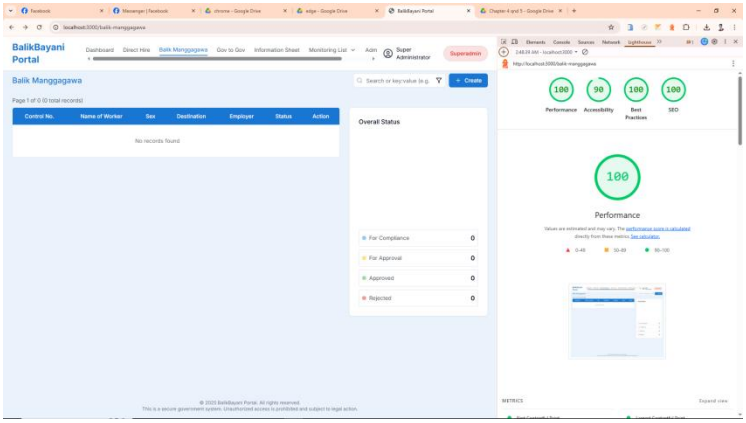


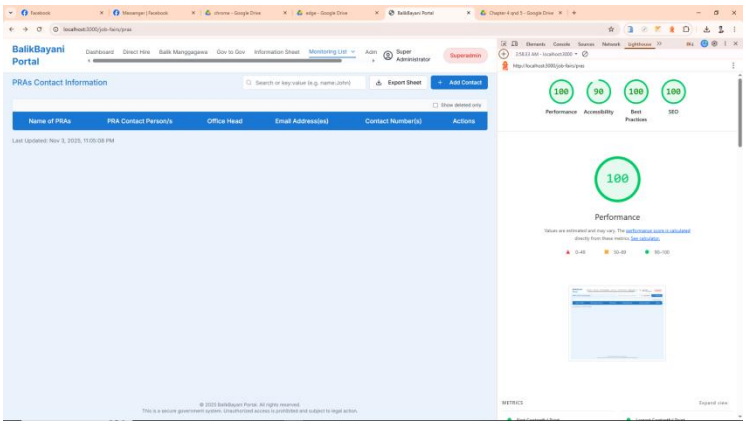


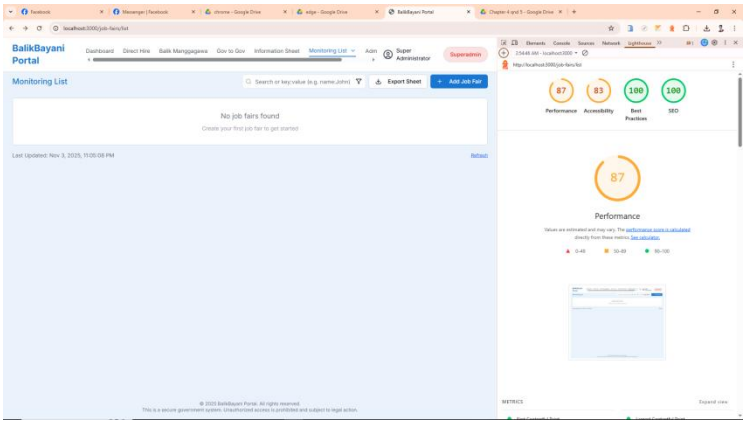


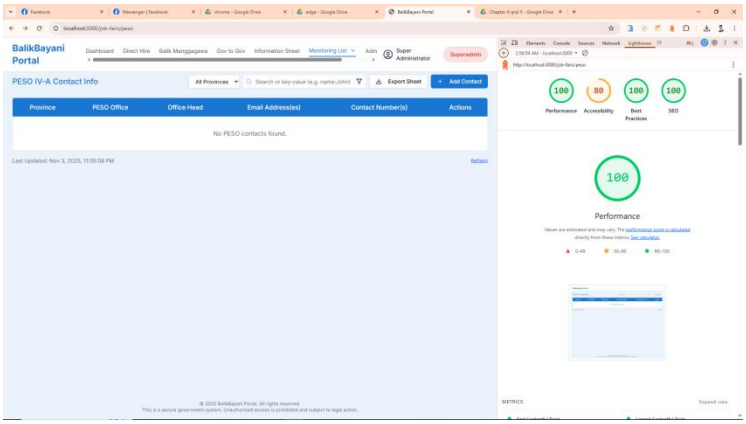


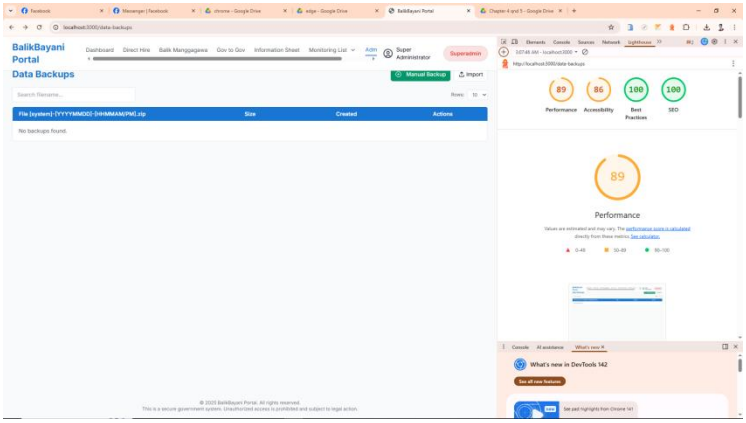

BalikBayani Portal
Information Sheet Requests
Export Sheet
Create
Last Name
First Name
Gender
Purpose
Requested Record
Actions
Overall Purpose
Overall Requested Records
Performance
Accessibility
Best Practices
SEO
96
Performance

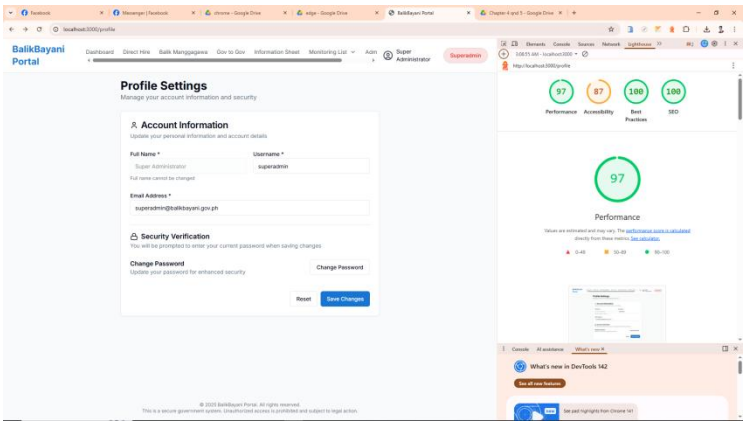

BalikBayani Portal
Balik Manggagawa
Create
Control No.
Name of Worker
Sex
Destination
Employer
Status
Action
No records found
Overall Status
For Compliance
For Approval
Approved
Rejected
Performance
Accessibility
Best Practices
SEO
100
Performance

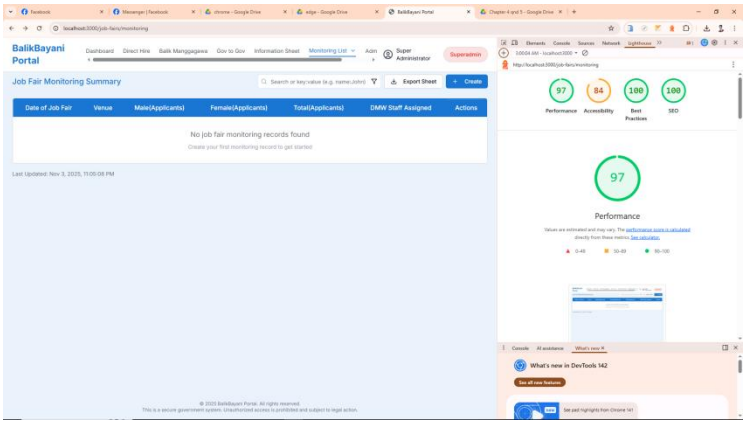

BalikBayani Portal
PRAs Contact Information
Export Sheet
Add Contact
Name of PRAs
PRA Contact Person/s
Office Head
Email Address(es)
Contact Number(s)
Actions
Performance
Accessibility
Best Practices
SEO
100
Performance

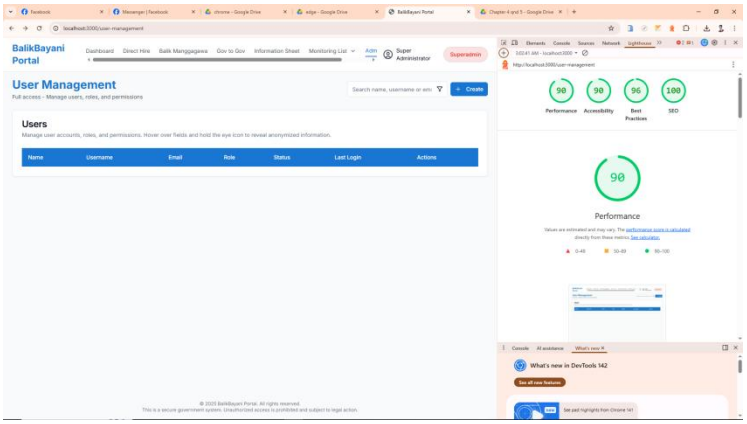

BalikBayani Portal
Monitoring List
Export Sheet
Add Job Fair
No job fairs found
Performance
Accessibility
Best Practices
SEO
87
83
100
100
87
Performance

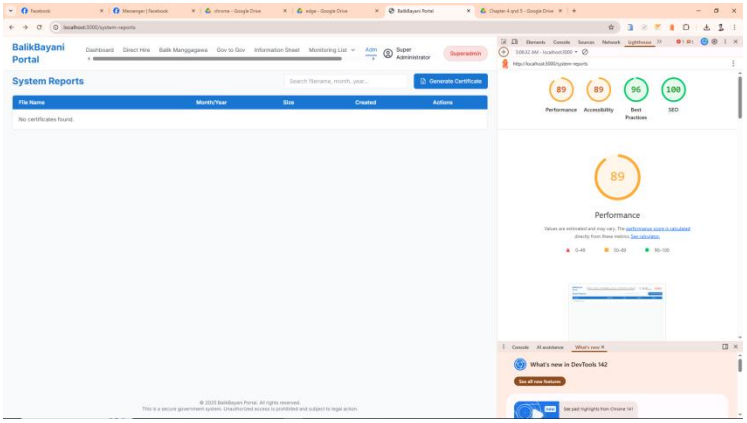

BalikBayani Portal
PESO IV-A Contact Info
Province
PESO Office
Office Head
Email Address(es)
Contact Number(s)
Actions
No PESO contacts found.
Performance
Accessibility
Best Practices
SEO


BalikBayani Portal
Data Backups
Manual Backup
Import
File (system) (YYYYMMDD-HHMMAM/PM).zip
Size
Created
Actions
No backups found.
Performance
Accessibility
Best Practices
SEO

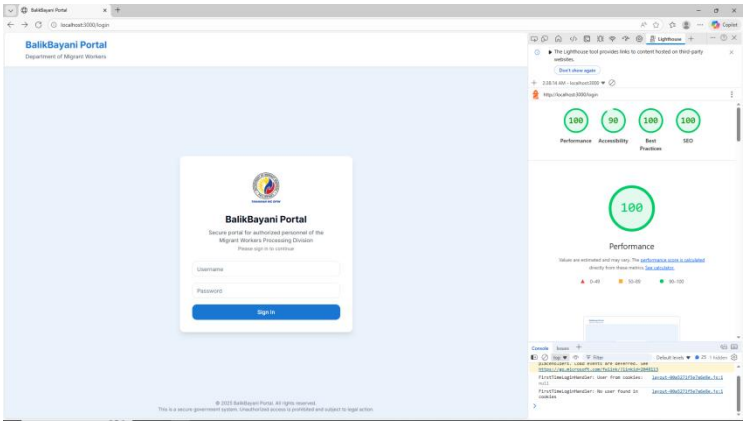

BalikBayani Portal
Profile Settings
Manage your account information and security
Account Information
Full Name *
Username *
Email Address *
Security Verification
Change Password
Reset
Save Changes
Performance
Accessibility
Best Practices
SEO

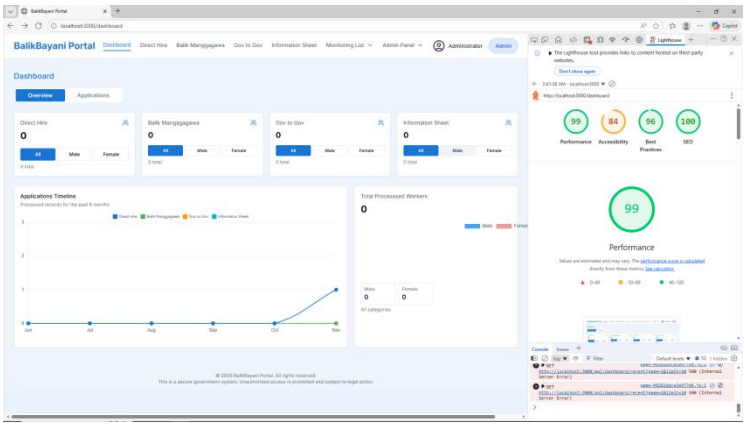

BalikBayani Portal
Job Fair Monitoring Summary
Export Sheet
Create
Date of Job Fair
Venue
Male(Applicants)
Female(Applicants)
Total(Applicants)
DMW Staff Assigned
Actions
No job fair monitoring records found
Performance
Accessibility
Best Practices
SEO

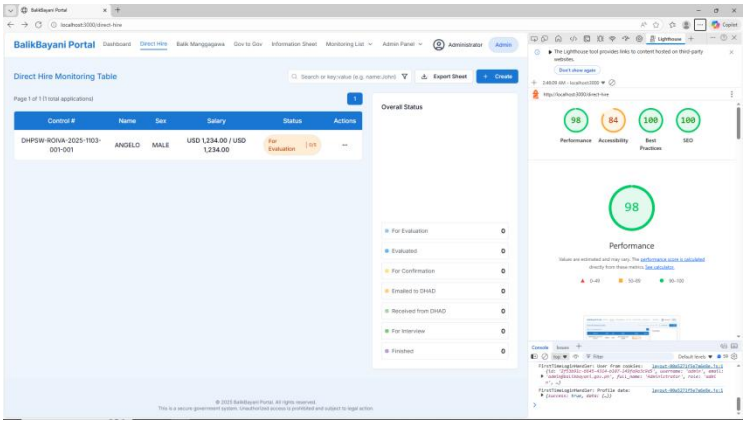


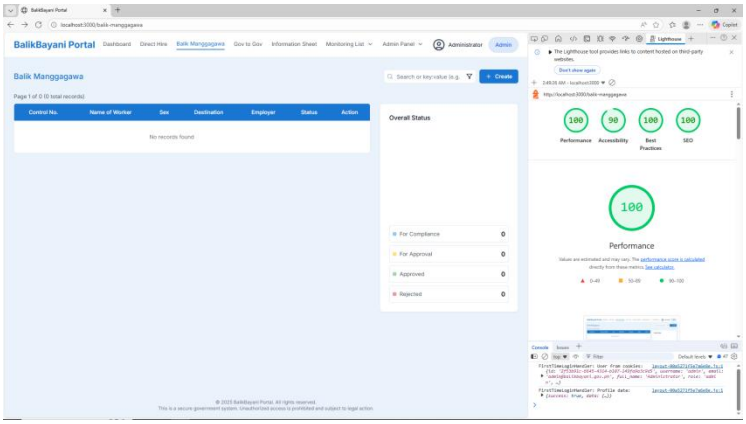


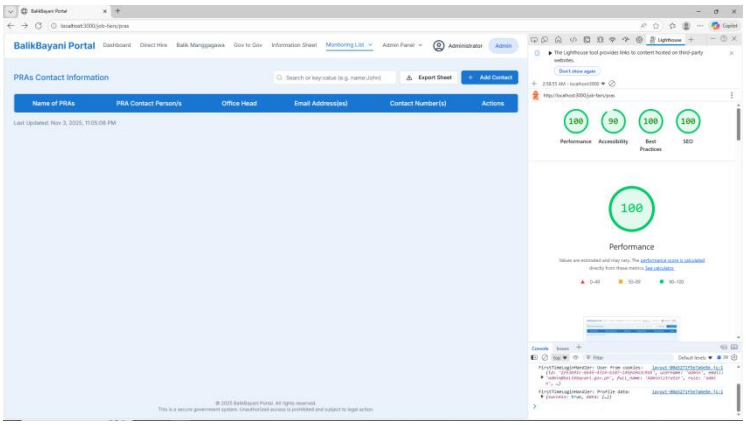
Micorsoft Edge Browser

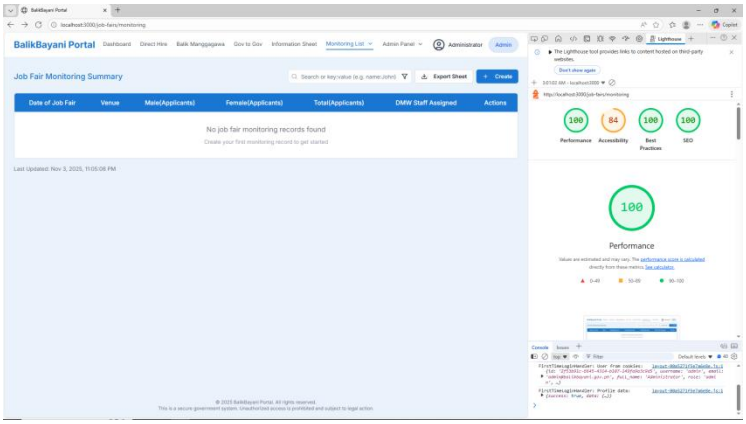


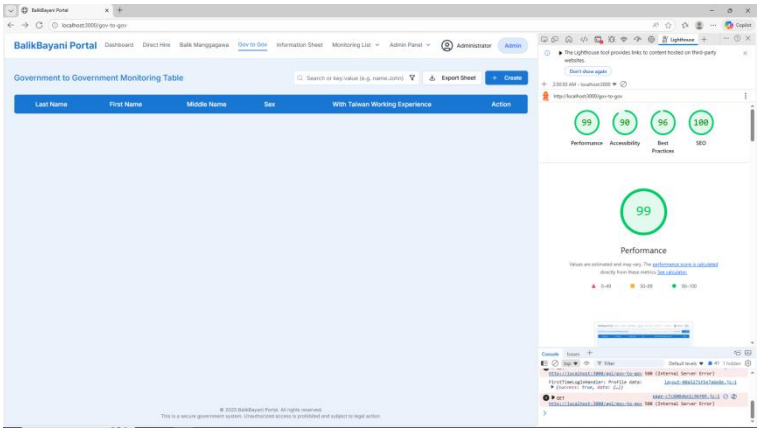


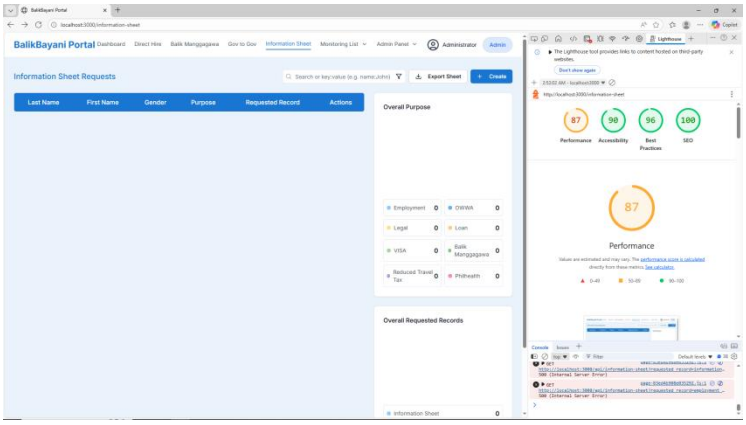

BalikBayani Portal
Direct Hire Monitoring Table
Overall Status
Performance
98

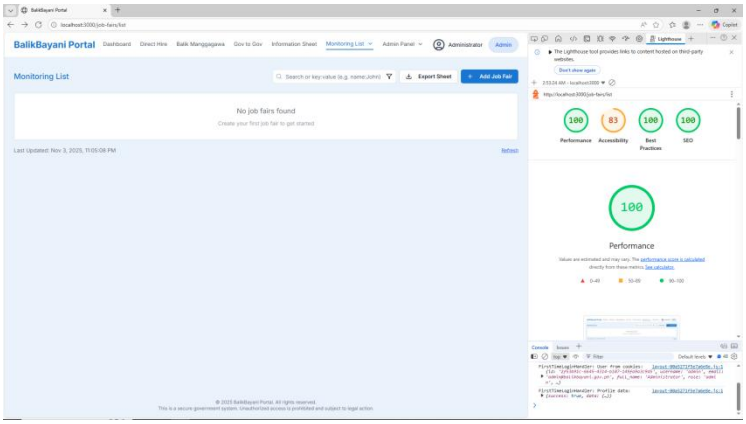

BalikBayani Portal
Balik Manggagawa
Overall Status
Performance
100

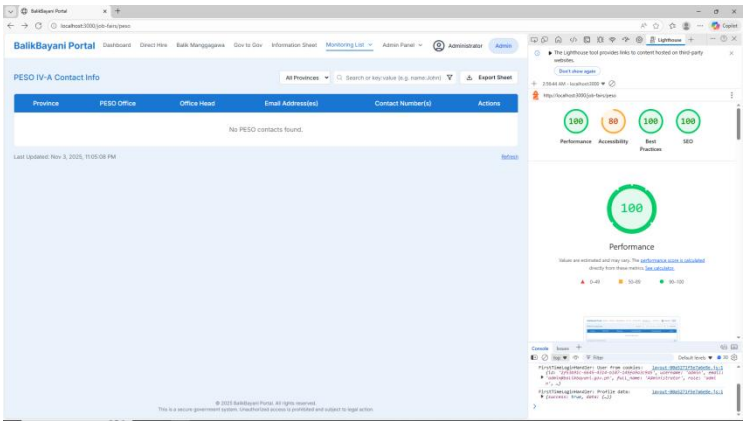

BalikBayani Portal
PRAs Contact Information
Performance
100

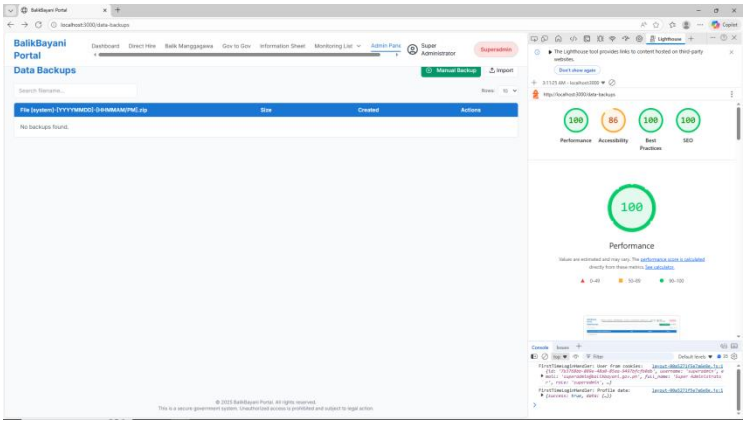

BalikBayani Portal
Job Fair Monitoring Summary
No job fair monitoring records found
Performance
100

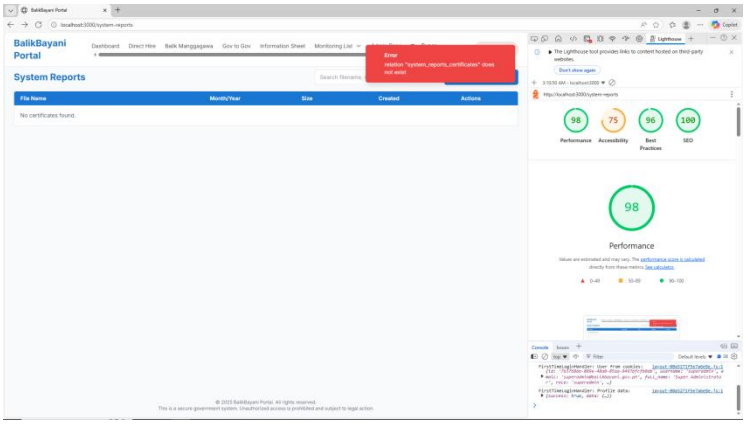

BalikBayani Portal
Government to Government Monitoring Table
Last Name
First Name
Middle Name
Sex
With Taiwan Working Experience
Action
Performance
Accessibility
Best Practices
SEO

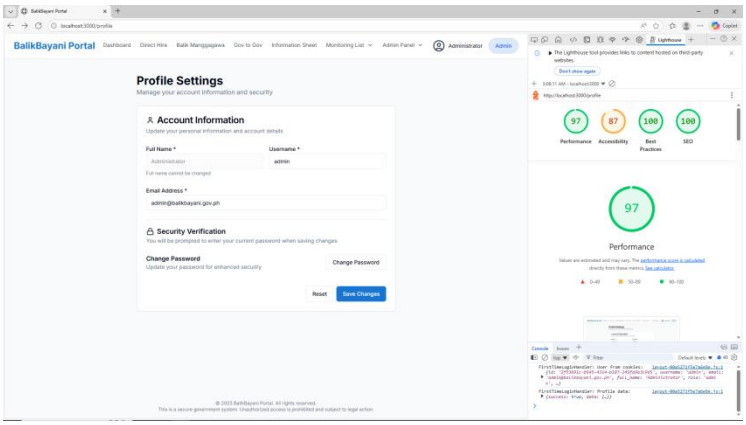

BalikBayani Portal
Information Sheet Requests
Last Name
First Name
Gender
Purpose
Requested Record
Actions
Overall Purpose
Overall Requested Records
Performance
Accessibility
Best Practices
SEO

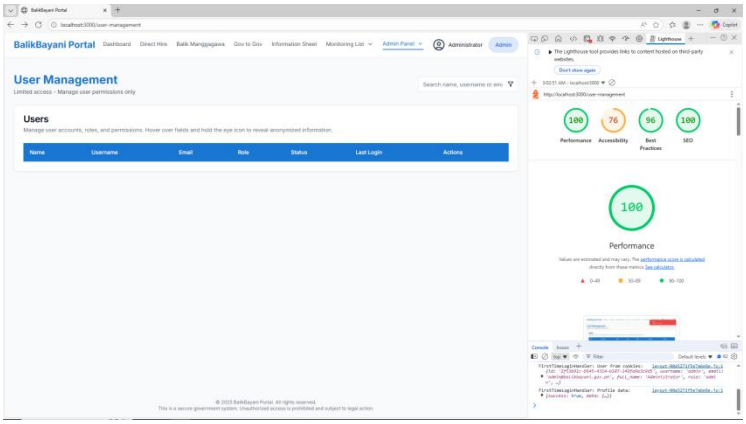

BalikBayani Portal
Monitoring List
No job fairs found
Performance
Accessibility
Best Practices
SEO


BalikBayani Portal
PESO IV-A Contact Info
Province
PESO Office
Office Head
Email Address(es)
Contact Number(s)
Actions
No PESO contacts found.
Performance
Accessibility
Best Practices
SEO

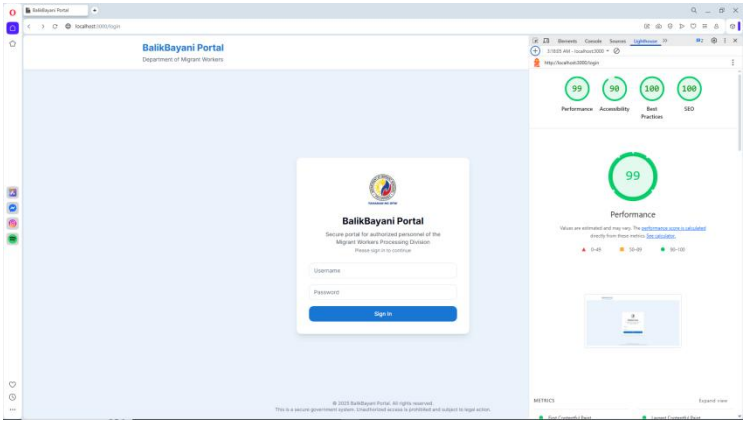

BalikBayani Portal
Data Backups
Manual Backup
Import
No backups found.
100
86
100
100
Performance
Accessibility
Best Practices
SEO
Performance

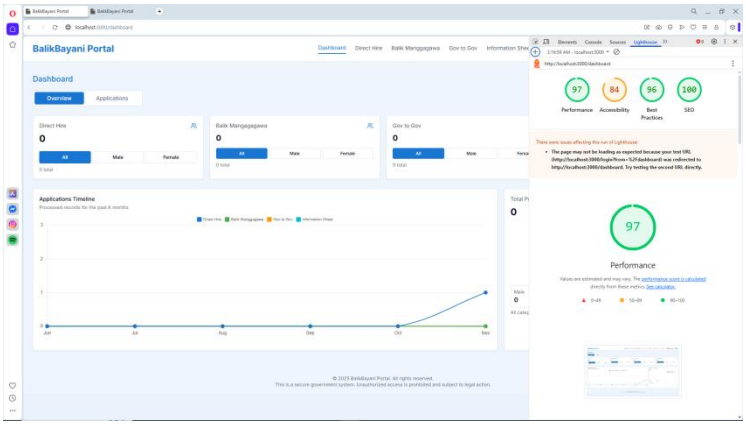

BalikBayani Portal
System Reports
Error
relation "system_reports_certificates" does not exist
No certificates found.
98
75
96
100
Performance
Accessibility
Best Practices
SEO
98
Performance

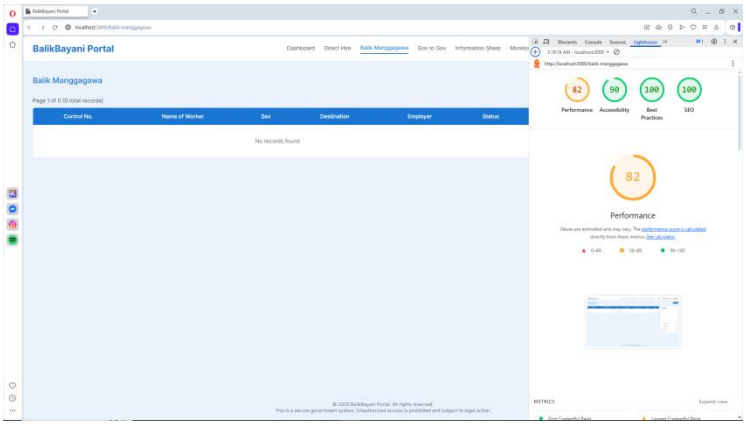

BalikBayani Portal
Profile Settings
Manage your account information and security
Account Information
Update your personal information and account details
Full Name *
Administrator
Full name cannot be changed
Username *
admin
Email Address *
admin@balikbayani.gov.ph
Security Verification
You will be prompted to enter your current password when saving changes
Change Password
Update your password for enhanced security
Change Password
Reset
Save Changes
97
87
100
100
Performance
Accessibility
Best Practices
SEO
97
Performance

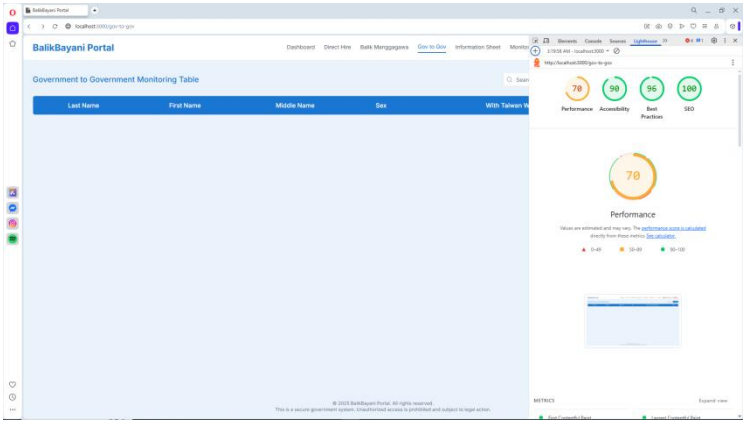

BalikBayani Portal
User Management
Limited access - Manage user permissions only
Users
Manage user accounts, roles, and permissions. Hover over fields and hold the eye icon to reveal anonymized information.
Name
Username
Email
Role
Status
Last Login
Actions
100
76
96
100
Performance
Accessibility
Best Practices
SEO
100
Performance

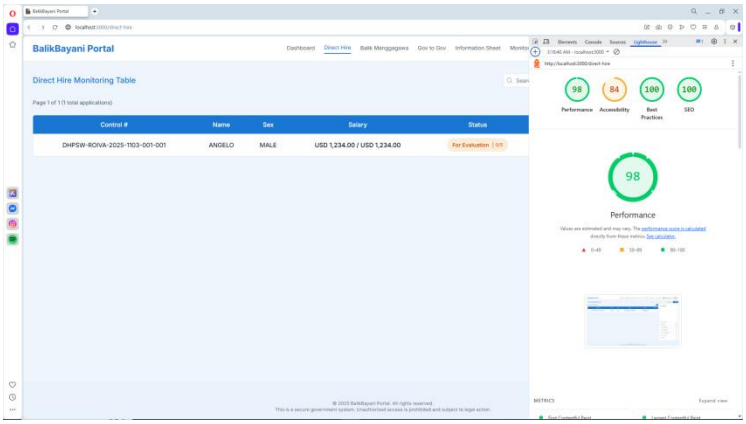
Opera GX Browser

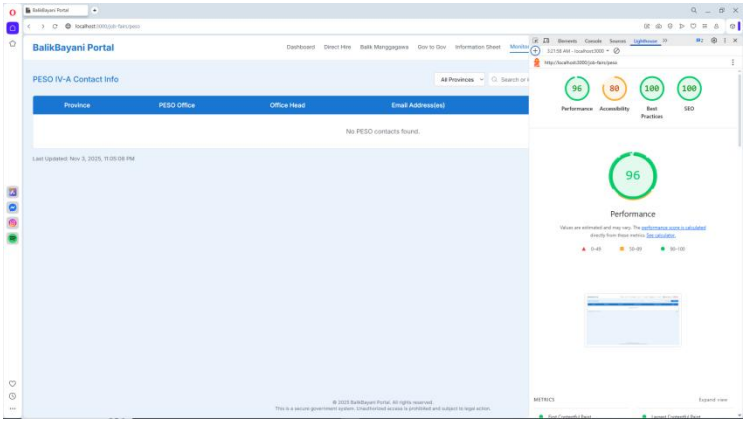


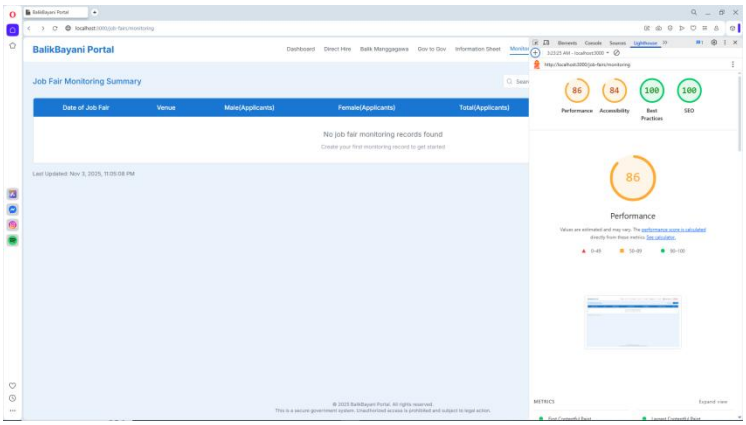


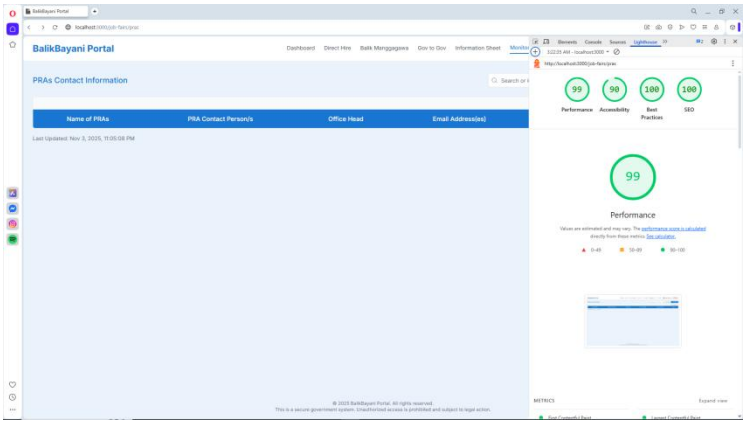


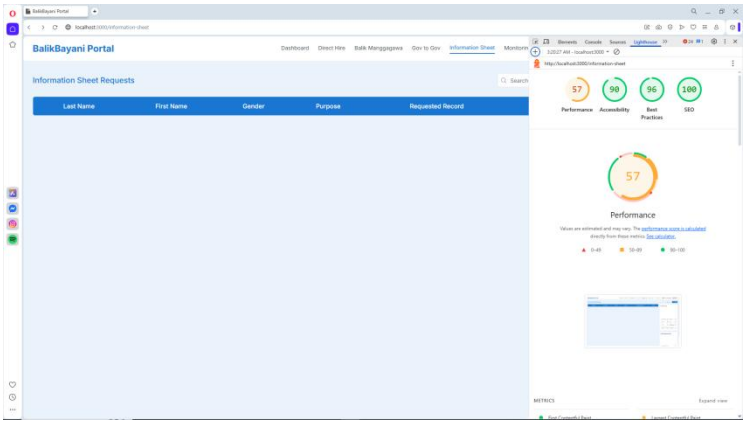


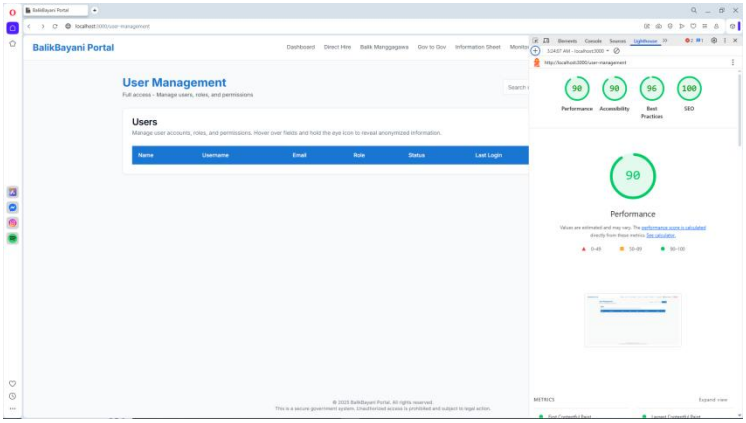

BalikBayani Portal
Dashboard
Direct Hire
Balik Manggagawa
Gov to Gov
Information Sheet
Monitor
Direct Hire Monitoring Table
Page 1 of 1 (1 total application)
Control #
Name
Sex
Salary
Status
DHPSW-ROIVA-2025-1103-001-001
ANGELO
MALE
USD 1,234.00 / USD 1,234.00
For Evaluation
98
84
100
100
Performance
Accessibility
Best Practices
SEO
98
Performance
METRICS
Expand view

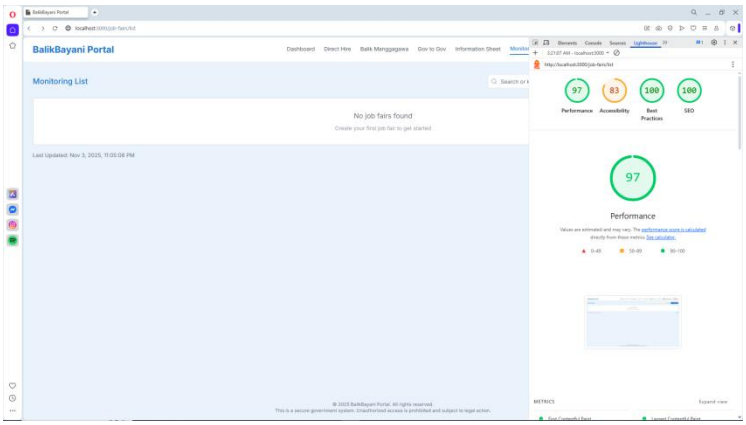

BalikBayani Portal
Dashboard
Direct Hire
Balik Manggagawa
Gov to Gov
Information Sheet
Monitor
PESO IV-A Contact Info
All Provinces
Province
PESO Office
Office Head
Email Address(es)
No PESO contacts found.
Last Updated: Nov 3, 2025, 11:05:08 PM
96
80
100
100
Performance
Accessibility
Best Practices
SEO
96
Performance
METRICS
Expand view

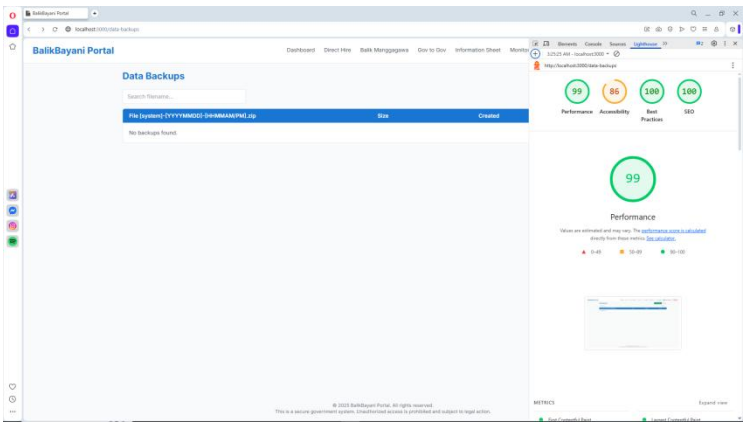

BalikBayani Portal
Dashboard
Direct Hire
Balik Manggagawa
Gov to Gov
Information Sheet
Monitor
Job Fair Monitoring Summary
Date of Job Fair
Venue
Male(Applicants)
Female(Applicants)
Total(Applicants)
No job fair monitoring records found
Create your first monitoring record to get started
Last Updated: Nov 3, 2025, 11:05:08 PM
86
84
100
100
Performance
Accessibility
Best Practices
SEO
86
Performance
METRICS
Expand view

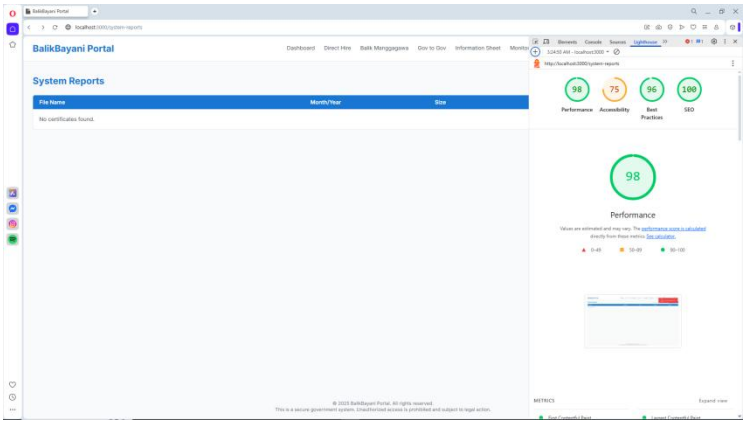

BalikBayani Portal
Dashboard
Direct Hire
Balik Manggagawa
Gov to Gov
Information Sheet
Monitor
PRAs Contact Information
Name of PRAs
PRA Contact Person/s
Office Head
Email Address(es)
Last Updated: Nov 3, 2025, 11:05:08 PM
99
90
100
100
Performance
Accessibility
Best Practices
SEO
99
Performance
METRICS
Expand view

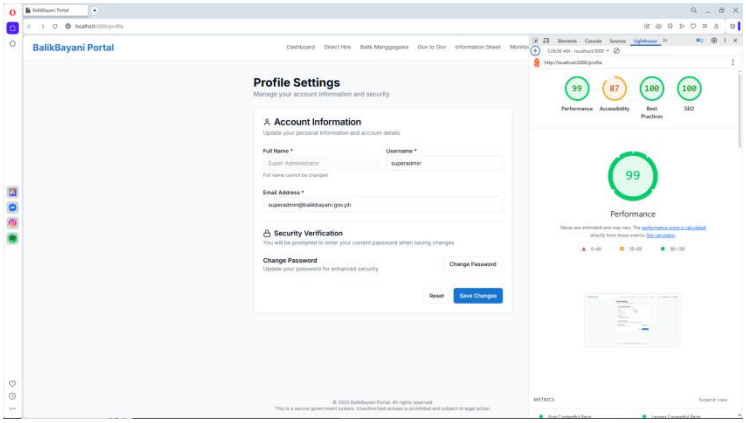

BalikBayani Portal
Dashboard
Direct Hire
Balik Manggagawa
Gov to Gov
Information Sheet
Monitoring
Information Sheet Requests
Last Name
First Name
Gender
Purpose
Requested Record
57
90
96
100
Performance
Accessibility
Best Practices
SEO
Performance
METRICS
Expand view


BalikBayani Portal
Dashboard
Direct Hire
Balik Manggagawa
Gov to Gov
Information Sheet
Monitoring
User Management
Full access - Manage users, roles, and permissions
Users
Manage user accounts, roles, and permissions. Hover over fields and hold the eye icon to reveal anonymized information.
Name
Username
Email
Role
Status
Last Login
90
90
96
100
Performance
Accessibility
Best Practices
SEO
Performance
METRICS
Expand view


BalikBayani Portal
Dashboard
Direct Hire
Balik Manggagawa
Gov to Gov
Information Sheet
Monitoring
Monitoring List
No job fairs found
Create your first job fair to get started
Last Updated: Nov 3, 2025, 11:05:08 PM
97
83
100
100
Performance
Accessibility
Best Practices
SEO
97
Performance
METRICS
Expand view


BalikBayani Portal
Dashboard
Direct Hire
Balik Manggagawa
Gov to Gov
Information Sheet
Monitoring
Data Backups
Search filename...
Size
Created
No backups found
99
86
100
100
Performance
Accessibility
Best Practices
SEO
99
Performance
METRICS
Expand view


BalikBayani Portal
Dashboard
Direct Hire
Balik Manggagawa
Gov to Gov
Information Sheet
Monitor
System Reports
File Name
Month/Year
Size
No certificates found.
98
75
96
100
Performance
Accessibility
Best Practices
SEO
98
Performance
METRICS
Expand view


BalikBayani Portal
Dashboard
Direct Hire
Balik Manggagawa
Gov to Gov
Information Sheet
Monitor
Profile Settings
Manage your account information and security
Account Information
Update your personal information and account details
Full Name *
Super Administrator
Full name cannot be changed
Username *
superadmin
Email Address *
superadmin@balikbayani.gov.ph
Security Verification
You will be prompted to enter your current password when saving changes
Change Password
Update your password for enhanced security
Change Password
Reset
Save Changes
99
87
100
100
Performance
Accessibility
Best Practices
SEO
99
Performance
METRICS
Expand view

APPENDIX D

Survey Questionnaires

LAGUNA STATE POLYTECHNIC UNIVERSITY- LOS BAÑOS CAMPUS
COLLEGE OF COMPUTER STUDIES

TECHNOLOGY AND ACCEPTANCE MODEL

Dear respondents,

We are students of Laguna State Polytechnic University - Los Banos Campus under the College of Computer Studies, currently conducting a research study on the project entitled **"BALIKBAYANI PORTAL: A FILING SYSTEM FOR MIGRANT RECORDS IN THE PROCESSING DIVISION OF THE DEPARTMENT OF MIGRANT WORKERS."** We are seeking your valuable opinion regarding the acceptability of the proposed project. Kindly evaluate each statement below by checking the number that best represents your level of agreement, using the following scale:

4 - High
3 - Moderate
2 - Low
1 - Very Low

Please be assured that all information provided will be treated with the utmost confidentiality, in accordance with RA no. 10173 (Data Privacy Act of 2012), and will be used strictly for academic purposes only. Thank you very much for your cooperation.

Regards,
Limpiada, Andrei Nikolai
Evangelista, Fatima L.
Mancera, Angelo M.

Name of Respondent: _____

Main Criteria	Sub-Criteria	Description/ Indicators	High (4)	Moderate (3)	Low (2)	Very Low (1)
Perceived Usefulness	Task Efficiency	The system enables users to complete tasks faster and minimizes the time and effort required for document processing.	Tasks completed quickly with minimal effort.	Noticeable improvement in task speed.	Slightly improvement in task completion.	System causes delays and errors in task performance.
	Productivity Improvement	The system contributes to improved work productivity within the processing division.	Clears improvement in staff productivity.	Moderate improvement in task management.	Minimal increase in productivity.	No noticeable productivity improvement.
	Quality of Output	The BalikBayani Portal produces accurate and reliable outputs.	Outputs are accurate and dependable.	Minor errors occasionally occur.	Frequent inconsistencies in results.	Outputs are often incorrect or unreliable.
Perceived Ease of Use	System Navigation	The layout is clear and easy to navigate.	Layout is clear and easy to follow.	Navigation is fairly smooth.	Some parts are confusing to navigate.	Layout is disorganized and hard to use.
	Learnability	Users can easily remember system functions without frequent assistance, making it user-friendly for first-time users.	Users easily recall system functions.	Some assistance needed occasionally.	Users often forget functions.	Users always need help to operate the system.
	Task Simplicity	The process of completing tasks in the BalikBayani Portal is simple and minimal.	Task process is straightforward and efficient.	Process is generally easy to complete.	Process is somewhat complicated.	Process is confusing and time-consuming.
User Satisfaction	System Performance	The BalikBayani Portal responds quickly and runs smoothly without crashes.	System responds fast and runs smoothly.	Occasional minor delays.	System is slow or sometimes unresponsive.	System often crashes or freezes.
	User Experience	The system design and interface are visually appealing.	Interface is visually appealing and well-organized.	Design is simple but acceptable.	Interface looks dull or outdated.	Interface is cluttered and unappealing.
	Overall Satisfaction	The system meets the user expectations.	Fully meets and exceed user expectations.	Meets most user expectations.	Partially meets user expectations.	Fails to meet user expectations.

Willingness to Adopt	Future Use	The BalikBayani Portal is beneficial for long-term use.	Highly beneficial and reliable for long-term use.	Useful for mederate-term application.	Limited long-term benefits.	Not suitable for long-term use.
	Technology Acceptance	Users perceive the BalikBayani Portal as essential for record management.	Considered essential for record management.	Helpful for record-related tasks.	Occasionally useful for records.	Not useful for record management.
	Recommendation	Users believe that others will benefit from using the system and are willing to recommend the BalikBayani Portal to others.	Users highly recommend the system.	Users likely to recommend the system.	Users hesitant to recommend.	Users would not recommend the system.

_____

Signature

LAGUNA STATE POLYTECHNIC UNIVERSITY- LOS BAÑOS CAMPUS
COLLEGE OF COMPUTER STUDIES

BALIKBAYANI PORTAL: A FILING SYSTEM FOR MIGRANT RECORDS IN THE PROCESSING DIVISION OF THE DEPARTMENT OF MIGRANT WORKERS

Good Day! We are 4th year students, currently pursuing Bachelor of Science in Information Technology specializing in Web and Mobile Application Development. As part of our academic endeavors, we have develop a web application adhering to the ISO 25010 standards.

To assess the effectiveness and user satisfaction of our system, we have devised a set of questions. Your responses, rated from 1 to 5, will help us evaluate various aspects of the systems, from functional suitability to overall user experience.

5 - Excellent
4 - Good
3 - Fair
2 - Needs Improvement
1 - Poor

Regards,
Limpiada, Andrei Nikolai
Evangelista, Fatima L.
Mancera, Angelo M.

Name of Evaluator: _____

RUBRIC FOR 5-Point Likert Scale to evaluate product/system quality using ISO 25010

Main Criteria	Sub-Criteria	Description/ Indicators	Excellent (5)	Good (4)	Fair (3)	Needs Improvement (2)	Poor (1)
Functional Suitability	Functional Completeness	The BalikBayani Portal provides all necessary features for migrant record filing and tracking.	All required functions are implemented and fully operational.	Most functions are implemented with minor gaps.	Some functions are missing, but core features work.	Many features are missing or incomplete.	Major functions fail or are absent.
	Functional Correctness	Accuracy of record handling, generation, and updates.	All outputs and records are accurate and error-free.	Minor inconsistencies but generally accurate.	Some errors found in generated data.	Frequent data inconsistencies.	Outputs are unreliable or incorrect.
	Functional Appropriateness	Relevance and effectiveness of functions in supporting DMW process.	Functions directly support all user tasks efficiently.	Functions generally meet user needs.	Functions somewhat align with user goals.	Functions often feels unnecessary or confusing.	Functions do not meet user needs.
Performance Efficiency	Time Behaviour	Response time and loading speed of the BalikBayani Portal.	Pages and processes load instantly.	Slight delays but acceptable.	Noticeable delays occasionally.	Slow performance affects usability.	System frequently lags or times out.
	Resource Utilization	Efficiency in using system resources (storage, memory).	Uses minimal resources efficiently.	Slightly high usage but acceptable.	Moderate resource use.	High usage impacts performance.	Extremely inefficient or causes system crashes.
	Capacity	Ability to handle large numbers of users or records.	Handles heavy loads without issues.	Manages moderate loads smoothly.	Handles small loads adequately.	Slows with many users/records.	Crashes with multiple users or large datasets.
Compatibility	Co-Existence	Ability to run smoothly with other system.	Works perfectly alongside with other applications.	Minor issues with other software.	Occasional confllicts occurs.	Frequent issues with co-existing systems.	Cannot run with other systems.
	Interoperability	Capability to exchange data with other systems.	Seamlessly integrates and exchanges data.	Data exchange works with minimal errors.	Limited integration features.	Integration possible but error prone.	No interoperability functions.
Interaction Capability	Appropriateness Recognizability	How well the system's interface and functions match user expectations.	Interface is intuitive and instantly recognizable.	Mostly intuitive; minor visual inconsistencies.	Some elements are unclear or misplaced.	Interface often confuces users.	Very difficult to recognize functions.
	Learnability	Ease for new users to understand and operate the system.	Very easy to learn; minimal instruction require.	Easy to use with brief guidance.	Requires some practice to learn.	Requires significant practice.	Very difficult to use.
	Operability	Ease of operation and task completion.	Smooth operation and fast task flow.	Generally smooth with minor issues.	Usable but requires effort.	Frequent interruption or errors.	Difficult to operate or navigate.
	User Error Protection	Ability to prevent and correct user mistakes.	Effectively prevents all user errors.	Prevents most errors.	Provides only basic error protection.	Lacks helpful error prevention.	No error prevention or guidance.

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	User Engagement	Level of satisfaction and engagement when using the BalikBayani Portal.	Very engaging and satisfying to use.	Generally pleasant and clear.	Neutral experience.	Unattractive or tiring interface.	Frustrating or unappealing interface.
	Inclusivity	Accessibility for all users regardless of ability or device.	Fully accessible on all devices and for all users.	Accessible with minor device issues.	Works for most users but with limited accessibility.	Often inaccessible for some users.	Not accessible at all.
	User Assistance	Help features and guidance for users.	Clear, context-aware help available anytime.	Help section available but limited.	Basic help instructions provided.	Help options rarely useful.	No assistance features available.
	Self-Descriptiveness	How clearly the system communicates its purpose and functions.	Every feature is self-explanatory.	Most features are understandable.	Some require trial and error.	Often unclear or confusing.	No indication of purpose or need.
Reliability	Faultness	Frequency and impact of system errors or bugs.	System runs perfectly without errors	Rare, non-critical errors.	Occasional minor bugs.	Frequent issues affecting workflow.	Constant system malfunctions.
	Availability	System uptime and accessibility.	Always accessible; zero downtime.	Almost always accessible with rare short downtimes.	Sometimes unavailable but recovers quickly.	Frequent downtimes affect users.	System often unavailable.
	Fault Tolerance	Ability to continue functioning after partial failure.	System automatically handles faults smoothly.	Recovers with minimal delay.	Some disruption during fault recovery.	Requires frequent manual intervention.	Cannot handle faults.
	Recoverability	Ability to restore data and functions after failure.	Full data recovery with no loss.	Minor data loss but recoverable.	Partial recovery possible.	Inconsistent recovery process.	No recovery system in place.
Secuirty	Confidentiality	Ensuring that only authorized users access sensitive data.	All user data fully protected and encrypted.	Strong protection with rare vulnerabilities.	Adequate but needs improved access control.	Weak encryption or controls.	No data privacy protection.
	Integrity	Maintaining accurate, consistent data throughout operations.	Data always accurate and unchanged.	Rare integrity issues.	Occasional inconsistencies .	Frequent data inaccuracies.	Data corruption common.
	Non-Repudiation	Ability to prove actions or transactions were performed by authorized users.	All actions traceable and verifiable.	Logs available with minor gaps.	Some transactions not traceable.	Weak audit trail.	No proof of user actions.
	Accountability	Tracking and monitoring user actions responsibly.	Full audit trail and activity monitoring.	Good tracking with minimal errors.	Partial user monitoring.	Weak accountability measures.	No activity tracking or logs.
	Authenticity	Verifying legitimate access and valid data.	All users and data fully verified.	Most verification processes succeed.	Occasional verification failure.	Weak verification procedures.	No verification checks.
	Resistance	System's ability to resist attacks or unauthorized access.	Fully resistant to all known vulnerabilities.	Resistant to most threats.	Moderate defense level.	Often vulnerable to threats.	Highly exposed to attacks.
Maintainability	Modularity	The BalikBayani Portal components are organized into independent modules that cannot be modified without affecting others.	Highly modular, changes do not affect other modules.	Mostly modular with few dependencies.	Moderately modular; some components are linked.	Difficult to modify without affecting others.	Poorly structured; changes causes major issues.
	Reusability	System components or code can be reused in other parts of the system or future projects.	Components are reusable across different modules or systems.	Mostly reusable with small adjustments.	Some code reusable with effort.	Rarely usable.	No reusable components.
	Analysability	Ease of diagnosing system errors or understanding the system's structure for maintenance purposes.	Errors easily traced; structure well-documented.	Problems identifiable with minimal effort.	Some effort needed to analyze errors.	Difficult to analyze system issues.	Almost impossible to diagnose errors.
	Modifiability	The BalikBayani Portal allows easy modifications to adapt to new requirements or correct defects.	Modifications are quick and risk-free.	Modifications require minimal testing.	Modifications possible but time-consumung.	Changes often cause other issues.	Hard or impossible to modify.
	Testability	The BalikBayani Portal can be effectively tested after changes or updates are made.	Easily testable; automated testing available.	Test process efficient with few issues.	Manual testing possible but limited.	Difficult to test changes.	No testing framework or method available.
Flexibility	Adaptability	The system can easily adapt to changing user needs or environment.	Fully adaptable to new process or policies.	Adapts with minor configuration.	Requires moderate effort to adapt.	Adaptation difficult or requires development.	Cannot adapt to any changes.
	Scalability	The BalikBayani Portal performs efficiently even as the workload, users, or data volume increases.	Seamlessly handles high traffic or data loads.	Performs well with moderate load.	Acceptable at small scale but slows under pressure.	Struggles with high load.	Fails when usage increases.
	Installability	The system can be installed or deployed smoothly	Installation and setup are quick	Installation easy with few steps.	Installation manageable	Installation complicated or	Installation fails or often crashes.

		without technical difficulties.	and straightforward,		but requires guidance.	time-consuming.	
	Replaceability	Components or modules can be replaced without affecting overall system performance.	Components easily replaceable with no issues.	Replacement causes minor adjustments.	Replacement possible with testing.	Replacement disrupts other modules.	Replacement causes system failures.
Safety	Operational Constraint	The system follows operational limits to prevent unsafe conditions or misuse.	Operates safely under all conditions.	Minor safe-operation issues under stress.	Occasionally exceeds operational limits.	Frequent unsafe operations requiring manual correction.	Unsafe operations cause data or system failures.
	Risk Identification	The system can identify, assess, and minimize potential risks or hazards.	All risks proactively identified and mitigated.	Most risks identified with proper actions.	Common risks identified but missed rare ones.	Risks identified only after problem occur.	No risk identification in place.
	Fail Safe	The system maintains a safe state during failure or malfunction.	Automatically reverts to safe state.	Recovers safely with minimal loss.	Basic fail-safe features in place.	System recovery inconsistent.	Fails without any protection.
	Hazard Warning	The system provides timely alerts and warnings to prevent user harm or data loss.	Alerts displayed immediately and clearly.	Warnings appear but sometimes delayed.	Warnings present but unclear.	Alerts rarely appear or are confusing.	No warning alerts at all.
	Safe Integration	The system ensures safe interaction with external components or other systems.	Fully safe integration with other platforms.	Integrates safely with minor issues.	Occasional safety issues in integration.	Integration prone to data errors.	Unsafe integration causes major failures.

Signature

APPENDIX E

TAM and ISO 25010 Evaluation Results

TAM Result

	A	B	C	D	E	F	G	H	I	J	K	L
1	Personal Usefulness			Perceived Ease of Use			User Satisfaction			Willingness to Adopt		
2	Task Effiency	Productivity Improvement	Quality of Output.	System Navigation	Learnability	Task Simplicity	System Performance	User Experience	Overall Satisfaction	Future Use	Technology Acceptance	Recommendation
3	4	3	4	4	3	4	4	3	4	3	4	4
4	3	4	4	4	3	3	3	4	3	3	4	3
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106	4	4	4	4	4	4	4	4	4	4	4	4	
107	3.84	3.74	3.88	3.75	3.67	3.75	3.88	3.72	3.77	3.76	3.81	3.80	MEAN OF SUB CRITERIA
108	0.40	0.48	0.32	0.46	0.57	0.52	0.35	0.45	0.47	0.49	0.44	0.40	SD
109													
110													

111	CRITERIA	MEAN	SD
112	Personal Usefulness	3.82	0.08
113	Perceived Ease of Use	3.72	0.05
114	User Satisfaction	3.79	0.06
115	Willingness to Adopt	3.79	0.04
116	OVERALL	**3.78**	**0.02**

117			
118			

ISO 25010 Result

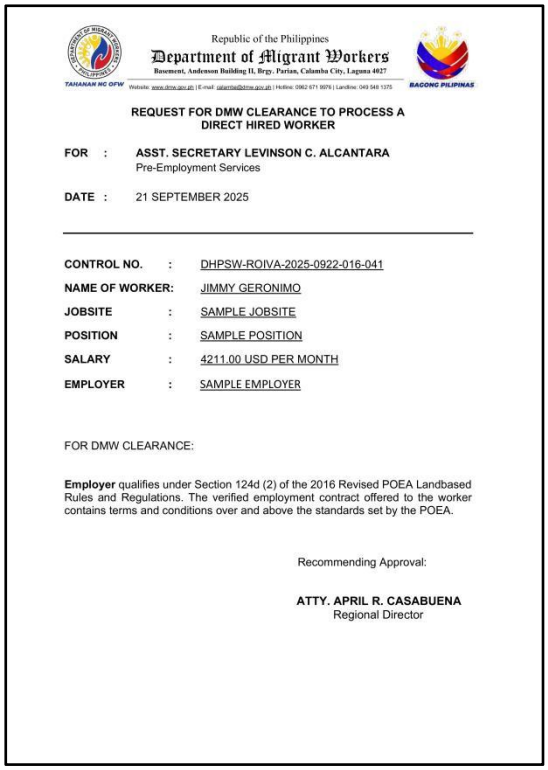
A	B	C	D	E	F
CRITERIA	RESPONDENT 1 - Sir Mike	RESPONDENT 2 - Sir AJ	RESPONDENT 3 - Paolo	Mean	SD
Functional Suitability					
Functional Completeness	5	5	5	5.00	0.00
Functional Correctness	4	5	4	4.33	0.58
Functional Appropriateness	5	5	5	5.00	0.00
Performance Efficiency					
Time Behaviour	4	5	5	4.67	0.58
Resource Utilization	4	5	5	4.67	0.58
Capacity	4	5	4	4.33	0.58
Compatibility					
Co-Existence	5	4	5	4.67	0.58
Interoperability	4	4	5	4.33	0.58
Interaction Capability					
Appropriateness Recognizability	5	4	5	4.67	0.58
Learnability	4	4	4	4.00	0.00
Operability	5	4	5	4.67	0.58
User Error Protection	5	5	5	5.00	0.00
User Engagement	4	5	5	4.67	0.58
Inclusivity	4	5	5	4.67	0.58
User Assistance	4	5	5	4.67	0.58
Self-Descriptiveness	5	5	4	4.67	0.58
Reliability					
Faultness	4	4	5	4.33	0.58
Availability	4	5	5	4.67	0.58
Fault Tolerance	4	5	5	4.67	0.58
Recoverability	4	4	5	4.33	0.58
Security					
Confidentiality	5	4	5	4.67	0.58
Integrity	5	5	5	5.00	0.00
Non-Repudiation	5	4	5	4.67	0.58
Accountability	5	5	5	5.00	0.00
Authenticity	5	4	5	4.67	0.58
Resistance	5	5	4	4.67	0.58
Maintainability					
Modularity	5	5	5	5.00	0.00
Reusability	5	4	5	4.67	0.58
Analysability	4	5	5	4.67	0.58
Modifiability	5	4	5	4.67	0.58
Testability	5	5	4	4.67	0.58
Flexibility					
Adaptability	5	5	4	4.67	0.58
Scalability	4	4	5	4.33	0.58
Installability	4	5	5	4.67	0.58
Replaceability	5	5	5	5.00	0.00
Safety					
Operational Constraint	5	5	5	5.00	0.00
Risk Identification	5	5	4	4.67	0.58
Fail Safe	5	5	5	5.00	0.00
Hazard Warning	5	5	5	5.00	0.00
Safe Integration	5	5	5	5.00	0.00
OVERALL					

CRITERIA	COMPOSITE MEAN	SD		
Functional Suitability	4.78	0.33		
Performance Efficiency	4.56	0		
Compatibility	4.50	0		
Interaction Capability	4.63	0.27		
Reliability	4.50	0		
Security	4.78	0.30		
Maintainability	4.733	0.26		
Flexibility	4.67	0.29		
Safety	4.93	0.26		
OVERALL MEAN	4.67	0.14	**OVERALL SD**	

APPENDIX F

Sample of Output

Direct Hire Clearance



Republic of the Philippines
Department of Migrant Workers
Basement, Andenson Building II, Brgy. Parian, Calamba City, Laguna 4027
Website: www.dmw.gov.ph | E-mail: calarmie@dmw.gov.ph | Hotline: 0962 671 9976 | Landline: 049 548 1375

TAHANAN NG OFW

BAGONG PILIPINAS

REQUEST FOR DMW CLEARANCE TO PROCESS A DIRECT HIRED WORKER

FOR : **ASST. SECRETARY LEVINSON C. ALCANTARA**
Pre-Employment Services

DATE : 21 SEPTEMBER 2025

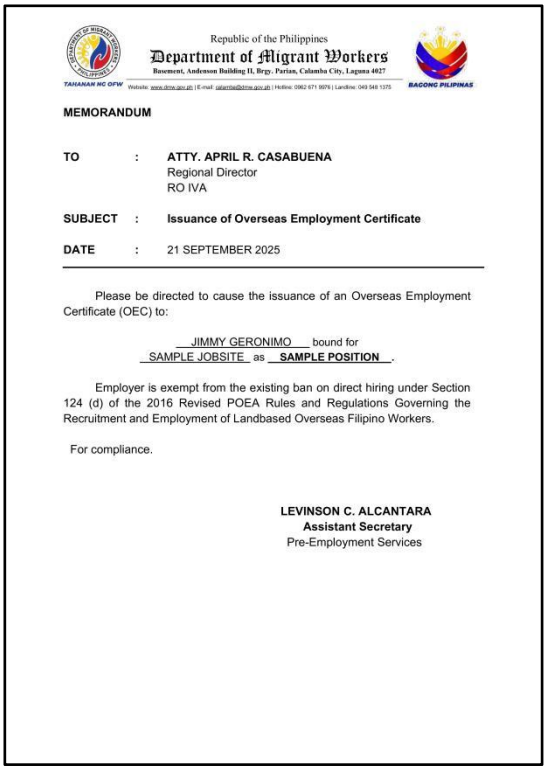
CONTROL NO. : DHPSW-ROIVA-2025-0922-016-041
NAME OF WORKER: JIMMY GERONIMO
JOBSITE : SAMPLE JOBSITE
POSITION : SAMPLE POSITION
SALARY : 4211.00 USD PER MONTH
EMPLOYER : SAMPLE EMPLOYER

FOR DMW CLEARANCE:

Employer qualifies under Section 124d (2) of the 2016 Revised POEA Landbased Rules and Regulations. The verified employment contract offered to the worker contains terms and conditions over and above the standards set by the POEA.

Recommending Approval:

ATTY. APRIL R. CASABUENA
Regional Director



Republic of the Philippines
Department of Migrant Workers
Basement, Andenson Building II, Brgy. Parian, Calamba City, Laguna 4027
Website: www.dmw.gov.ph | E-mail: calarmie@dmw.gov.ph | Hotline: 0962 671 9976 | Landline: 049 548 1375

TAHANAN NG OFW

BAGONG PILIPINAS

MEMORANDUM

TO : **ATTY. APRIL R. CASABUENA**
Regional Director
RO IVA

SUBJECT : **Issuance of Overseas Employment Certificate**

DATE : 21 SEPTEMBER 2025

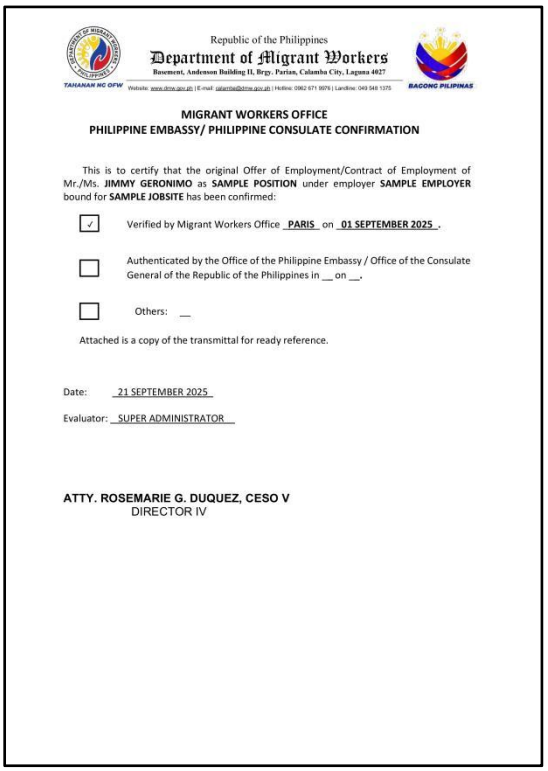
Please be directed to cause the issuance of an Overseas Employment Certificate (OEC) to:

JIMMY GERONIMO bound for
SAMPLE JOBSITE as **SAMPLE POSITION**.

Employer is exempt from the existing ban on direct hiring under Section 124 (d) of the 2016 Revised POEA Rules and Regulations Governing the Recruitment and Employment of Landbased Overseas Filipino Workers.

For compliance.

LEVINSON C. ALCANTARA
Assistant Secretary
Pre-Employment Services



Republic of the Philippines
Department of Migrant Workers
Basement, Andenson Building II, Brgy. Parian, Calamba City, Laguna 4027
Website: www.dmw.gov.ph | E-mail: calarmie@dmw.gov.ph | Hotline: 0962 671 9976 | Landline: 049 548 1375

TAHANAN NG OFW

BAGONG PILIPINAS

MIGRANT WORKERS OFFICE
PHILIPPINE EMBASSY/ PHILIPPINE CONSULATE CONFIRMATION

This is to certify that the original Offer of Employment/Contract of Employment of Mr./Ms. **JIMMY GERONIMO** as **SAMPLE POSITION** under employer **SAMPLE EMPLOYER** bound for **SAMPLE JOBSITE** has been confirmed:

☑ Verified by Migrant Workers Office **PARIS** on **01 SEPTEMBER 2025**.

☐ Authenticated by the Office of the Philippine Embassy / Office of the Consulate General of the Republic of the Philippines in __ on __.

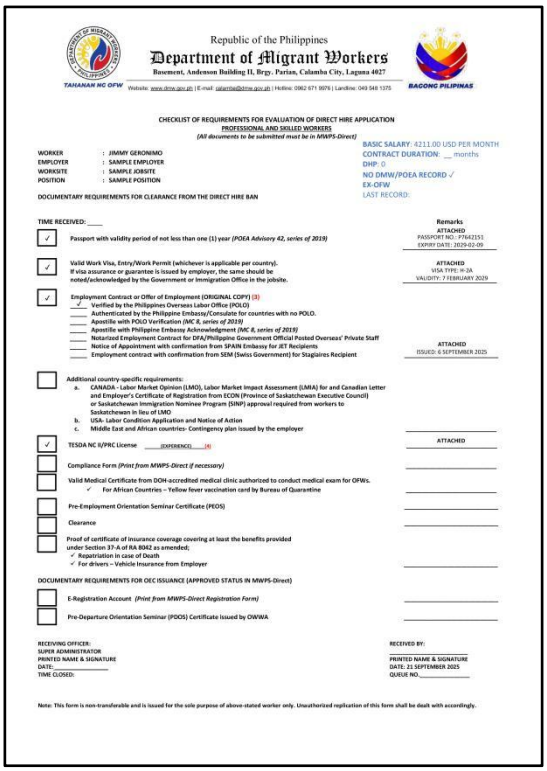
☐ Others: __

Attached is a copy of the transmittal for ready reference.

Date: 21 SEPTEMBER 2025

Evaluator: SUPER ADMINISTRATOR

ATTY. ROSEMARIE G. DUQUEZ, CESO V
DIRECTOR IV



Republic of the Philippines
Department of Migrant Workers
Basement, Andenson Building II, Brgy. Parian, Calamba City, Laguna 4027
Website: www.dmw.gov.ph | E-mail: calarmie@dmw.gov.ph | Hotline: 0962 671 9976 | Landline: 049 548 1375

TAHANAN NG OFW

BAGONG PILIPINAS

CHECKLIST OF REQUIREMENTS FOR EVALUATION OF DIRECT HIRE APPLICATION
PROFESSIONAL AND SKILLED WORKERS
(All documents to be submitted must be in MWPS-Direct)

WORKER : JIMMY GERONIMO
EMPLOYER : SAMPLE EMPLOYER
WORKSITE : SAMPLE JOBSITE
POSITION : SAMPLE POSITION

BASIC SALARY: 4211.00 USD PER MONTH
CONTRACT DURATION: __ months
DHP: 0
NO DMW/POEA RECORD ✓
EX-OFW
LAST RECORD:

DOCUMENTARY REQUIREMENTS FOR CLEARANCE FROM THE DIRECT HIRE BAN

TIME RECEIVED: ___

		Remarks
✓	**Passport with validity period of not less than one (1) year *(POEA Advisory 42, series of 2019)***	ATTACHED PASSPORT NO.: P7642151 EXPIRY DATE: 2029-02-09
✓	**Valid Work Visa, Entry/Work Permit (whichever is applicable per country). If visa assurance or guarantee is issued by employer, the same should be noted/acknowledged by the Government or Immigration Office in the jobsite.**	ATTACHED VISA TYPE: H-2A VALIDITY: 7 FEBRUARY 2029
✓	**Employment Contract or Offer of Employment (ORIGINAL COPY) (3)** ✓ Verified by the Philippines Overseas Labor Office (POLO) Authenticated by the Philippine Embassy/Consulate for countries with no POLO. Apostille with POLO Verification *(MC II, series of 2019)* Apostille with Philippine Embassy Acknowledgment *(MC II, series of 2019)* Notarized Employment Contract for DFA/Philippine Government Official Posted Overseas' Private Staff Notice of Appointment with confirmation from SPAIN Embassy for JET Recipients Employment contract with confirmation from SEM (Swiss Government) for Stagiaires Recipient	ATTACHED ISSUED: 6 SEPTEMBER 2025
☐	**Additional country-specific requirements:** a. CANADA - Labor Market Opinion (LMO), Labor Market Impact Assessment (LMIA) for and Canadian Letter and Employer's Certificate of Registration from ECON (Province of Saskatchewan Executive Council) or Saskatchewan Immigration Nominee Program (SINP) approval required from workers to Saskatchewan in lieu of LMO b. USA- Labor Condition Application and Notice of Action c. Middle East and African countries- Contingency plan issued by the employer	
✓	**TESDA NC II/PRC License** ____(EXPERIENCE)____ (4)	ATTACHED
☐	**Compliance Form *(Print from MWPS-Direct if necessary)***	
☐	**Valid Medical Certificate from DOH-accredited medical clinic authorized to conduct medical exam for OFWs.** ✓ For African Countries – Yellow fever vaccination card by Bureau of Quarantine	
☐	**Pre-Employment Orientation Seminar Certificate (PEOS)**	
☐	**Clearance**	
☐	**Proof of certificate of insurance coverage covering at least the benefits provided under Section 37-A of RA 8042 as amended;** ✓ Repatriation in case of Death ✓ For drivers – Vehicle Insurance from Employer	

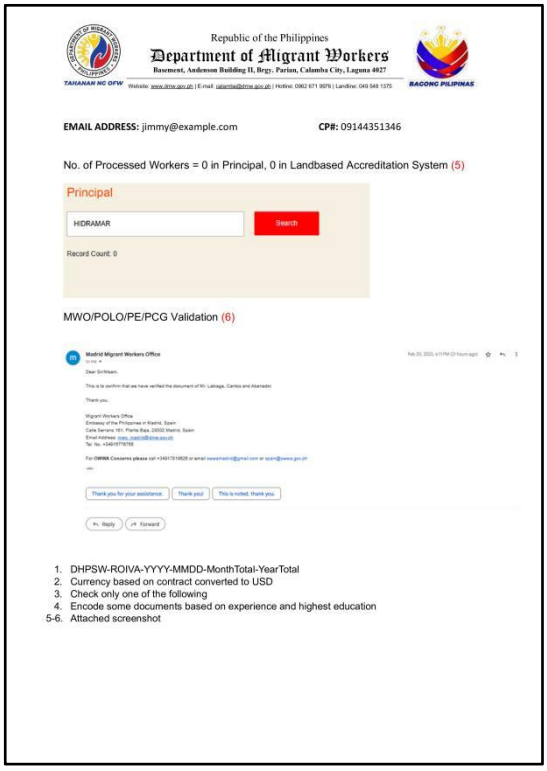
DOCUMENTARY REQUIREMENTS FOR OEC ISSUANCE (APPROVED STATUS IN MWPS-Direct)

☐	**E-Registration Account *(Print from MWPS-Direct Registration Form)***	
☐	**Pre-Departure Orientation Seminar (PDOS) Certificate issued by OWWA**	

RECEIVING OFFICER:
SUPER ADMINISTRATOR
PRINTED NAME & SIGNATURE
DATE:
TIME CLOSED:

RECEIVED BY:
PRINTED NAME & SIGNATURE
DATE: 21 SEPTEMBER 2025
QUEUE NO.

Note: This form is non-transferable and is issued for the sole purpose of above-stated worker only. Unauthorized replication of this form shall be dealt with accordingly.



Republic of the Philippines
Department of Migrant Workers
Basement, Anderson Building II, Brgy. Parian, Calamba City, Laguna 4027
TAHANAN NG OFW | BAGONG PILIPINAS

EMAIL ADDRESS: jimmy@example.com **CP#:** 09144351346

No. of Processed Workers = 0 in Principal, 0 in Landbased Accreditation System (5)

Principal

HIDRAMAR Search

Record Count: 0

MWO/POLO/PE/PCG Validation (6)

Madrid Migrant Workers Office

Thank you for your assistance. Thank you! This is noted, thank you.

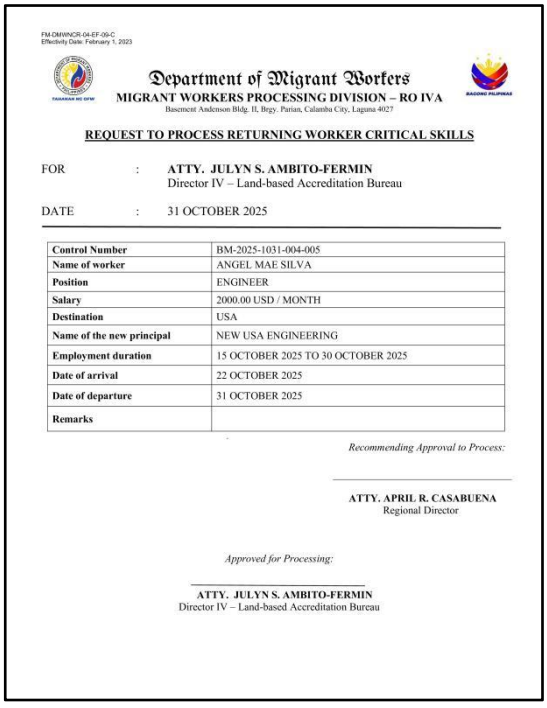
Reply Forward

1. DHPSW-ROIVA-YYYY-MMDD-MonthTotal-YearTotal
2. Currency based on contract converted to USD
3. Check only one of the following
4. Encode some documents based on experience and highest education

5-6. Attached screenshot

Balik Mnaggagawa

BM - For Critical Skills



FM-DMWNCR-04-EF-09-C
Effectivity Date: February 1, 2023

Department of Migrant Workers
MIGRANT WORKERS PROCESSING DIVISION – RO IVA
Basement Anderson Bldg. II, Brgy. Parian, Calamba City, Laguna 4027

REQUEST TO PROCESS RETURNING WORKER CRITICAL SKILLS

FOR : **ATTY. JULYN S. AMBITO-FERMIN**
Director IV – Land-based Accreditation Bureau

DATE : 31 OCTOBER 2025

Control Number	BM-2025-1031-004-005
Name of worker	ANGEL MAE SILVA
Position	ENGINEER
Salary	2000.00 USD / MONTH
Destination	USA
Name of the new principal	NEW USA ENGINEERING
Employment duration	15 OCTOBER 2025 TO 30 OCTOBER 2025
Date of arrival	22 OCTOBER 2025
Date of departure	31 OCTOBER 2025
Remarks	

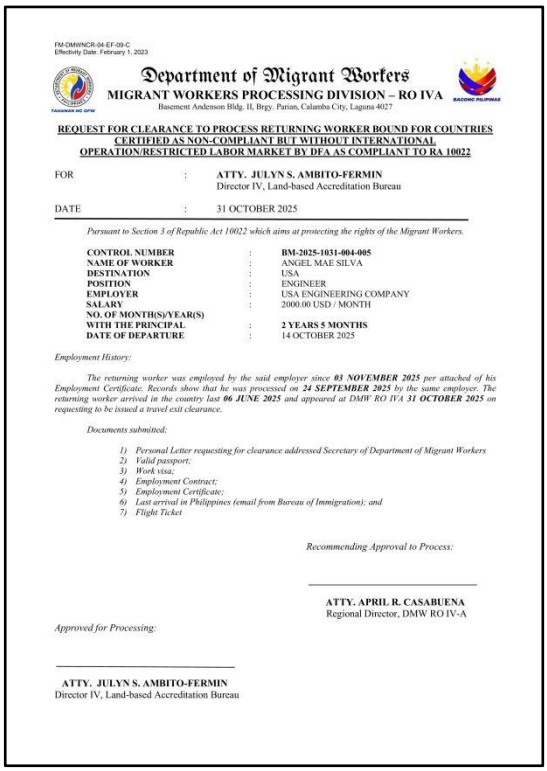
Recommending Approval to Process:

ATTY. APRIL R. CASABUENA
Regional Director

Approved for Processing:

ATTY. JULYN S. AMBITO-FERMIN
Director IV – Land-based Accreditation Bureau

BM - Non Complaint Country



FM-DMWNCR-04-EF-09-C
Effectivity Date: February 1, 2023

Department of Migrant Workers
MIGRANT WORKERS PROCESSING DIVISION – RO IVA
Basement Anderson Bldg. II, Brgy. Parian, Calamba City, Laguna 4027

REQUEST FOR CLEARANCE TO PROCESS RETURNING WORKER BOUND FOR COUNTRIES CERTIFIED AS NON-COMPLIANT BUT WITHOUT INTERNATIONAL OPERATION/RESTRICTED LABOR MARKET BY DFA AS COMPLIANT TO RA 10022

FOR : **ATTY. JULYN S. AMBITO-FERMIN**
Director IV, Land-based Accreditation Bureau

DATE : 31 OCTOBER 2025

Pursuant to Section 3 of Republic Act 10022 which aims at protecting the rights of the Migrant Workers.

CONTROL NUMBER	:	**BM-2025-1031-004-005**
NAME OF WORKER	:	ANGEL MAE SILVA
DESTINATION	:	USA
POSITION	:	ENGINEER
EMPLOYER	:	USA ENGINEERING COMPANY
SALARY	:	2000.00 USD / MONTH
NO. OF MONTH(S)/YEAR(S) WITH THE PRINCIPAL	:	**2 YEARS 5 MONTHS**
DATE OF DEPARTURE	:	14 OCTOBER 2025

Employment History:

*The returning worker was employed by the said employer since **03 NOVEMBER 2025** per attached of his Employment Certificate. Records show that he was processed on **24 SEPTEMBER 2025** by the same employer. The returning worker arrived in the country last **06 JUNE 2025** and appeared at DMW RO IVA **31 OCTOBER 2025** on requesting to be issued a travel exit clearance.*

Documents submitted;

1) *Personal Letter requesting for clearance addressed Secretary of Department of Migrant Workers*
2) *Valid passport;*
3) *Work visa;*
4) *Employment Contract;*
5) *Employment Certificate;*
6) *Last arrival in Philippines (email from Bureau of Immigration); and*
7) *Flight Ticket*

Recommending Approval to Process:

ATTY. APRIL R. CASABUENA
Regional Director, DMW RO IV-A

Approved for Processing:

ATTY. JULYN S. AMBITO-FERMIN
Director IV, Land-based Accreditation Bureau

BM - Seafarer's Position



FM-DMWNCR-04-EF-09-C
Effectivity Date: February 1, 2023

Department of Migrant Workers
MIGRANT WORKERS PROCESSING DIVISION – RO IVA
Basement Anderson Bldg. II, Brgy. Parian, Calamba City, Laguna 4027

REQUEST TO PROCESS RETURNING WORKER WITH SEAFERER'S POSITION

FOR : **ATTY. JULYN S. AMBITO-FERMIN**
Director IV – Land-based Accreditation Bureau

DATE : 31 OCTOBER 2025

Control Number	BM-2025-1031-004-005
Name of worker	ANGEL MAE SILVA
Position	ENGINEER
Salary	2000.00 USD / MONTH
Destination	USA
Name of the new principal	SAMSUNG
Employment duration	10 NOVEMBER 2025 TO 10 NOVEMBER 2027
Date of arrival	16 OCTOBER 2025
Date of departure	07 NOVEMBER 2025
Remarks	

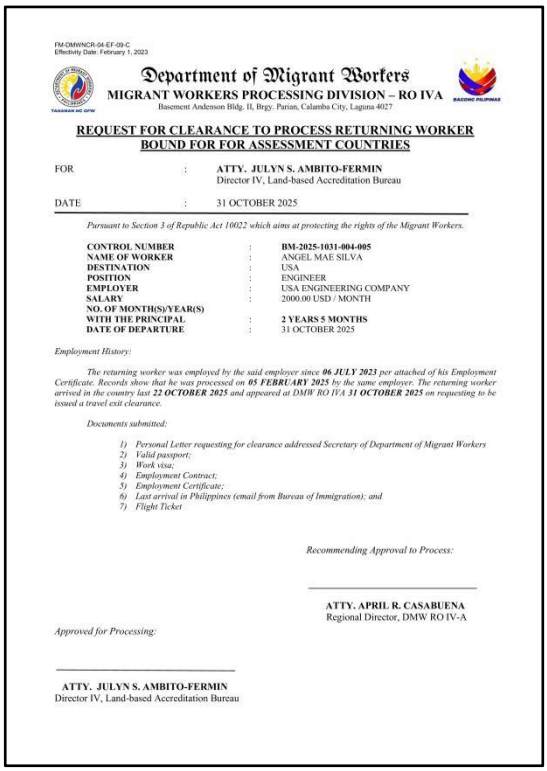
Recommending Approval to Process:

ATTY. APRIL R. CASABUENA
Regional Director

Approved for Processing:

ATTY. JULYN S. AMBITO-FERMIN
Director IV – Land-based Accreditation Bureau

BM - For Assessment Country



FM-DMWNCR-04-EF-09-C
Effectivity Date: February 1, 2023

Department of Migrant Workers
MIGRANT WORKERS PROCESSING DIVISION – RO IVA
Basement Anderson Bldg. II, Brgy. Parian, Calamba City, Laguna 4027

REQUEST FOR CLEARANCE TO PROCESS RETURNING WORKER BOUND FOR FOR ASSESSMENT COUNTRIES

FOR : **ATTY. JULYN S. AMBITO-FERMIN**
Director IV, Land-based Accreditation Bureau

DATE : 31 OCTOBER 2025

Pursuant to Section 3 of Republic Act 10022 which aims at protecting the rights of the Migrant Workers.

CONTROL NUMBER	:	**BM-2025-1031-004-005**
NAME OF WORKER	:	ANGEL MAE SILVA
DESTINATION	:	USA
POSITION	:	ENGINEER
EMPLOYER	:	USA ENGINEERING COMPANY
SALARY	:	2000.00 USD / MONTH
NO. OF MONTH(S)/YEAR(S) WITH THE PRINCIPAL	:	**2 YEARS 5 MONTHS**
DATE OF DEPARTURE	:	31 OCTOBER 2025

Employment History:

*The returning worker was employed by the said employer since **06 JULY 2023** per attached of his Employment Certificate. Records show that he was processed on **05 FEBRUARY 2025** by the same employer. The returning worker arrived in the country last **22 OCTOBER 2025** and appeared at DMW RO IVA **31 OCTOBER 2025** on requesting to be issued a travel exit clearance.*

Documents submitted:

1) *Personal Letter requesting for clearance addressed Secretary of Department of Migrant Workers*
2) *Valid passport;*
3) *Work visa;*
4) *Employment Contract;*
5) *Employment Certificate;*
6) *Last arrival in Philippines (email from Bureau of Immigration); and*
7) *Flight Ticket*

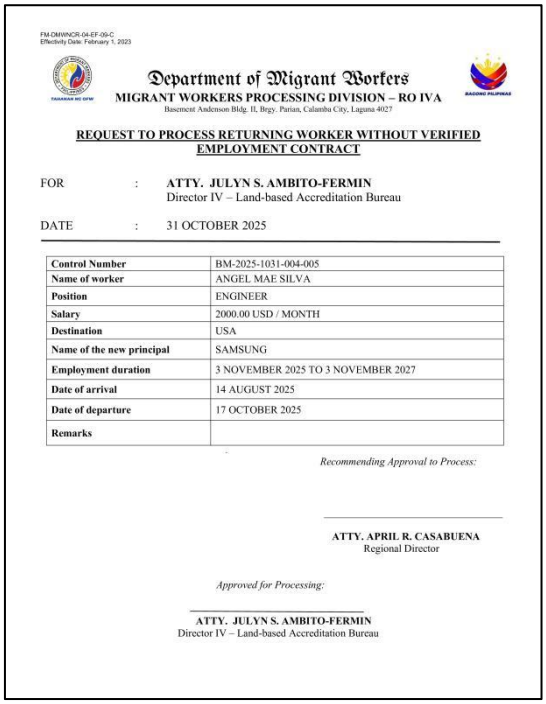
Recommending Approval to Process:

ATTY. APRIL R. CASABUENA
Regional Director, DMW RO IV-A

Approved for Processing:

ATTY. JULYN S. AMBITO-FERMIN
Director IV, Land-based Accreditation Bureau

BM - No Verified Contract



FM-DMWNCR-04-EF-09-C
Effectivity Date: February 1, 2023

Department of Migrant Workers
MIGRANT WORKERS PROCESSING DIVISION – RO IVA
Basement Anderson Bldg. II, Brgy. Parian, Calamba City, Laguna 4027

REQUEST TO PROCESS RETURNING WORKER WITHOUT VERIFIED EMPLOYMENT CONTRACT

FOR : **ATTY. JULYN S. AMBITO-FERMIN**
Director IV – Land-based Accreditation Bureau

DATE : 31 OCTOBER 2025

Control Number	BM-2025-1031-004-005
Name of worker	ANGEL MAE SILVA
Position	ENGINEER
Salary	2000.00 USD / MONTH
Destination	USA
Name of the new principal	SAMSUNG
Employment duration	3 NOVEMBER 2025 TO 3 NOVEMBER 2027
Date of arrival	14 AUGUST 2025
Date of departure	17 OCTOBER 2025
Remarks	

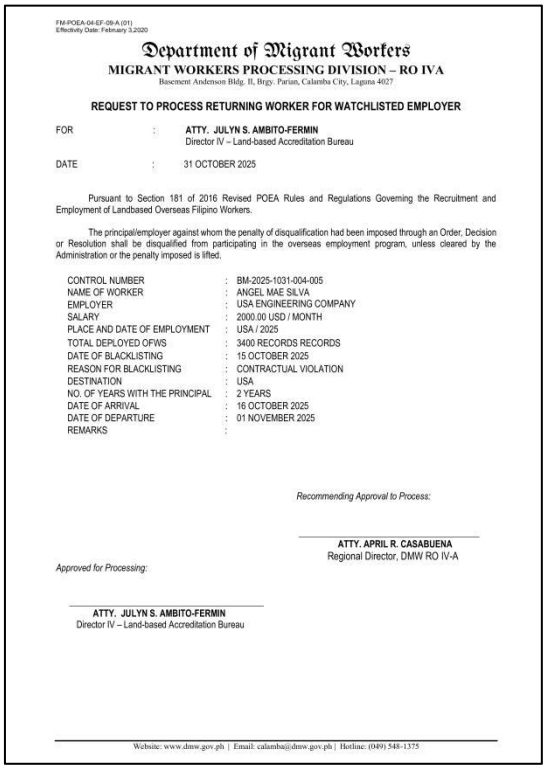
Recommending Approval to Process:

ATTY. APRIL R. CASABUENA
Regional Director

Approved for Processing:

ATTY. JULYN S. AMBITO-FERMIN
Director IV – Land-based Accreditation Bureau

BM - Watchlisted Employer



FM-POEA-04-EF-09-A (01)
Effectivity Date: February 3,2020

Department of Migrant Workers
MIGRANT WORKERS PROCESSING DIVISION – RO IVA
Basement Andenson Bldg. II, Brgy. Parian, Calamba City, Laguna 4027

REQUEST TO PROCESS RETURNING WORKER FOR WATCHLISTED EMPLOYER

FOR : **ATTY. JULYN S. AMBITO-FERMIN**
Director IV – Land-based Accreditation Bureau

DATE : 31 OCTOBER 2025

Pursuant to Section 181 of 2016 Revised POEA Rules and Regulations Governing the Recruitment and Employment of Landbased Overseas Filipino Workers.

The principal/employer against whom the penalty of disqualification had been imposed through an Order, Decision or Resolution shall be disqualified from participating in the overseas employment program, unless cleared by the Administration or the penalty imposed is lifted.

CONTROL NUMBER	:	BM-2025-1031-004-005
NAME OF WORKER	:	ANGEL MAE SILVA
EMPLOYER	:	USA ENGINEERING COMPANY
SALARY	:	2000.00 USD / MONTH
PLACE AND DATE OF EMPLOYMENT	:	USA / 2025
TOTAL DEPLOYED OFWS	:	3400 RECORDS RECORDS
DATE OF BLACKLISTING	:	15 OCTOBER 2025
REASON FOR BLACKLISTING	:	CONTRACTUAL VIOLATION
DESTINATION	:	USA
NO. OF YEARS WITH THE PRINCIPAL	:	2 YEARS
DATE OF ARRIVAL	:	16 OCTOBER 2025
DATE OF DEPARTURE	:	01 NOVEMBER 2025
REMARKS	:	

Recommending Approval to Process:

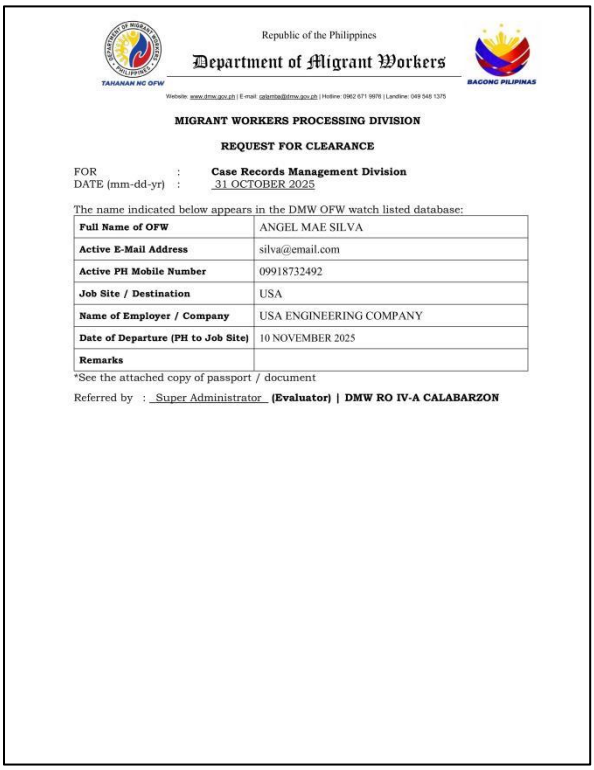
ATTY. APRIL R. CASABUENA
Regional Director, DMW RO IV-A

Approved for Processing:

ATTY. JULYN S. AMBITO-FERMIN
Director IV – Land-based Accreditation Bureau

Website: www.dmw.gov.ph | Email: calamba@dmw.gov.ph | Hotline: (049) 548-1375

BM - Watchlisted OFW



Republic of the Philippines
Department of Migrant Workers

TAHANAN NG OFW

BAGONG PILIPINAS

Website: www.dmw.gov.ph | E-mail: calamba@dmw.gov.ph | Hotline: 0962 671 9978 | Landline: 049 548 1375

MIGRANT WORKERS PROCESSING DIVISION

REQUEST FOR CLEARANCE

FOR : **Case Records Management Division**
DATE (mm-dd-yr) : 31 OCTOBER 2025

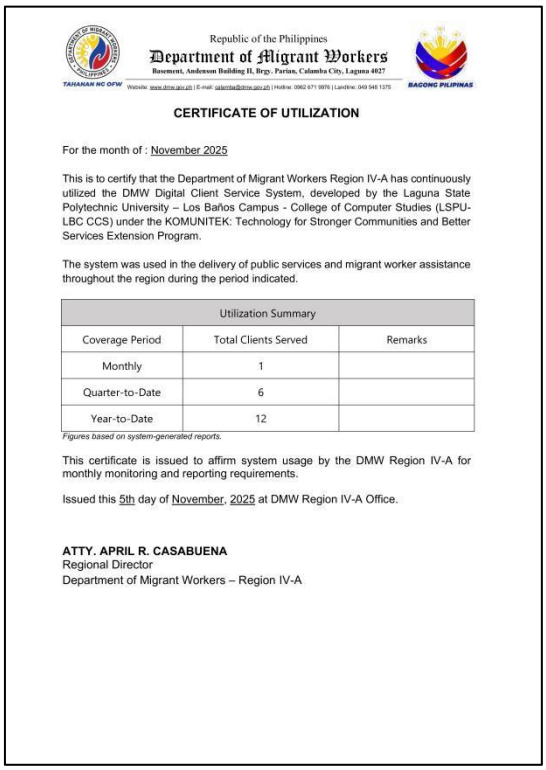
The name indicated below appears in the DMW OFW watch listed database:

Full Name of OFW	ANGEL MAE SILVA
Active E-Mail Address	silva@email.com
Active PH Mobile Number	09918732492
Job Site / Destination	USA
Name of Employer / Company	USA ENGINEERING COMPANY
Date of Departure (PH to Job Site)	10 NOVEMBER 2025
Remarks	

*See the attached copy of passport / document

Referred by : Super Administrator **(Evaluator) | DMW RO IV-A CALABARZON**

Certificate of Utilization



Republic of the Philippines
Department of Migrant Workers
Basement, Andesson Building II, Brgy. Parian, Calamba City, Laguna 4027
TAHANAN NG OFW Website: www.dmw.gov.ph | E-mail: calamba@dmw.gov.ph | Hotline: 0962 671 9976 | Landline: 049 545 1375 BAGONG PILIPINAS

CERTIFICATE OF UTILIZATION

For the month of : November 2025

This is to certify that the Department of Migrant Workers Region IV-A has continuously utilized the DMW Digital Client Service System, developed by the Laguna State Polytechnic University – Los Baños Campus - College of Computer Studies (LSPU-LBC CCS) under the KOMUNITEK: Technology for Stronger Communities and Better Services Extension Program.

The system was used in the delivery of public services and migrant worker assistance throughout the region during the period indicated.

Utilization Summary		
Coverage Period	Total Clients Served	Remarks
Monthly	1	
Quarter-to-Date	6	
Year-to-Date	12	

Figures based on system-generated reports.

This certificate is issued to affirm system usage by the DMW Region IV-A for monthly monitoring and reporting requirements.

Issued this 5th day of November, 2025 at DMW Region IV-A Office.

ATTY. APRIL R. CASABUENA
Regional Director
Department of Migrant Workers – Region IV-A

APPENDIX G

Source Code

Source Code

Frontend:

Applications-table.tsx

```
"use client"
import { Button } from "@/components/ui/button"
import { MoreHorizontal } from "lucide-react"
import { DropdownMenu, DropdownMenuContent, DropdownMenuItem, DropdownMenuTrigger } from "@/components/ui/dropdown-menu"
type Application = {
  id: string | number
  controlNumber: string
  name: string
  category: string
  jobsite: string
  sex: string
  status: string
  created_at?: string}
interface ApplicationsTableProps {
  applications: Application[]
  search?: string
  filterQuery?: string
  pagination?: {
    page: number
    limit: number
    total: number
    totalPages: number}
  onPageChange?: (page: number) => void
  onRowDoubleClick?: (application: Application) => void}
const getCategoryColor = (category: string) => {
  const categoryLower = category.toLowerCase()
  if (categoryLower.includes('direct hire') || categoryLower.includes('direct-hire')) {
    return 'bg-blue-100 text-blue-800 border-blue-200'}
  if (categoryLower.includes('balik manggagawa') || categoryLower.includes('clearance')) {
    return 'bg-green-100 text-green-800 border-green-200'}
  if (categoryLower.includes('gov to gov') || categoryLower.includes('gov-to-gov')) {
    return 'bg-purple-100 text-purple-800 border-purple-200'}
  if (categoryLower.includes('information sheet') || categoryLower.includes('info sheet')) {
    return 'bg-orange-100 text-orange-800 border-orange-200'}
  if (categoryLower.includes('job fair') || categoryLower.includes('job-fair')) {
    return 'bg-pink-100 text-pink-800 border-pink-200'}
  if (categoryLower.includes('processing')) {
    return 'bg-yellow-100 text-yellow-800 border-yellow-200'}
 return 'bg-gray-100 text-gray-800 border-gray-200'}
export default function ApplicationsTable({ applications, search = "", filterQuery = "", pagination, onPageChange, onRowDoubleClick }: ApplicationsTableProps) {
  return (
    <>
      {/* Pagination Controls */}
      {pagination && pagination.total > 0 && (
        <div className="mb-4 flex items-center justify-between text-sm text-gray-600">
          <div className="flex items-center gap-4">
            <span>
              Page {pagination.page} of {pagination.totalPages} ({pagination.total} total applications)
            </span>
          </div>
          <div className="flex items-center gap-1">
            {(() => {
              const pages = [];
              const totalPages = pagination.totalPages;
              const currentPage = pagination.page;
              if (totalPages <= 7) {
                for (let i = 1; i <= totalPages; i++) {
                  pages.push(
                    <Button
                      key={i}
                      variant={i === currentPage ? "default" : "outline"}
                      size="sm"
                      onClick={() => onPageChange?.(i)}
                      className="min-w-[40px] h-8"
                    >
                      {i}
                    </Button>
                  );
                }
              } else {
```

```
        // Dynamic pagination: show 5 pages around current page
        let startPage = Math.max(1, currentPage - 2);
        let endPage = Math.min(totalPages, startPage + 4);

        // Adjust if we're near the end
        if (endPage - startPage < 4) {
          startPage = Math.max(1, endPage - 4);
        }
        if (startPage > 1) {
          pages.push(
            <Button
              key={1}
              variant={1 === currentPage ? "default" : "outline"}
              size="sm"
              onClick={() => onPageChange?.(1)}
              className="min-w-[40px] h-8"
            >
              1
            </Button>
          );
          if (startPage > 2) {
            pages.push(
              <span key="ellipses-start" className="px-2 text-gray-500">
                ...
              </span>
            );
          }
        }
        for (let i = startPage; i <= endPage; i++) {
          pages.push(
            <Button
              key={i}
              variant={i === currentPage ? "default" : "outline"}
              size="sm"
              onClick={() => onPageChange?.(i)}
              className="min-w-[40px] h-8"
            >
              {i}
            </Button>
          );
        }
        if (endPage < totalPages) {
          if (endPage < totalPages - 1) {
            pages.push(
              <span key="ellipses-end" className="px-2 text-gray-500">
                ...
              </span>
            ); }

          pages.push(
            <Button
              key={totalPages}
              variant={totalPages === currentPage ? "default" : "outline"}
              size="sm"
              onClick={() => onPageChange?.(totalPages)}
              className="min-w-[40px] h-8"
            >
              {totalPages}
            </Button>
          );
        }
      }

      return pages;
    })()}
  </div>
 </div>
)}

<div className="bg-white rounded-md border overflow-hidden">
 <div className="overflow-x-auto max-h-[calc(100vh-280px)] overflow-y-auto">
 <table className="w-full">
  <thead>
   <tr className="bg-[#1976D2] text-white sticky top-0 z-10 bg-[#1976D2]">
    <th className="py-3 px-4 font-medium text-center">Control #</th>
```

```
            <th className="py-3 px-4 font-medium text-center">Name</th>
            <th className="py-3 px-4 font-medium text-center">Category</th>
            <th className="py-3 px-4 font-medium text-center">Jobsite</th>
            <th className="py-3 px-4 font-medium text-center">Sex</th>
          </tr>
        </thead>
        <tbody className="divide-y divide-gray-200">
          {applications.map((application: Application) => (
            <tr
              key={application.id}
              className="hover:bg-gray-150 transition-colors duration-75 cursor-pointer select-none"
              onDoubleClick={(e) => {
                e.preventDefault()
                onRowDoubleClick?.(application)
              }}
            >
              <td className="py-3 px-4 text-center">{application.controlNumber}</td>
              <td className="py-3 px-4 text-center">{application.name}</td>
              <td className="py-3 px-4 text-center">
                        <span className={`inline-flex items-center px-2.5 py-0.5 rounded-full text-xs font-medium border
${getCategoryColor(application.category)}`}>
                  {application.category}
                </span>
              </td>
              <td className="py-3 px-4 text-center">{application.jobsite}</td>
              <td className="py-3 px-4 text-center">{application.sex}</td>
            </tr>
          ))}
        </tbody>
      </table>
    </div>
   </div>
   </>
  )
}
```

dashboard-page.tsx

```
"use client"
import DirectHireTable from "@/components/direct-hire-table"
import Header from "@/components/shared/header"
import { useLoginSuccessToast } from "@/hooks/use-login-success-toast"
export default function DashboardPage() {
  // Handle login success toast
  useLoginSuccessToast()
  return (
    <div className="flex flex-col bg-[#EEF5FD]">
      <Header />

      {/* Main Content */}
      <main className="flex-1 px-6 pt-24 pb-12">
        <DirectHireTable />
      </main>
    </div>
  )
}
```

Backend:

filehandling.ts

```
// utils/fileHandling.ts
import { DocFiles, DocMetadata } from './formValidation'
export const handleDragOver = (e: React.DragEvent, key: string, setDragOver: (key: string | null) => void) => {
  e.preventDefault()
  setDragOver(key)}
export const handleDragLeave = (e: React.DragEvent, setDragOver: (key: string | null) => void) => {
  e.preventDefault()
  setDragOver(null)}
export const handleDrop = (
  e: React.DragEvent,
  key: string,
  setDragOver: (key: string | null) => void,
  handleDocChange: (key: keyof DocFiles, file: File | null) => void,
  toast: (options: any) => void
) => {
  e.preventDefault()
  setDragOver(null)
  const files = Array.from(e.dataTransfer.files)
```

```
  if (files.length > 0) {
    const file = files[0]
    const isAllowedType = file.type.startsWith('image/') || file.type === 'application/pdf'
    if (isAllowedType) {
      handleDocChange(key as keyof DocFiles, file)
    } else {
      toast({
        title: 'Invalid File Type',
        description: 'Please upload only PDF, PNG, or JPEG files',
        variant: 'destructive'})
    }
  }
}
export const handlePasteFromClipboard = async (
  screenshotKey: 'screenshot1' | 'screenshot2',
  setDocFiles: (fn: (prev: DocFiles) => DocFiles) => void,
  setDocMetadata: (fn: (prev: DocMetadata) => DocMetadata) => void,
  toast: (options: any) => void
) => {
  try {
    if (!navigator.clipboard || typeof navigator.clipboard.read !== 'function') {
      toast({
        title: 'Clipboard Not Supported',
        description: 'Your browser does not support clipboard access. Please use the file upload instead.',
        variant: 'destructive'
      })
      return}
    const clipboardItems = await navigator.clipboard.read()
    let imageFound = false

    for (const clipboardItem of clipboardItems) {
      for (const type of clipboardItem.types) {
        if (type.startsWith('image/')) {
          const blob = await clipboardItem.getType(type)
          const file = new File([blob], `screenshot-${Date.now()}.${type.split('/')[1]}`, { type })

          // Store the file in docFiles
          setDocFiles(prev => ({ ...prev, [screenshotKey]: file }))

          // Create a preview URL for the screenshot
          const url = URL.createObjectURL(file)
          setDocMetadata(prev => ({ ...prev, [`${screenshotKey}_url`]: url }))

          toast({
            title: 'Screenshot Pasted',
            description: 'Image has been pasted from clipboard successfully.',
          })
          imageFound = true
          break
        }
      }
      if (imageFound) break
    }

    if (!imageFound) {
      toast({
        title: 'No Image Found',
        description: 'No image found in clipboard. Please copy an image first.',
        variant: 'destructive'
      })
    }
  } catch (error) {
    console.error('Error pasting from clipboard:', error)
    toast({
      title: 'Paste Failed',
      description: 'Failed to paste image from clipboard. Please try using file upload instead.',
      variant: 'destructive'
    })
  }
}
export const checkClipboardSupport = (): boolean => {
  return !!(navigator.clipboard && typeof navigator.clipboard.read === 'function')
}
// File type validation
export const isValidFileType = (file: File): boolean => {
```

```
  const allowedTypes = [
    'application/pdf',
    'image/jpeg',
    'image/png',
    'image/gif',
    'application/msword',
    'application/vnd.openxmlformats-officedocument.wordprocessingml.document',
    'application/vnd.ms-excel',
    'application/vnd.openxmlformats-officedocument.spreadsheetml.sheet'
  ]
  return allowedTypes.includes(file.type)
}
export const validateVisaNumber = (value: string): string => {
  return value.toUpperCase().replace(/[^A-Z0-9]/g, '')
}
```

formvalidation.ts

```
// utils/formValidation.ts
import { Currency } from "@/lib/currency-converter"

export interface FormData {
  name: string
  email: string
  cellphone: string
  sex: 'male' | 'female' | ''
  jobsite: string
  position: string
  job_type: 'household' | 'professional' | ''
  salary: string
  employer: string
  salaryCurrency: Currency | ''
  processed_workers_principal: string
  processed_workers_las: string
  verifier_type: 'MWO' | 'PEPCG' | 'OTHERS' | ''
  verifier_office: string
  pe_pcg_city: string
  others_text: string
  verified_date: string
  time_received: string
  time_released: string
}

export interface DocMetadata {
  passport_number: string
  passport_expiry: string
  visa_category: string
  visa_type: string
  visa_validity: string
  visa_number: string
  ec_issued_date: string
  ec_verification: string
  screenshot1_url: string
  screenshot2_url: string
}

export interface DocFiles {
  passport: File | null
  work_visa: File | null
  employment_contract: File | null
  tesda_license: File | null
  country_specific: File | null
  compliance_form: File | null
  medical_certificate: File | null
  peos_certificate: File | null
  clearance: File | null
  insurance_coverage: File | null
  eregistration: File | null
  pdos_certificate: File | null
  screenshot1: File | null
  screenshot2: File | null
}

// Date validation helpers
```

```
export const getMinDate = () => {
  const today = new Date()
  const oneYearFromNow = new Date(today.getFullYear() + 1, today.getMonth(), today.getDate())
  return oneYearFromNow.toISOString().split('T')[0]
}

export const getMaxDate = () => {
  const today = new Date()
  return today.toISOString().split('T')[0]
}

export const getPassportMinDate = () => {
  const today = new Date()
  const oneYearFromNow = new Date(today.getFullYear() + 1, today.getMonth(), today.getDate())
  return oneYearFromNow.toISOString().split('T')[0]
}

export const getVisaValidityMinDate = () => {
  const today = new Date()
  const tomorrow = new Date(today.getTime() + 24 * 60 * 60 * 1000)
  return tomorrow.toISOString().split('T')[0]
}

// Basic form validation
export const validateForm = (formData: FormData): string[] => {
  const errors: string[] = []
  const emailRegex = /^[^\s@]+@[^\s@]+\.[^\s@]{2,}$/i

  // Form 1 validation
  if (!formData.name.trim() || !formData.jobsite.trim() || !formData.position.trim() ||
      !formData.salary.trim() || isNaN(parseFloat(formData.salary)) || parseFloat(formData.salary) <= 0 ||
      !formData.sex || !formData.job_type || !formData.salaryCurrency) {
    errors.push("Form 1 is incomplete");
  }
  // Optional but must be valid if provided
  if (formData.email && !emailRegex.test(formData.email)) {
    errors.push("Email address is invalid")
  }
  if (formData.cellphone) {
    const digits = formData.cellphone.replace(/\D/g, '')
    if (!/^09\d{9}$/.test(digits)) {
      errors.push("Cellphone number must start with 09 and be 11 digits")
    }
  }

  return errors;
}

// Draft validation (only requires name)
export const validateDraftForm = (formData: FormData): string[] => {
  const errors: string[] = []

  if (!formData.name.trim()) {
    errors.push("Worker name is required");
  }

  return errors;
}

// Missing field validation function - identifies specific unfilled required fields
export const getMissingFields = (formData: FormData, docMetadata: DocMetadata, docFiles: DocFiles): string[] => {
  const missingFields: string[] = []
  const emailRegex = /^[^\s@]+@[^\s@]+\.[^\s@]{2,}$/i

  // Basic Information (Form 1) - Required fields
  if (!formData.name.trim()) missingFields.push("Worker Name");
  if (!formData.jobsite.trim()) missingFields.push("Jobsite");
  if (!formData.position.trim()) missingFields.push("Position");
  if (!formData.salary.trim() || isNaN(parseFloat(formData.salary)) || parseFloat(formData.salary) <= 0) {
    missingFields.push("Salary");
  }
  if (!formData.sex) missingFields.push("Sex");
  if (!formData.job_type) missingFields.push("Job Type");
```

```
if (!formData.salaryCurrency) missingFields.push("Salary Currency");

// Basic Information (Form 1) - Optional but must be valid if provided
if (formData.email && !emailRegex.test(formData.email)) {
  missingFields.push("Valid Email Address");
}
if (formData.cellphone) {
  const digits = formData.cellphone.replace(/\D/g, '')
  if (!/^09\d{9}$/.test(digits)) {
    missingFields.push("Valid Cellphone Number (starts with 09, 11 digits)");
  }
}

// Additional Details (Form 3) - Required fields
if (!formData.processed_workers_principal || formData.processed_workers_principal.trim() === '') {
  missingFields.push("No. of Processed Workers in Principal");
}
if (!formData.processed_workers_las || formData.processed_workers_las.trim() === '') {
  missingFields.push("No. of Processed Workers in LAS");
}
if (!formData.verifier_type) missingFields.push("Employment Contract Verified by");
if (!formData.verified_date) missingFields.push("Verification Date");

// Conditional required fields based on verifier type
if (formData.verifier_type === 'MWO' && (!formData.verifier_office || formData.verifier_office.trim() === '')) {
  missingFields.push("MWO Office");
}
if (formData.verifier_type === 'PEPCG' && (!formData.pe_pcg_city || formData.pe_pcg_city.trim() === '')) {
  missingFields.push("PE/PCG City/Office");
}
if (formData.verifier_type === 'OTHERS' && (!formData.others_text || formData.others_text.trim() === '')) {
  missingFields.push("Specify Other Verifier");
}

// Document metadata requirements - check if file is uploaded and metadata is filled
if (docFiles.passport || (docFiles.passport && (docFiles.passport as any).uploaded)) {
  if (!docMetadata.passport_number || docMetadata.passport_number.trim() === '') {
    missingFields.push("Passport Number");
  }
  if (!docMetadata.passport_expiry || docMetadata.passport_expiry.trim() === '') {
    missingFields.push("Passport Expiry Date");
  }
}

if (docFiles.work_visa || (docFiles.work_visa && (docFiles.work_visa as any).uploaded)) {
  if (!docMetadata.visa_category || docMetadata.visa_category.trim() === '') {
    missingFields.push("Visa Category");
  }
  if (!docMetadata.visa_type || docMetadata.visa_type.trim() === '') {
    missingFields.push("Visa Type");
  }
  if (!docMetadata.visa_number || docMetadata.visa_number.trim() === '') {
    missingFields.push("Visa Number");
  }
  if (!docMetadata.visa_validity || docMetadata.visa_validity.trim() === '') {
    missingFields.push("Visa Validity Date");
  }
}

if (docFiles.employment_contract || (docFiles.employment_contract && (docFiles.employment_contract as any).uploaded)) {
  if (!docMetadata.ec_issued_date || docMetadata.ec_issued_date.trim() === '') {
    missingFields.push("Employment Contract Issued Date");
  }
  if (!docMetadata.ec_verification || docMetadata.ec_verification.trim() === '') {
    missingFields.push("Employment Contract Verification Type");
  }
}

// Screenshot requirements - check both new uploads and previously uploaded
const hasScreenshot1 = docMetadata.screenshot1_url || (docFiles.screenshot1 && (docFiles.screenshot1 as any).uploaded)
const hasScreenshot2 = docMetadata.screenshot2_url || (docFiles.screenshot2 && (docFiles.screenshot2 as any).uploaded)

if (!hasScreenshot1) {
  missingFields.push("Screenshot for No. of Processed Workers");
}
```

```
  if (!hasScreenshot2) {
    missingFields.push("MWO/POLO/PE/PCG Validation Screenshot");
  }

  return missingFields;
}
export const getFieldErrors = (formData: FormData, isDraft: boolean = false): {[key: string]: string} => {
  const errors: {[key: string]: string} = {}
  const emailRegex = /^[^\s@]+@[^\s@]+\.[^\s@]{2,}$/i

  if (!formData.name.trim()) {
    errors.name = "Name is required";
  }

  if (!isDraft) {
    if (!formData.sex) {
      errors.sex = "Sex is required";
    }
    if (!formData.job_type) {
      errors.job_type = "Job type is required";
    }
    if (!formData.jobsite.trim()) {
      errors.jobsite = "Jobsite is required";
    }
    if (!formData.position.trim()) {
      errors.position = "Position is required";
    }
    if (!formData.salary.trim() || isNaN(parseFloat(formData.salary)) || parseFloat(formData.salary) <= 0) {
      errors.salary = "Valid salary is required";
    }
    if (!formData.salaryCurrency) {
      errors.salaryCurrency = "Currency is required";
    }
    if (formData.email && !emailRegex.test(formData.email)) {
      errors.email = "Enter a valid email"
    }
    if (formData.cellphone) {
      const digits = formData.cellphone.replace(/\D/g, '')
      if (!/^09\d{9}$/.test(digits)) {
        errors.cellphone = "Must start with 09 and be 11 digits"
      }
    }
  }

  return errors;
}
export const getDocumentMetadataErrors = (docFiles: DocFiles, docMetadata: DocMetadata): {[key: string]: string} => {
  const errors: {[key: string]: string} = {}

  if (docFiles.passport || (docFiles.passport && (docFiles.passport as any).uploaded)) {
    if (!docMetadata.passport_number || docMetadata.passport_number.trim() === '') {
      errors.passport_number = "Passport number is required";
    }
    if (!docMetadata.passport_expiry || docMetadata.passport_expiry.trim() === '') {
      errors.passport_expiry = "Passport expiry date is required";
    }
  }

  if (docFiles.work_visa || (docFiles.work_visa && (docFiles.work_visa as any).uploaded)) {
    if (!docMetadata.visa_category || docMetadata.visa_category.trim() === '') {
      errors.visa_category = "Visa category is required";
    }
    if (!docMetadata.visa_type || docMetadata.visa_type.trim() === '') {
      errors.visa_type = "Visa type is required";
    }
    if (!docMetadata.visa_number || docMetadata.visa_number.trim() === '') {
      errors.visa_number = "Visa number is required";
    }
    if (!docMetadata.visa_validity || docMetadata.visa_validity.trim() === '') {
      errors.visa_validity = "Visa validity date is required";
    }
  }
  if (docFiles.employment_contract || (docFiles.employment_contract && (docFiles.employment_contract as any).uploaded)) {
    if (!docMetadata.ec_issued_date || docMetadata.ec_issued_date.trim() === '') {
      errors.ec_issued_date = "Issued date is required";
```

```
    }
    if (!docMetadata.ec_verification || docMetadata.ec_verification.trim() === '') {
      errors.ec_verification = "Verification type is required";
    }
  }
  return errors;
}
export const validateFormWithMissingFields = (formData: FormData, docMetadata: DocMetadata, docFiles: DocFiles): string[]
=> {
  const missingFields = getMissingFields(formData, docMetadata, docFiles);
  if (missingFields.length > 0) {
    return [`Please complete the following required fields: ${missingFields.join(', ')}`];
  }
  return [];
}
```

APPENDIX H

Formula

To get the mean:

$$\bar{x} = \frac{\sum X}{N}$$

$\bar{x}$ - the mean (average)

$\sum$- means “sum of”

X - each individual rating

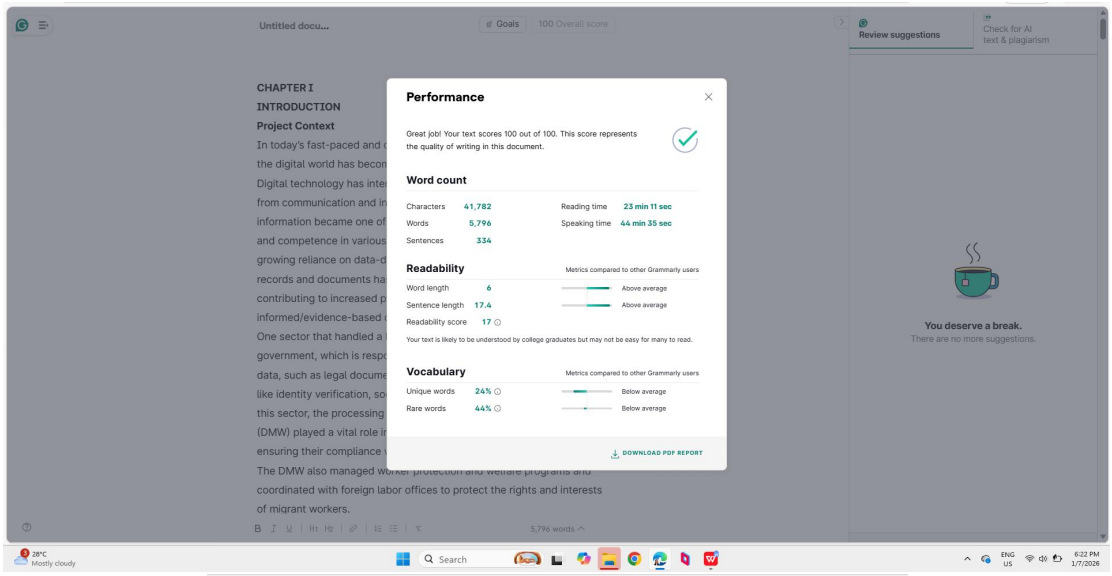
N - the total number of respondents

Standard Deviation

$$s = \sqrt{\frac{\Sigma(\bar{x} - x)^2}{(n-1)}}$$

APPENDIX I

Grammarly Writing Assistant Report



Report: Untitled

Untitled

by Ronah joy Ramirez

General metrics

41,783	5,796	334	23 min 11 sec	44 min 35 sec
characters	words	sentences	reading time	speaking time

Writing Issues

No issues found

Plagiarism

This text hasn't been checked for plagiarism

Report: Untitled

Unique Words

Measures vocabulary diversity by calculating the percentage of words used only once in your document

24% unique words

Rare Words

Measures depth of vocabulary by identifying words that are not among the 5,000 most common English words.

44% rare words

Word Length

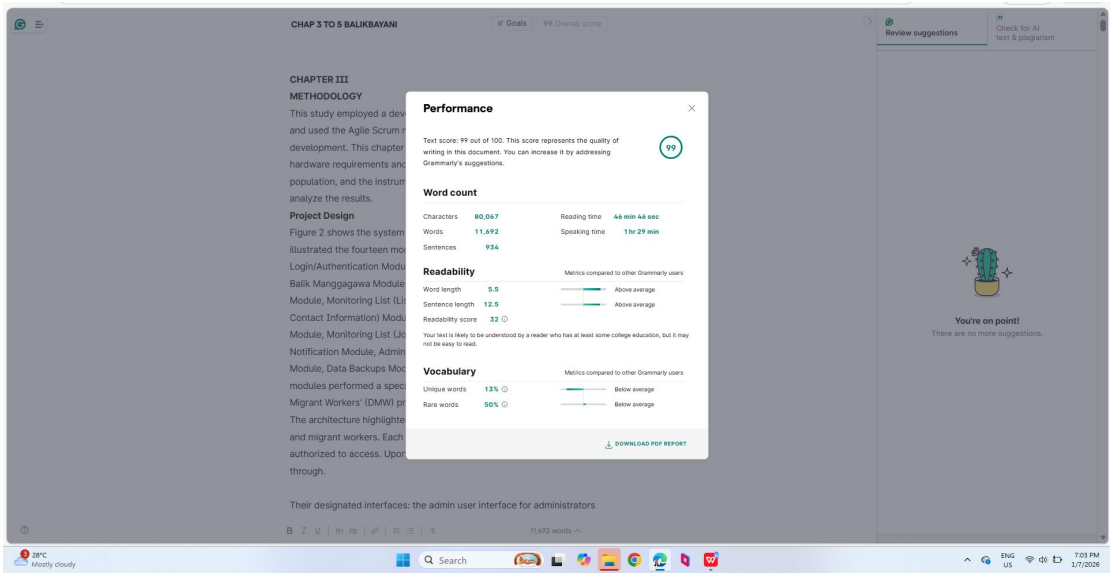
Measures average word length

6 characters per word

Sentence Length

Measures average sentence length

17.4 words per sentence



Report: CHAP 3 TO 5 BALIKBAYANI

CHAP 3 TO 5 BALIKBAYANI

by Ronah joy Ramirez

General metrics

80,068	11,692	934	46 min 46 sec	1 hr 29 min
characters	words	sentences	reading time	speaking time

Writing Issues

✓ No issues found

Plagiarism

This text hasn't been checked for plagiarism

Report: CHAP 3 TO 5 BALIKBAYANI

Unique Words

Measures vocabulary diversity by calculating the percentage of words used only once in your document

13%

unique words

Rare Words

Measures depth of vocabulary by identifying words that are not among the 5,000 most common English words.

50%

rare words

Word Length

Measures average word length

5.5

characters per word

Sentence Length

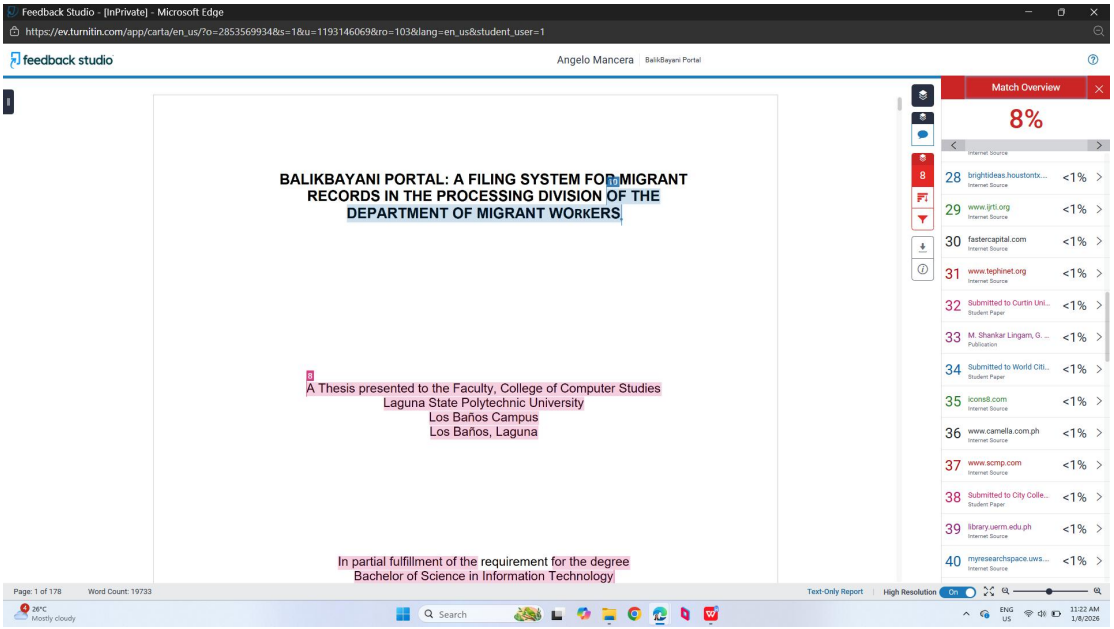
Measures average sentence length

12.5

words per sentence

APPENDIX J

Turnitin Similarity Report


Feedback Studio - [InPrivate] - Microsoft Edge
https://ev.turnitin.com/app/carta/en_us/?o=2853569934&s=1&u=1193146069&ro=103&lang=en_us&student_user=1
feedback studio
Angelo Mancera | BalikBayani Portal
BALIKBAYANI PORTAL: A FILING SYSTEM FOR MIGRANT RECORDS IN THE PROCESSING DIVISION OF THE DEPARTMENT OF MIGRANT WORKERS
A Thesis presented to the Faculty, College of Computer Studies
Laguna State Polytechnic University
Los Baños Campus
Los Baños, Laguna
In partial fulfillment of the requirement for the degree
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APPENDIX K

Curriculum Vitae

APPENDIX L

Publishable Paper

BalikBayani Portal: A Filing System for Migrant Records in the Processing Division of the Department of Migrant Workers

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ABSTRACT

The Processing Division of the Department of Migrant Workers (DMW) Region IV-A manages thousands of records for Overseas Filipino Workers (OFWs), with document handling that often involves slow and time-consuming procedures. To address this challenge, the study titled "BalikBayani Portal: A Filing System for Migrant Records in the Processing Division of the Department of Migrant Workers" aimed to digitize the filing process by integrating document generation and data visualization. The project was developed using the Agile Scrum methodology. System evaluation was conducted using the Technology Acceptance Model (TAM) to assess user perception and ISO 25010 to evaluate technical quality. Results from 104 TAM respondents yielded an overall mean score of 3.78, which was interpreted as high, while evaluations from 3 IT professionals using ISO 25010 received an overall mean score of 4.67, interpreted as excellent, indicating that the system met essential requirements for performance, usability, reliability, and security. Positive user feedback confirms the system's alignment with user needs, demonstrating its potential as an effective platform for record management. In conclusion, the BalikBayani Portal meets its technical and functional goals and aligns with the Department of Migrant Workers' mission to promote digital transformation and improve service delivery. With continued improvement and institutional support, the BalikBayani Portal can serve as a model for digital innovation in migrant worker governance.

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Email: andreilimpiada15@gmail.com

I. Introduction

Overseas Filipino Workers (OFWs) refer to a Filipino migrant workers who leave their homes to work overseas to provide a better standard of living for their family [1]. According to the Philippine Statistics Authority (PSA, 2022), the total number of OFWs was approximately 1.77 million from April to September 2020, a decrease from previous year due to the pandemic [2]. Subsequently, the number of OFWs increased to an estimated 2.16 million between April to September 2023. and out of the 17 regions in the country, CALABARZON served the largest number of OFWs in recent years (PSA, 2024) [3].

As the number of OFWs continues to grow, proper management of records and documents has become important, contributing to increased productivity, legal compliance, and informed/evidence-based decision-making strategies [4]. The processing division of the Department of Migrant Workers (DMW) holds responsibility for handling migrant workers' data and ensuring compliance with both local and international regulations [5]. However, the DMW still relies on traditional, manual processes for filing and maintaining documents. According to Bowers (2022), manual document filing can become cumbersome as it takes up a lot of space, is prone to damage, and misplacement, is difficult to modify, is very time consuming, and lacks security measures [6]. Currently, the DMW maintains these records through applications such as Excel and Word. While these applications are a widely accessible and familiar tool, Excel has several disadvantages when used for the long term, and it is not designed for large-scale record management, especially regarding limited scalability, security, features, and efficiency capabilities [7].

Given these traditional manual processes, the researchers aim to digitize the filing of records by developing a system called the *BalikBayani Portal.* This study also seeks to determine how the

BalikBayani Portal can improve document processing speed, minimize human error, and improve the overall management of migrant worker records. The proposed system is expected to be beneficial and advantageous to the Department of Migrant Workers, migrant workers, employees, researchers, and future researchers by providing a more structure and reliable document management process. This study focuses on the development and evaluation of the *BalikBayani Portal* and would be implemented within the processing division of the DMW. However, its scope is limited to internal use and does not cover integration with external agencies or departments outside the DMW.

Objectives

The main objective of this capstone project is to develop a BalikBayani Portal, a system designed to digitize the filing process of migrant worker records in the processing division of DMW. The system aims to improve workflow by reducing manual document processing. Moreover, this study aims to:

1. Develop a filing system that integrates document generation into printable format and supports quick, accurate retrieval of information.
2. Integrate a data visualization feature that provides representation across the main categories.
3. Test and evaluate user acceptance of the system using the Technology Acceptance Model (TAM) and the system's effectiveness based on the ISO 25010 software quality model focusing on performance, interaction capability, reliability, and security.

II. Related Literature

Department of Migrant Workers

Republic Act No. 11641 created the Department of Migrant Workers (DMW) to regulate overseas employment, protect the rights of Overseas Filipino Workers (OFWs), and handle necessary legal documents [8][9]. The documents, such as passports, work visas, medical clearances, and employment contracts, serve to verify a worker's identity, qualifications, and employment terms [10][11]. Insufficient documentation puts migrant workers at risk of trafficking, forced labor, and labor violations [12]. This shows how important the DMW is in making sure that workers are deployed safely and legally.

Migrant Worker

Migration continues to rise globally, with international migrants reaching 281 million in 2020 [13]. The Philippines remains a major labor-exporting country, recording 1.83 million OFWs in 2021 and around 800,000 more deployments in 2022, with CALABARZON sending the most workers [14]. OFWs strongly support the economy through $32.14 billion in remittances in 2022, contributing 8.9% to GDP [15]. Although the pandemic caused a 75% decline in deployment, Filipino workers continue to be highly in demand in sectors like healthcare, domestic work, and construction [16].

Filing System

Fenwick and Brownstone (2003) describe electronic filing as the submission of records in digital rather than paper form, while a filing system serves as an organized method for storing and managing information [17]. Filing systems may be traditional or electronic: traditional filing offers familiarity and direct interaction but is prone to slow retrieval and misplaced documents. While, electronic filing provides faster processing, fewer errors, convenience, and cost efficiency, allowing digital files to be easily sorted, retrieved, and shared [18][19].

Record Management

A record "lifecycle" refers to the stages a record goes through from creation to disposal [20]. As digital records become more common, robust records management strategies are needed to ensure their trustworthiness, reliability, and usefulness [21]. In the public sector, effective records management supports good governance by addressing issues such as corruption, fraud, and record alteration. Proper management, including the use of electronic records management (ERM) systems, enhances organizational efficiency, information flow, and compliance with legal and regulatory requirements [22].

Web-Based

Computer-based information technology has advanced significantly, improving information delivery and organizational performance, such as in inventory management and operational efficiency [23]. Web-based applications are increasingly used in education and business, particularly for program execution, marketing, and online learning [24]. During the pandemic, web-based systems became crucial for education, reducing the reliance on attendance as a measure of engagement. Integrated platforms, such as the SIESE system, help assess, track, and enhance student engagement while providing automated reporting and interventions to support positive learning outcomes [25].

III. Methodology

Project Design

Figure 1 shows the system architecture of BalikBayani Portal. It shows the interaction between the thirteen modules of the system. The thirteen main modules of the system are the Login/Authentication Module, Dashboard Module, Direct Hire Module, Balik Manggagawa Module, Gov to Gov Module, Information Sheet Module, Monitoring List (List of

Job Fairs) Module, Monitoring List (PESO Contact Info) Module, Monitoring List (PRAs Contact Info) Module, Monitoring List (Job Fair Monitoring Summary) Module, Super Admin Panel (User Management) Module, Super Admin Panel (Back Up) Module and User Profile Module.

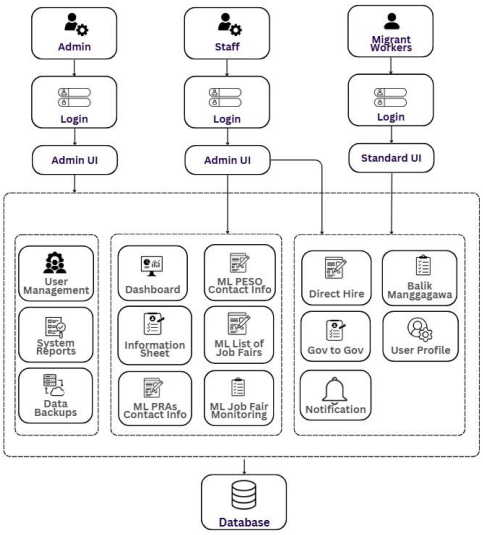


Figure 1. System Architecture

Project Development

Figure 2 shows the BalikBayani Portal's life cycle, from idea to implementation, using the Agile Scrum Framework. This method involves an iterative process consisting of planning, designing, developing, testing, and evaluating the system in the following sprints. Each sprint starts with planning and backlog refinement, followed by development until the system achieves its objectives [26].

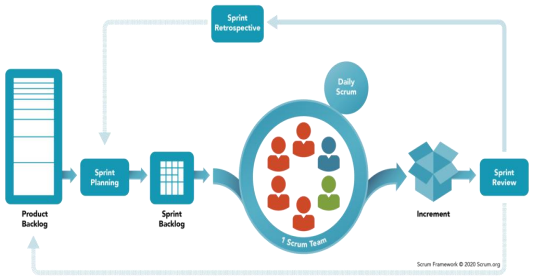


Figure 2. Agile Scrum Framework

Evaluation Procedure

As shown in Figure 3 and 4, the system was evaluated using two different approaches: the ISO 25010 Software Quality Model and the Technology Acceptance Model (TAM).

Figure 3 shows the Technology Acceptance Model (TAM), which was used for evaluating the user acceptance of the system. A questionnaire survey was distributed to chosen participants who served as end users, and the collected responses helped determine how well users accepted and interacted with the system [27].

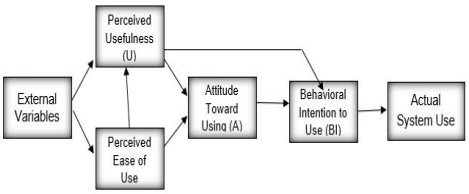


Figure 3. Technology Acceptance Model

Figure 4 shows the ISO 25010 Software Quality Model, a standard framework for evaluating and measuring software quality using measurable characteristics and sub-characteristics. IT professionals evaluated the system based on its nine key characteristics to ensure it meets the essential quality standard [28].

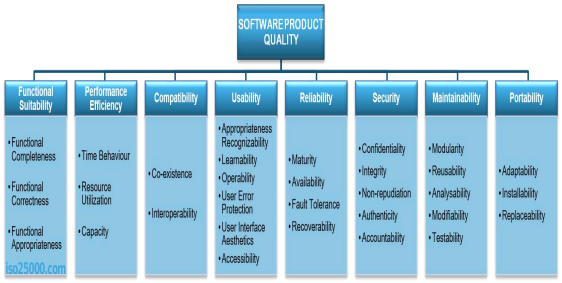


Figure 4. ISO/IEC 25010 Software Product Quality Characteristics

Population of the Study

The population of the study consist of (102) users and (3) IT professionals, for a total of 105 respondents. These respondents would evaluate and test the system and would complete the survey questionnaire provided by the researchers.

Statistical Treatment of Data

The study uses a purposive sampling technique to select respondents and to collect and analyze data from the target respondents. The researchers uses a Likert scale to evaluate the respondents' viewpoint. The Likert scale is rated as follows: 1, with a range of 1.00 - 1.74, interpreted as very low; 2, with a range of 1.75 - 2.49, interpreted as low; 3, with a range of 2.50 - 3.24, interpreted as moderate; and 4, with a range of 3.25 - 4.00, as interpreted as high.

IV. Result and Discussion

Develop a filing system that integrates document generation into printable format and supports quick, accurate retrieval of information.

The Record Details View feature was designed to improve the BalikBayani Portal's ability to properly manage and generate documents. It served as an essential interface for displaying all encoded information, uploaded requirements, and system-generated documents for each migrant worker. Instead of manually searching through multiple spreadsheets, users were able to retrieve a

complete record through a single query. The document generation capability automatically produced official forms in printable formats directly from validated data. This reduced redundant encoding, minimized the chance of human error, and accelerated document preparation. These improvements directly addressed limitations in the previous process, where record lookup and form preparation required multiple manual steps.

Integrate a data visualization feature that provides representation across the main categories.

The BalikBayani Portal achieved the second objective by integrating data visualization across the main categories. The system's visual analytics, in the form of charts and graphs, summarized records and activities stored in the database. These visual representations made it easier for admins and staff to understand and interpret data without the need to manually review large quantities of raw records, enabling users to accurately monitor activities and identify patterns that could be used for performance monitoring and annual reports.

Results of TAM

The evaluation of the survey results, as presented in Table 1, aligns with the Technology Acceptance Model (TAM), which evaluates factors including Perceived Usefulness (PU), Perceived Ease of Use (PEOU), User Satisfaction, and Willingness to Adopt a system. The mean score of 3.78 indicates favorable approval across all TAM criteria, indicating that the system successfully meets users' needs and is likely to be adopted and used consistently.

Table 1. Evaluation of survey results

Item	Mean	Standard Deviation	Interpretation
Perceived Usefulness	3.82	±0.08	High
Perceived Ease of Use	3.72	±0.05	High
User Satisfaction	3.79	±0.06	High
Willingness to Adopt	3.79	±0.04	High
Overall Mean	**3.78**	**±0.02**	**High**

***Legends**: 1.00–1.74 (Very Low), 1.75–2.49 (Low), 2.50–3.24 (Moderate), 3.25–4.00 (High).*

Results of ISO25010

The evaluation results presented in Table 2, adapted from the ISO/IEC 25010 software quality model, show outstanding performance across all quality attributes. The overall mean of 4.67, interpreted as 'Excellent,' indicates that the system fulfills its intended objectives, reflecting its readiness for real-world applications and potential for long-term use

Table 2. Evaluation of survey results

Item	Mean	Standard Deviation	Interpretation
Functional Suitability	4.78	±0.33	Excellent
Performance Efficiency	4.56	0	Excellent
Compatibility	4.50	0	Excellent
Interaction Capability	4.63	±0.27	Excellent
Reliability	4.50	0	Excellent
Security	4.78	±0.30	Excellent
Maintainability	4.73	±0.26	Excellent
Flexibility	4.67	±0.29	Excellent
Safety	4.93	±0.26	Excellent
Overall Mean	**4.67**	**±0.14**	**Excellent**

***Legend:** 4.21 – 5.00 = Excellent, 3.41 – 4.20 = Good, 2.61 – 3.40 = Fair, 1.81 – 2.60 = Needs Improvement, 1.00 – 1.80 = Poor*

V. Conclusion and Future Work

The development and implementation of the BalikBayani Portal demonstrate the positive impact of digital innovation on migrant worker record management. The system successfully digitizes filing processes, generates official documents, and integrates data visualization to support efficient decision-making. Evaluations using TAM and ISO 25010 standards confirmed the system's performance, usability, reliability, and security, while user feedback highlighted its alignment with organizational needs. The BalikBayani Portal supports the Department of Migrant Workers' mission to promote digital transformation and aligns with UN Sustainable Development Goals by enhancing workplace efficiency, digital infrastructure, fair processing, and equitable access to services. With continued support, the BalikBayani Portal can serve as a model for digital innovation in migrant worker governance.

Future improvements include fully paperless transactions, automated calculation of process cycle times, integration of scanning devices with OCR for automatic data entry, and continuous use by the Processing Division to sustain the system. Further enhancements could involve adopting advanced technologies to improve functionality and maintainability and integrating a mobile interface to allow applicants to access and monitor applications in real-time.

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